

2FH4 Cheat Sheet

Vector Algebra

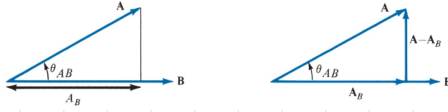
- $\vec{A} = (A_x, A_y, A_z) = A_x \hat{a}_x + A_y \hat{a}_y + A_z \hat{a}_z = A_x \hat{i} + A_y \hat{j} + A_z \hat{k}$
- $|\vec{A}| = \sqrt{A_x^2 + A_y^2 + A_z^2}$
- $a_A = \frac{\vec{A}}{|\vec{A}|}$
- Vector Addition & Subtraction Properties
 - commutative
 - associative
 - distributive
- Position & Displacement
 - $\vec{PQ} = \vec{r}_{PQ} = \vec{r}_Q - \vec{r}_P$
- Dot Product :
 - $A \cdot B = |A||B| \cos \theta_{AB}$
 - Properties:
 - commutative : $\vec{A} \cdot \vec{B} = \vec{B} \cdot \vec{A}$
 - distributive : $\vec{A} \cdot (\vec{B} + \vec{C}) = \vec{A} \cdot \vec{B} + \vec{A} \cdot \vec{C}$
 - $A \cdot A = |A|^2$
- Cross Product :
 - $A \times B = a_n |A||B| \sin \theta_{AB}$
 - Properties:
 - Anticommutative $\rightarrow$ order matters
 - Distributive : $\vec{A} \times (\vec{B} + \vec{C}) = \vec{A} \times \vec{B} + \vec{A} \times \vec{C}$
 - Not associative
 - Scaling : $k(\vec{A} \times \vec{B}) = k\vec{A} \times \vec{B} = \vec{A} \times k\vec{B}$
 - $A \times A = 0$
 - $x \xrightarrow{-z} y \xrightarrow{-x} z \xrightarrow{-y} x$
- Scalar triple product
 - $A \cdot (B \times C)$
 - results in scalar
- Cross triple product
 - $A \times (B \times C) = B(A \cdot C) - C(A \cdot B)$
 - results in vector

• Components of a Vector

Scalar product can be used to find the projection of $\mathbf{A}$ along $\mathbf{B}$. For this purpose, we need to do dot multiplication between $\mathbf{A}$ and the unit vector in the direction of $\mathbf{B}$, $\mathbf{a}_B$.

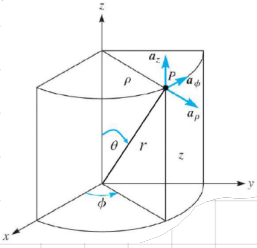
$$A_B = A \cos \theta_{AB} = |\mathbf{A}| |\mathbf{a}_B| \cos \theta_{AB} = \mathbf{A} \cdot \mathbf{a}_B$$

$$\mathbf{A}_B = A_B \mathbf{a}_B = (\mathbf{A} \cdot \mathbf{a}_B) \mathbf{a}_B$$



• Coordinate Systems

- Cylindrical Coordinates - $\vec{A} = (A_\rho, A_\phi, A_z)$
 - best for rotational symmetry around z-axis



- ρ - distance from z-axis
- ϕ - $\angle$ between x & y from +x
- z - height

$$\mathbf{R} = \rho \mathbf{a}_\rho + z \mathbf{a}_z$$

* no a_ϕ term

↳ $a_\phi \perp$ to plane of $\mathbf{R}$ & $\mathbf{a}_z$

$$\mathbf{a}_\rho = \cos \phi \mathbf{a}_x + \sin \phi \mathbf{a}_y$$

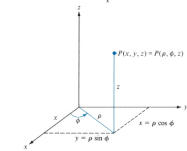
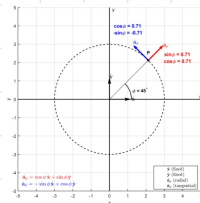
$$x = \rho \cos \phi$$

$$y = \rho \sin \phi$$

$$\rho = \sqrt{x^2 + y^2}$$

$$\rho = r \sin \theta$$

$$r = |\mathbf{R}| = \sqrt{x^2 + y^2 + z^2}$$



$0 \leq \rho < \infty$ $0 \leq \phi < 2\pi$ $-\infty < z < \infty$	$A_\rho \mathbf{a}_\rho + A_\phi \mathbf{a}_\phi + A_z \mathbf{a}_z$ $\mathbf{a}_\rho \cdot \mathbf{a}_\rho = \mathbf{a}_\phi \cdot \mathbf{a}_\phi = \mathbf{a}_z \cdot \mathbf{a}_z = 1$ $\mathbf{a}_\rho \cdot \mathbf{a}_\phi = \mathbf{a}_\phi \cdot \mathbf{a}_z = \mathbf{a}_z \cdot \mathbf{a}_\rho = 0$
$\mathbf{a}_\rho \times \mathbf{a}_\phi = \mathbf{a}_z$ $\mathbf{a}_\phi \times \mathbf{a}_z = \mathbf{a}_\rho$ $\mathbf{a}_z \times \mathbf{a}_\rho = \mathbf{a}_\phi$	➤ Cylindrical to Rectangular $x = \rho \cos \phi, \quad y = \rho \sin \phi, \quad z = z$
◆ form a right-handed coordinate system.	➤ Rectangular to Cylindrical $\rho = \sqrt{x^2 + y^2}, \quad \phi = \tan^{-1} \frac{y}{x}, \quad z = z$
➤ Position vector and magnitude $\mathbf{R} = x \mathbf{a}_x + y \mathbf{a}_y + z \mathbf{a}_z$ $\mathbf{R} = \rho \mathbf{a}_\rho + z \mathbf{a}_z$	$ \mathbf{R} = \sqrt{x^2 + y^2 + z^2} = \sqrt{\rho^2 + z^2}$
➤ Cylindrical to Rectangular and vice-versa unit vectors $\mathbf{a}_\rho = \cos \phi \mathbf{a}_x + \sin \phi \mathbf{a}_y$ $\mathbf{a}_\phi = -\sin \phi \mathbf{a}_x + \cos \phi \mathbf{a}_y$	$\mathbf{a}_x = \cos \phi \mathbf{a}_\rho - \sin \phi \mathbf{a}_\phi$ $\mathbf{a}_y = \sin \phi \mathbf{a}_\rho + \cos \phi \mathbf{a}_\phi$

- Spherical Coordinates : $\vec{A} = (A_r, A_\theta, A_\phi)$

$$\mathbf{a}_r \cdot \mathbf{a}_r = \mathbf{a}_\theta \cdot \mathbf{a}_\theta = \mathbf{a}_\phi \cdot \mathbf{a}_\phi = 1$$

$$\mathbf{a}_r \cdot \mathbf{a}_\theta = \mathbf{a}_\theta \cdot \mathbf{a}_\phi = \mathbf{a}_\phi \cdot \mathbf{a}_r = 0$$

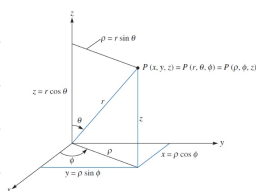
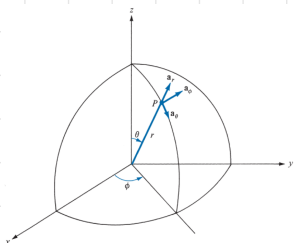
$$\mathbf{a}_r \times \mathbf{a}_\theta = \mathbf{a}_\phi$$

$$\mathbf{a}_\theta \times \mathbf{a}_\phi = \mathbf{a}_r$$

$$\mathbf{a}_\phi \times \mathbf{a}_r = \mathbf{a}_\theta$$

$$r = \sqrt{x^2 + y^2 + z^2}, \quad \theta = \tan^{-1} \frac{\sqrt{x^2 + y^2}}{z}, \quad \phi = \tan^{-1} \frac{y}{x}$$

$$x = r \sin \theta \cos \phi, \quad y = r \sin \theta \sin \phi, \quad z = r \cos \theta$$



$$\mathbf{a}_x = \sin \theta \cos \phi \mathbf{a}_r + \cos \theta \cos \phi \mathbf{a}_\theta - \sin \phi \mathbf{a}_\phi$$

$$\mathbf{a}_y = \sin \theta \sin \phi \mathbf{a}_r + \cos \theta \sin \phi \mathbf{a}_\theta + \cos \phi \mathbf{a}_\phi$$

$$\mathbf{a}_z = \cos \theta \mathbf{a}_r - \sin \theta \mathbf{a}_\theta$$

Vector Calculus

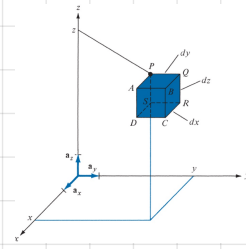
Rectangular Coordinates

- differential displacement/length:
- differential normal surface area:
- differential volume:

$$d\vec{l} = dx\vec{a}_x + dy\vec{a}_y + dz\vec{a}_z$$

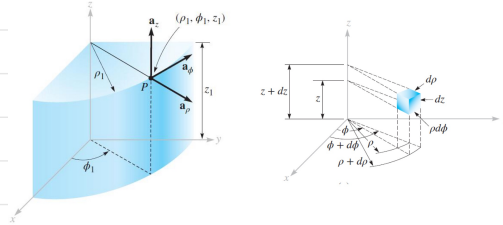
$$d\vec{S} = \vec{n}dS$$

$$dV = dx dy dz$$



Cylindrical Coordinates

- $\rho = \sqrt{x^2 + y^2}$
- $\phi = \tan^{-1}(y/x)$
- $z = z$



- differential displacement/length:
- differential normal surface area:
- differential volume:

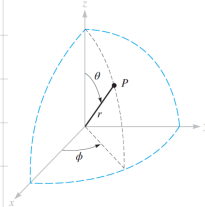
$$d\vec{l} = d\rho\vec{a}_\rho + \rho d\phi\vec{a}_\phi + dz\vec{a}_z$$

$$d\vec{S} = \vec{n}dS$$

$$dV = \rho d\rho d\phi dz$$

Spherical Coordinates

- $r = \sqrt{x^2 + y^2 + z^2}$
- $\theta = \cos^{-1}(z/r) = \cos^{-1}\left(\frac{z}{\sqrt{x^2 + y^2 + z^2}}\right)$
- $\phi = \tan^{-1}(y/x)$

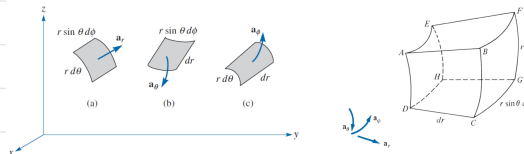


- differential displacement/length:
- differential normal surface area:
- differential volume:

$$d\vec{l} = dr\vec{a}_r + r d\theta\vec{a}_\theta + r \sin\theta d\phi\vec{a}_\phi$$

$$d\vec{S} = \vec{n}dS$$

$$dV = r^2 \sin\theta dr d\theta d\phi$$



- Line Integral: $\oint_L A \cdot d\vec{l}$
- Surface Integral: $\Psi = \int_S A \cdot d\vec{S}$
- Volume Integral: $\int_V \rho_v dV$

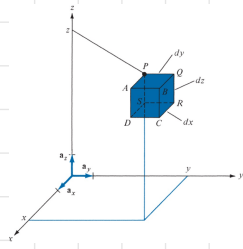
Vector Calculus

Rectangular Coordinates

$$d\vec{l} = dx\vec{a}_x + dy\vec{a}_y + dz\vec{a}_z$$

$$d\vec{S} = \vec{n}ds$$

$$dv = dx dy dz$$



Cylindrical Coordinates

$$d\vec{l} = d\rho\vec{a}_\rho + \rho d\phi\vec{a}_\phi + dz\vec{a}_z$$

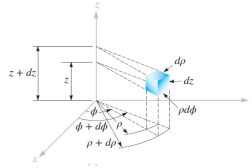
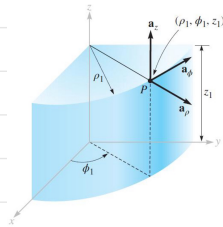
$$d\vec{S} = \vec{n}ds$$

$$dv = \rho d\rho d\phi dz$$

$$\rho = \sqrt{x^2 + y^2}$$

$$\phi = \tan^{-1}(y/x)$$

$$z = z$$



Spherical Coordinates

$$d\vec{l} = dr\vec{a}_r + r d\theta\vec{a}_\theta + r \sin\theta d\phi\vec{a}_\phi$$

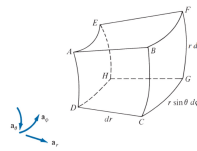
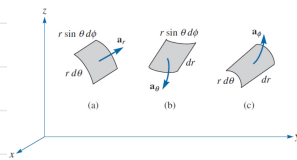
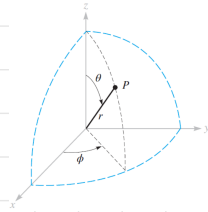
$$d\vec{S} = \vec{n}ds$$

$$dv = r^2 \sin\theta dr d\theta d\phi$$

$$r = \sqrt{x^2 + y^2 + z^2}$$

$$\theta = \cos^{-1}\left(\frac{z}{r}\right)$$

$$\phi = \tan^{-1}(y/x)$$



Line Integral : $\oint_L \vec{A} \cdot d\vec{l}$

Surface Integral : $\Psi = \int_S \vec{A} \cdot d\vec{S}$

Volume Integral : $\int_V \rho_v dv$

• Field

- physical quantity defined @ every pt in space

- $\phi(x, y, z)$, $f(x, y, z)$, $\vec{V}(x, y, z)$
- $\vec{A}(x, y, z)$, $\vec{E}(x, y, z)$
- $\vec{A}(x, y, z) = A_x(x, y, z)\hat{a}_x + A_y(x, y, z)\hat{a}_y + A_z(x, y, z)\hat{a}_z$

- Scalar field - only quantity, no direction

- Eg. Temp: $T = T(x, y, z)$
Electric Potential: $V = V(x, y, z)$
Charge Density: $\rho = \rho(x, y, z)$

- Vector field

- Del Operator - ∇ ie Nabla

- vector like form, components are differential operators, not physical vector components
- used to describe change
- tells how fields vary in space when applied to scalar & vector

- Rectangular: $\nabla = \frac{\partial}{\partial x}\hat{a}_x + \frac{\partial}{\partial y}\hat{a}_y + \frac{\partial}{\partial z}\hat{a}_z$
- Cylindrical: $\nabla = \hat{a}_\rho \frac{\partial}{\partial \rho} + \hat{a}_\phi \frac{1}{\rho} \frac{\partial}{\partial \phi} + \hat{a}_z \frac{\partial}{\partial z}$
- Spherical: $\nabla = \hat{a}_r \frac{\partial}{\partial r} + \hat{a}_\theta \frac{1}{r} \frac{\partial}{\partial \theta} + \hat{a}_\phi \frac{1}{r \sin \theta} \frac{\partial}{\partial \phi}$

- Gradient - determines ROC of scalar field along given direction

- result of ∇ operator on scalar field
- gives direction of max ROC/ change
↳ ie where change is the strongest

- Rectangular: $\nabla V = \frac{\partial V}{\partial x}\hat{a}_x + \frac{\partial V}{\partial y}\hat{a}_y + \frac{\partial V}{\partial z}\hat{a}_z$
- Cylindrical: $\nabla V = \frac{\partial V}{\partial \rho}\hat{a}_\rho + \frac{1}{\rho} \frac{\partial V}{\partial \phi}\hat{a}_\phi + \frac{\partial V}{\partial z}\hat{a}_z$
- Spherical: $\nabla V = \frac{\partial V}{\partial r}\hat{a}_r + \frac{1}{r} \frac{\partial V}{\partial \theta}\hat{a}_\theta + \frac{1}{r \sin \theta} \frac{\partial V}{\partial \phi}\hat{a}_\phi$

- Divergence - ∇ of vector field

- describes how much field spreads out / converges @ each pt of space
- + = diverge/spread out & - = converge
- eg. presence of source & sinks
- local quantity $\rightarrow$ os-by small region around a pt

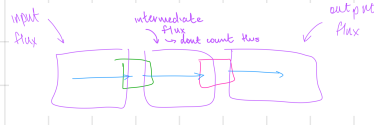
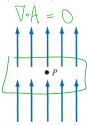
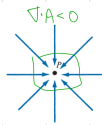
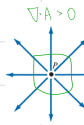
- Rectangular: $\nabla \cdot \vec{A} = \frac{\partial A_x}{\partial x} + \frac{\partial A_y}{\partial y} + \frac{\partial A_z}{\partial z}$
- Cylindrical: $\nabla \cdot \vec{A} = \frac{1}{\rho} \frac{\partial}{\partial \rho}(\rho A_\rho) + \frac{1}{\rho} \frac{\partial A_\phi}{\partial \phi} + \frac{\partial A_z}{\partial z}$
- Spherical: $\nabla \cdot \vec{A} = \frac{1}{r^2} \frac{\partial}{\partial r}(r^2 A_r) + \frac{1}{r \sin \theta} \frac{\partial}{\partial \theta}(A_\theta \sin \theta) + \frac{1}{r \sin \theta} \frac{\partial A_\phi}{\partial \phi}$

◦ Divergence Theorem

- net outward flux of $\vec{A}$ of surrounding closed space P ÷ by volume enclosed $\rightarrow 0$
- States that all flux flows out of a region's boundary = flux generated inside the region

$$\text{div } \mathbf{A} = \nabla \cdot \mathbf{A} = \lim_{\Delta V \rightarrow 0} \frac{\oint_S \mathbf{A} \cdot d\mathbf{S}}{\Delta V}$$

$$\oint_S \mathbf{A} \cdot d\mathbf{S} = \int_V \nabla \cdot \mathbf{A} \, dV$$



◦ Point form - @ single point : $\nabla \cdot \mathbf{A}$

Use this when

1. You want the divergence at **a point**
2. You are given $A(x, y, z)$ explicitly
3. You are analyzing sources and sinks locally

Applications

1. Charge density in electromagnetics
2. Fluid flow, compression or expansion at a point
3. Heat flow or diffusion problems

◦ Integral / Flux form - over finite region $\oint_S \mathbf{A} \cdot d\mathbf{S} = \int_V (\nabla \cdot \mathbf{A}) \, dV$

Use this when

1. You want the total flux through a surface
2. The surface has symmetry
3. Direct surface integration is hard, but volume behavior is simple

Applications

1. Gauss's law in electromagnetics
2. Total charge enclosed
3. Verifying conservation laws

◦ Curl - axis of rotation

$$\text{curl } \mathbf{A} = \nabla \times \mathbf{A} = \left(\lim_{\Delta S \rightarrow 0} \frac{\oint_L \mathbf{A} \cdot d\mathbf{l}}{\Delta S} \right)_{\max} \mathbf{a}_n$$

- zero curl $\neq$ zero field
- measures magnitude & direction field wants to rotate @ a point

◦ Rectangular :

$$\nabla \times \mathbf{A} = \begin{vmatrix} \mathbf{a}_x & \mathbf{a}_y & \mathbf{a}_z \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ A_x & A_y & A_z \end{vmatrix}$$

◦ Cylindrical :

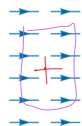
$$\nabla \times \mathbf{A} = \frac{1}{\rho} \begin{vmatrix} \mathbf{a}_\rho & \rho \mathbf{a}_\phi & \mathbf{a}_z \\ \frac{\partial}{\partial \rho} & \frac{\partial}{\partial \phi} & \frac{\partial}{\partial z} \\ A_\rho & \rho A_\phi & A_z \end{vmatrix}$$

◦ Spherical :

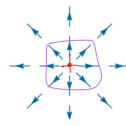
$$\nabla \times \mathbf{A} = \frac{1}{r^2 \sin \theta} \begin{vmatrix} \mathbf{a}_r & r \mathbf{a}_\theta & r \sin \theta \mathbf{a}_\phi \\ \frac{\partial}{\partial r} & \frac{\partial}{\partial \theta} & \frac{\partial}{\partial \phi} \\ A_r & r A_\theta & r \sin \theta A_\phi \end{vmatrix}$$

◦ Stokes Theorem - circulation of vector field = curl of field of any surface ↳ bounded by closed path / area

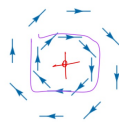
$$\oint_{L_{\text{enc}}} \mathbf{A} \cdot d\mathbf{l} = \int_{\text{Surface}} (\nabla \times \mathbf{A}) \cdot d\mathbf{S}$$



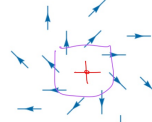
$$\nabla \cdot \mathbf{A} = 0, \nabla \times \mathbf{A} = 0$$



$$\nabla \cdot \mathbf{A} \neq 0, \nabla \times \mathbf{A} = 0$$



$$\nabla \cdot \mathbf{A} = 0, \nabla \times \mathbf{A} \neq 0$$



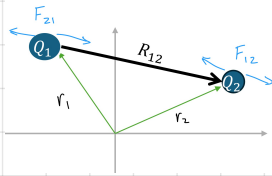
$$\nabla \cdot \mathbf{A} \neq 0, \nabla \times \mathbf{A} \neq 0$$

- Laplacian -

Electrostatics

- Coulomb's Law - $F = k \frac{Q_1 Q_2}{R^2}$, $m = \text{negligible}$

- $Q = 1C = 1.6 \times 10^{19} e^-$
- $R = \text{meters}$
- $F = N$
- $k = \frac{Nm^2}{C^2} = \frac{1}{4\pi\epsilon_0} = 9 \times 10^9 \text{ N/F}$
- $\epsilon_0 = F/m = \frac{1}{36\pi \times 10^9} = \text{Energy stored in defined space}$



$$F_{12} = \frac{Q_1 Q_2}{4\pi\epsilon_0 R^2} \hat{a}_{R_{12}} = \frac{Q_1 Q_2 \vec{R}_{12}}{4\pi\epsilon_0 R_{12}^3} = \frac{Q_1 Q_2}{4\pi\epsilon_0 R^2} \cdot \frac{\vec{R}_{12}}{|\vec{R}_{12}|}$$

$$\vec{F} = \frac{Q}{4\pi\epsilon_0} \sum_{k=1}^N \frac{Q_k (\vec{r} - \vec{r}_k)}{|\vec{r} - \vec{r}_k|^3}$$

- ϵ field due to a charge: $\vec{E} = \lim_{Q \rightarrow 0} \frac{\vec{F}}{Q} = \frac{\vec{F}}{Q}$

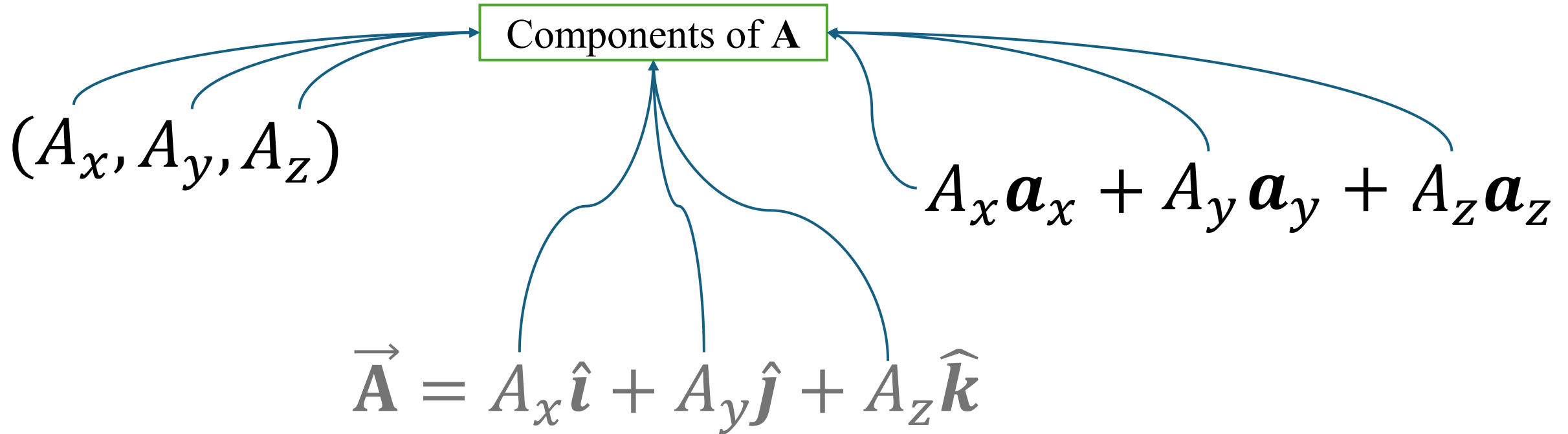
• line: $Q = \int_L \rho_L dl$ $E = \int_L \frac{\rho_L dl}{4\pi\epsilon_0 R^2} \hat{a}_R$ $\rho_L = \frac{Q}{L} = \frac{1}{L} \sum_{k=1}^N Q_k$: $Q = \int_L \rho_L dl$ & $E = \int_L \frac{\rho_L dl}{4\pi\epsilon_0 R^2} \hat{a}_R$

• surface: $dQ = \rho_S dS \rightarrow Q = \int_S \rho_S dS$
 $E = \int_S \frac{\rho_S dS}{4\pi\epsilon_0 R^2} \hat{a}_R$

• volume: $dQ = \rho_V dv \rightarrow Q = \int_V \rho_V dv$
 $E = \int_V \frac{\rho_V dv}{4\pi\epsilon_0 R^2} \hat{a}_R$

Vector Algebra – Unit Vector – Vectors in Rectangular System

- When we want to represent a vector in a rectangular system there are different notation we can use:



Vector Algebra – Unit Vector – Magnitude of a Vector

- The following formula gives the **magnitude** of a *vector* expressed in rectangular components:

$$A = |\mathbf{A}| = \sqrt{A_x^2 + A_y^2 + A_z^2}$$

- ❖ *The magnitude of a vector is coordinate-independent.*

Vector Algebra – Unit Vector

➤ Now that we know the **magnitude** of a vector, we can define its **unit vector**.

$$\mathbf{a}_A = \frac{\mathbf{A}}{|\mathbf{A}|} = \frac{A_x \mathbf{a}_x + A_y \mathbf{a}_y + A_z \mathbf{a}_z}{\sqrt{A_x^2 + A_y^2 + A_z^2}}$$

$$\mathbf{a}_A = \frac{A_x}{|\mathbf{A}|} \mathbf{a}_x + \frac{A_y}{|\mathbf{A}|} \mathbf{a}_y + \frac{A_z}{|\mathbf{A}|} \mathbf{a}_z$$

Vector Algebra – Position and Displacement Vectors

Position Vector: the position vector, $\mathbf{r}_P$, of point P is defined as **directed distance** from the origin $O(0,0,0)$ to $P(x,y,z)$:

$$\mathbf{r}_P = \overrightarrow{OP} = x\mathbf{a}_x + y\mathbf{a}_y + z\mathbf{a}_z$$

- ❖ *The position vector points from the origin to the point; its magnitude equals the shortest distance, but the vector itself includes direction.*

Vector Algebra – Vector Multiplication

- There are two types of multiplication defined for vectors,
 1. Dot product $\mathbf{A} \cdot \mathbf{B}$
 2. Cross product $\mathbf{A} \times \mathbf{B}$
- The result of each of these multiplications has a different physical meaning.
- The other name for **dot product** is **scalar product** → because the result is a scalar quantity.
- The other name for **cross product** is **vector product** → because the result is a vector quantity.
- ❖ *Dot product tells us how **aligned** two vectors are; cross product tells us how **perpendicular** they are.*

Vector Algebra – The Dot Product, Definition

- Dot product = scalar product = inner product
- Geometric definition of dot product

$$\mathbf{A} \cdot \mathbf{B} = |\mathbf{A}||\mathbf{B}|\cos\theta_{AB}$$

- Operational definition of dot product,

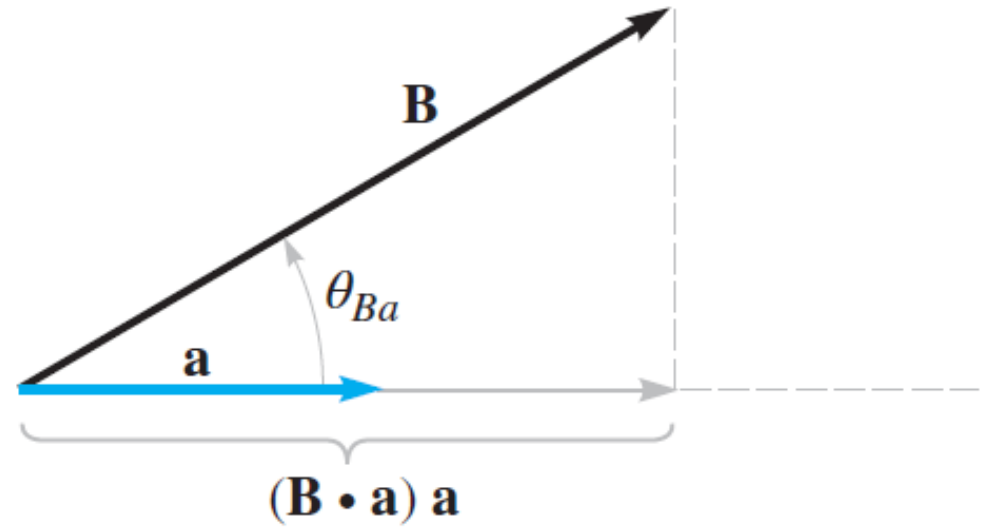
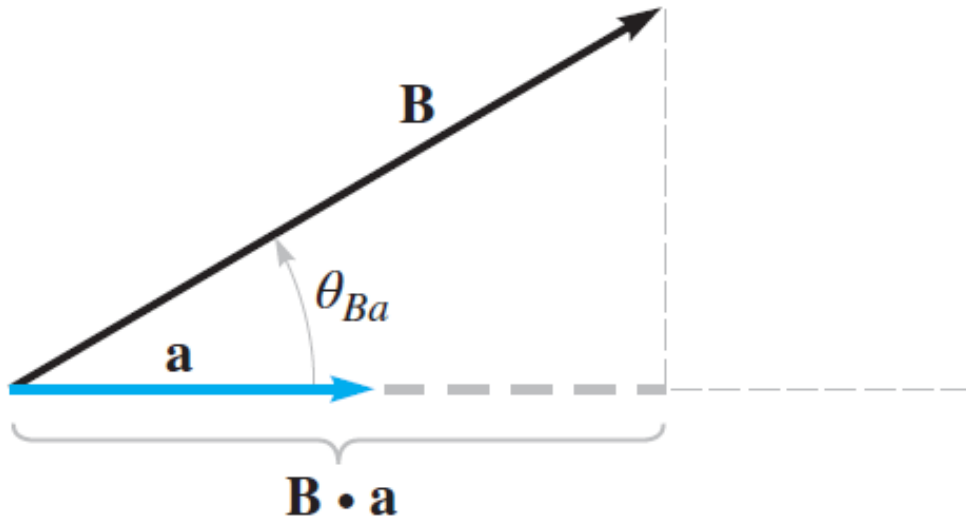
$$\mathbf{A} = A_x\mathbf{a}_x + A_y\mathbf{a}_y + A_z\mathbf{a}_z \quad \& \quad \mathbf{B} = B_x\mathbf{a}_x + B_y\mathbf{a}_y + B_z\mathbf{a}_z$$

- $\mathbf{A} \cdot \mathbf{B} =$

Vector Algebra – The Dot Product, Properties & Applications

1. Commutative
2. Distributive
3. Dot product of a vector by itself is its magnitude squared
4. As the application of dot product is that we can use it to find the components of a vector in the direction of unit vector.

Vector Algebra – The Dot Product, Properties & Applications



Vector Algebra – The Cross Product, Definition

- Cross product = vector product = outer product
- Geometric definition of cross product

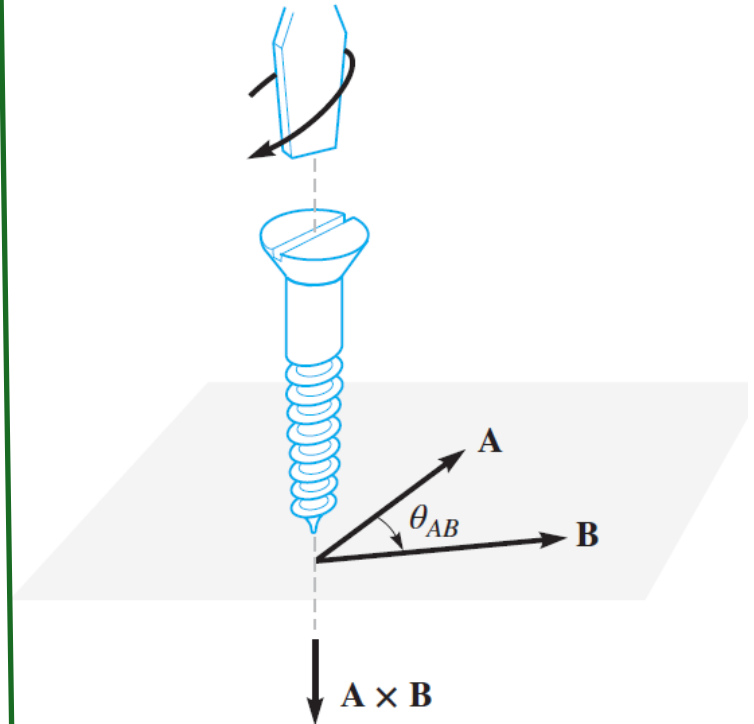
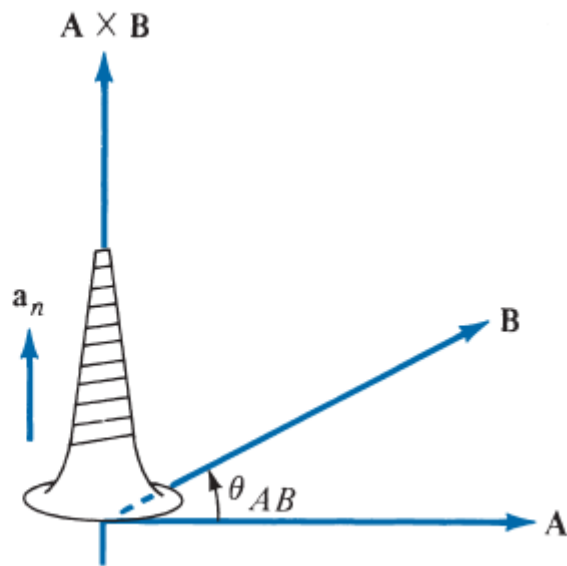
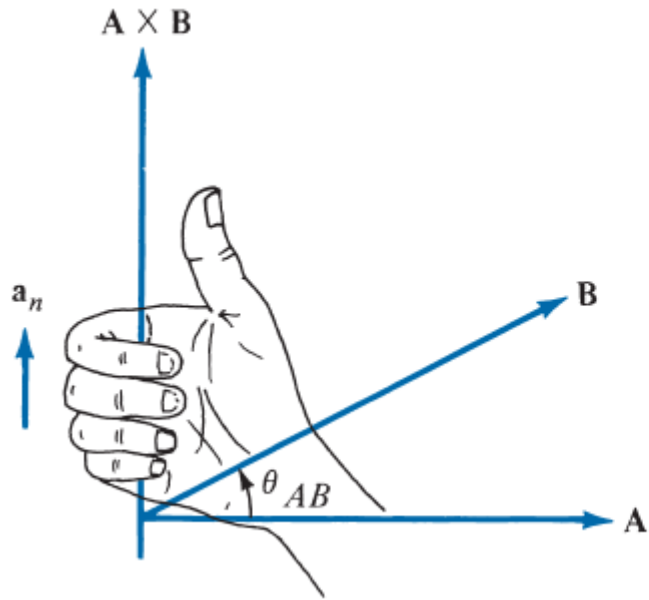
$$\mathbf{A} \times \mathbf{B} = \mathbf{a}_n |\mathbf{A}| |\mathbf{B}| \sin \theta_{AB}$$

- Operational definition of cross product,

$$\mathbf{A} = A_x \mathbf{a}_x + A_y \mathbf{a}_y + A_z \mathbf{a}_z \quad \& \quad \mathbf{B} = B_x \mathbf{a}_x + B_y \mathbf{a}_y + B_z \mathbf{a}_z$$

- $\mathbf{A} \times \mathbf{B} =$

Vector Algebra – The Cross Product, Definition



Vector Algebra – The Cross Product, Definition

$$\mathbf{A} \times \mathbf{B} = (A_y B_z - A_z B_y) \mathbf{a}_x + (A_z B_x - A_x B_z) \mathbf{a}_y + (A_x B_y - A_y B_x) \mathbf{a}_z$$

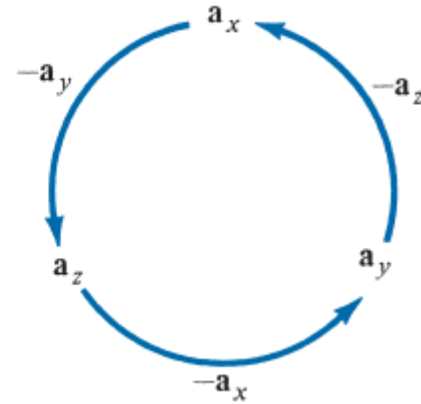
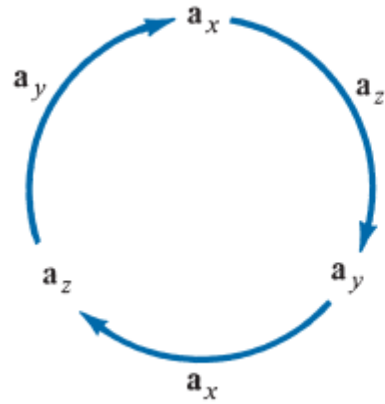
- Example $\mathbf{A} = 2\mathbf{a}_x - 3\mathbf{a}_y + \mathbf{a}_z$ & $\mathbf{B} = -4\mathbf{a}_x - 2\mathbf{a}_y + 5\mathbf{a}_z$

$$\mathbf{A} \times \mathbf{B} =$$

Vector Algebra – The Cross Product, Properties

1. Anticommutative
2. Distributive
3. Not associative
4. Scaling
5. A vector cross multiplication by itself is zero

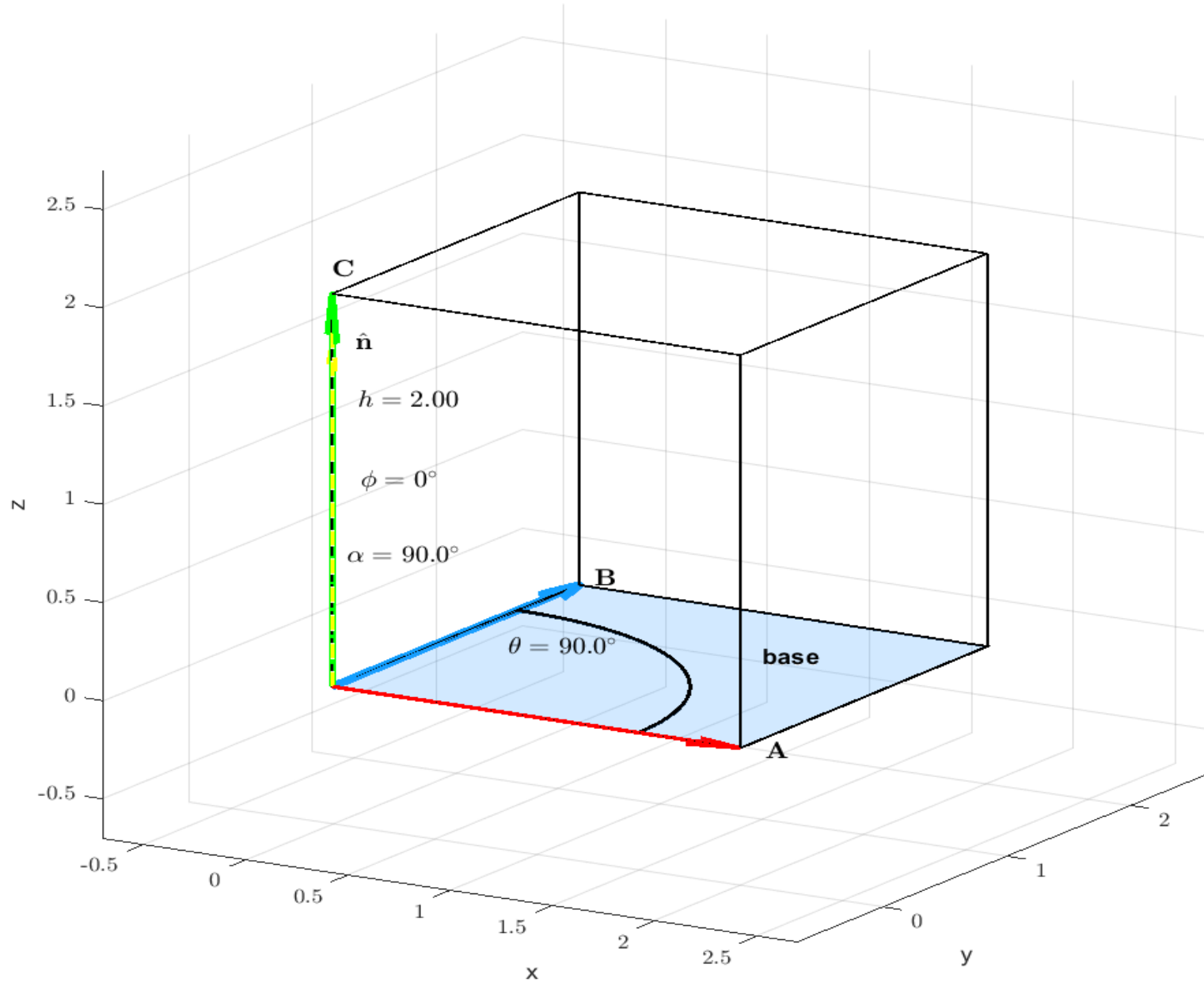
Vector Algebra – The Cross Product, Properties



Vector Algebra – Scalar Triple Product

$$\mathbf{A} \cdot (\mathbf{B} \times \mathbf{C}) =$$

Vector Algebra – Scalar Triple Product

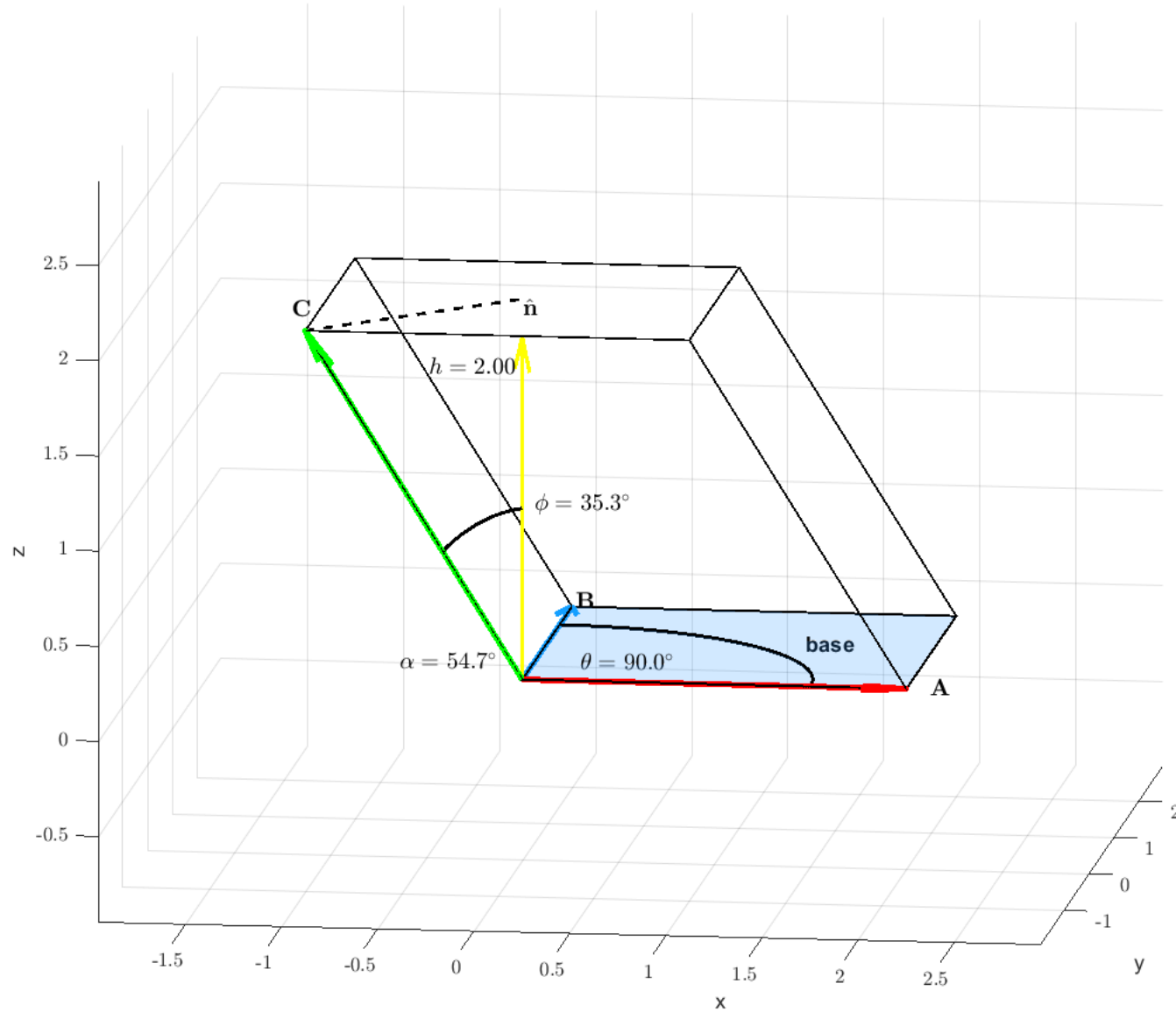


Vector Algebra – Scalar Triple Product

Triple Scalar Product = Volume

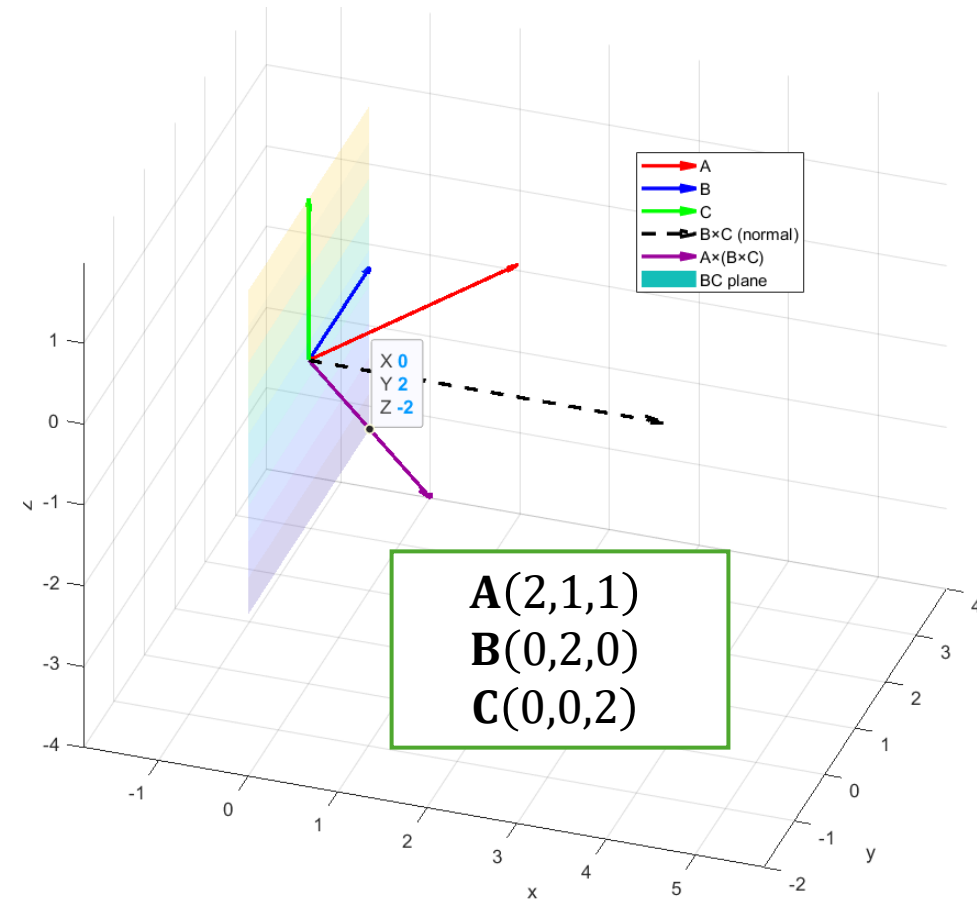
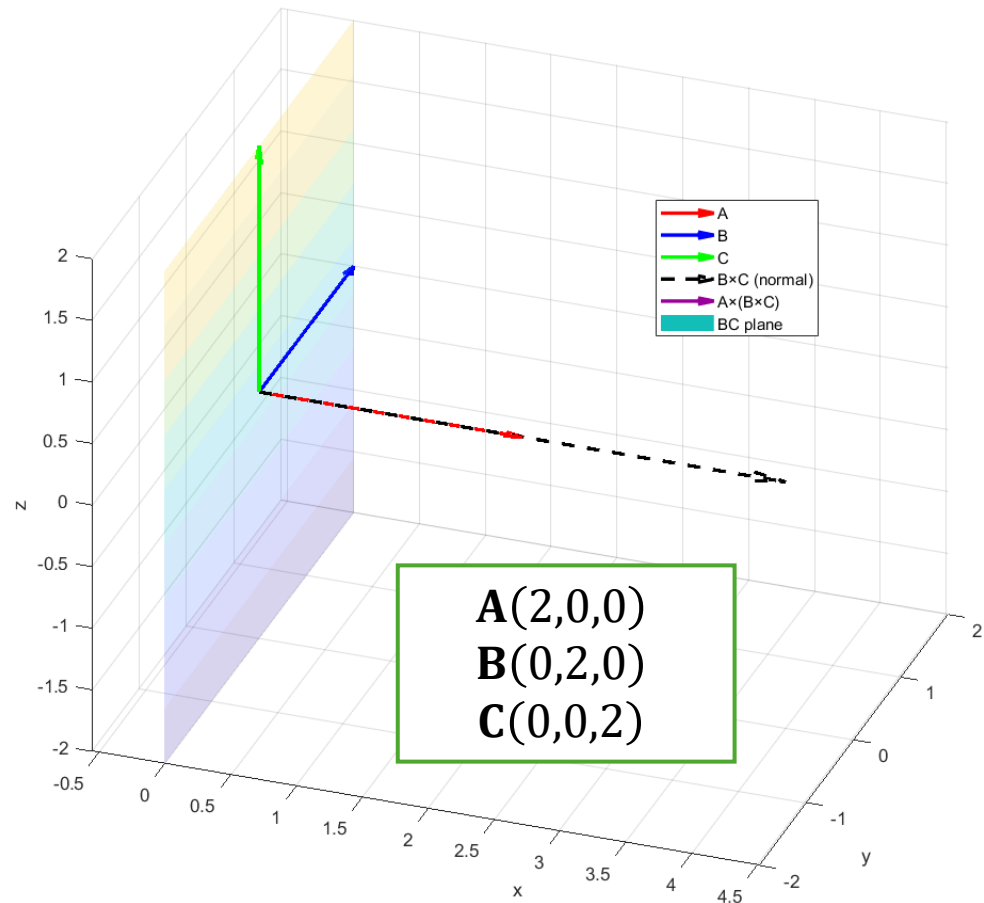
$$|\mathbf{C} \cdot (\mathbf{A} \times \mathbf{B})| = 8.000$$

$$|\mathbf{A} \times \mathbf{B}| = 4.000, \quad h = |\mathbf{C}| \cos \phi = 2.000$$

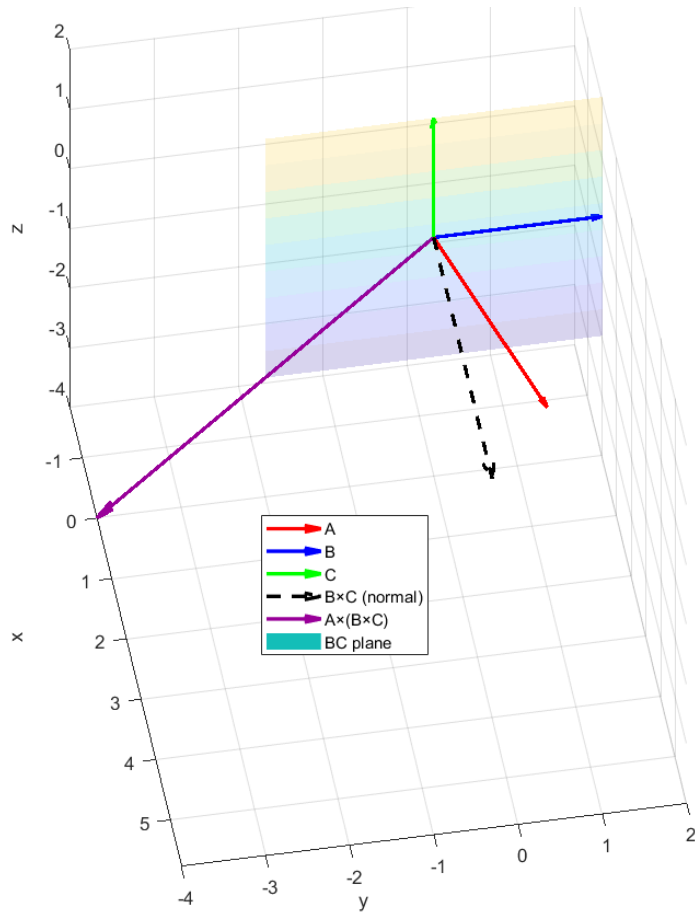


Vector Algebra – Cross Triple Product

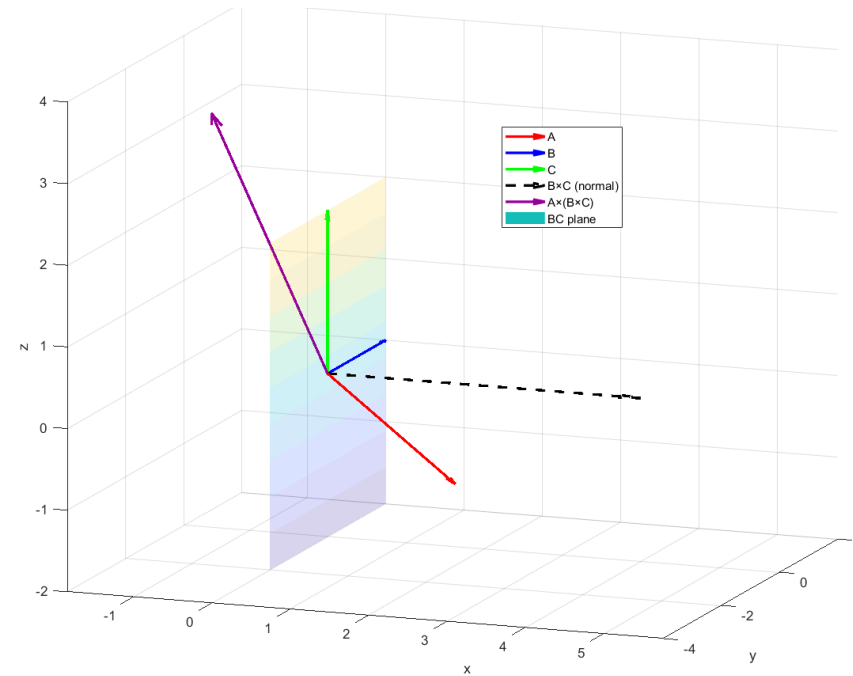
$$\mathbf{A} \times (\mathbf{B} \times \mathbf{C}) = \mathbf{B}(\mathbf{A} \cdot \mathbf{C}) - \mathbf{C}(\mathbf{A} \cdot \mathbf{B})$$



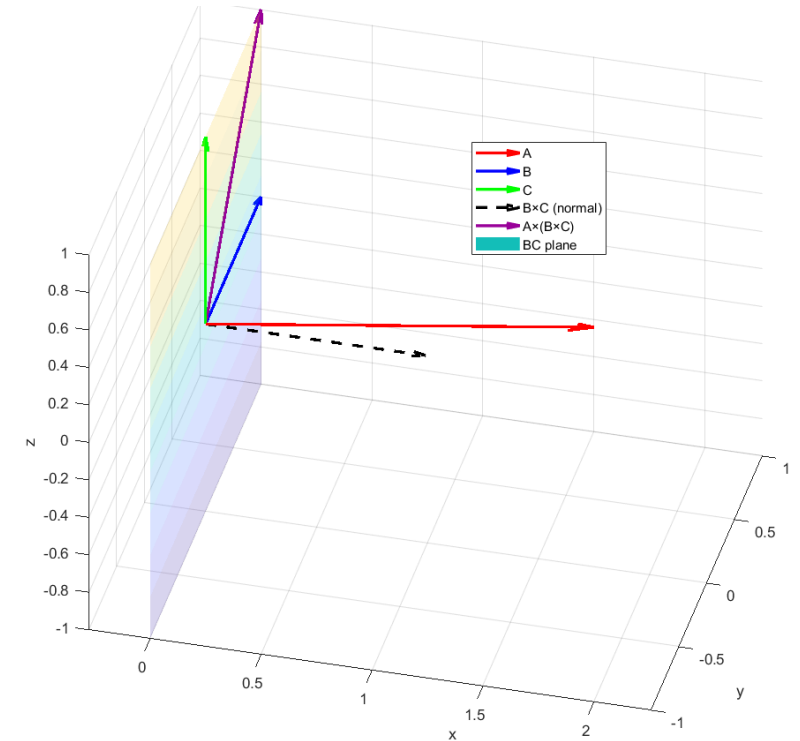
Vector Algebra – Cross Triple Product



$\mathbf{A}(2, 1, -1)$
 $\mathbf{B}(0, 1, 0)$
 $\mathbf{C}(0, 0, 1)$



$\mathbf{A}(2, -1, -1)$
 $\mathbf{B}(0, 2, 0)$
 $\mathbf{C}(0, 0, 2)$



$\mathbf{A}(2, -1, 1)$
 $\mathbf{B}(0, 1, 0)$
 $\mathbf{C}(0, 0, 1)$

Vector Algebra – Cross Triple Product –Example 2

Give the following vectors verify the BAC-CAB rule,

$$\vec{A} = \mathbf{a}_x + 2\mathbf{a}_y + 3\mathbf{a}_z$$

$$\vec{B} = -\mathbf{a}_x + 2\mathbf{a}_z$$

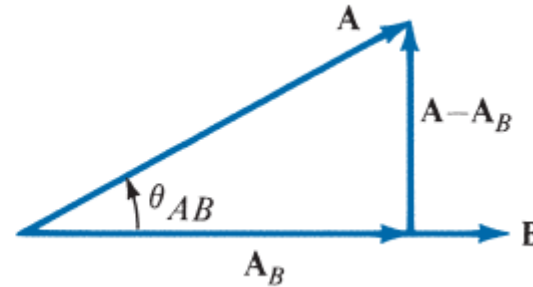
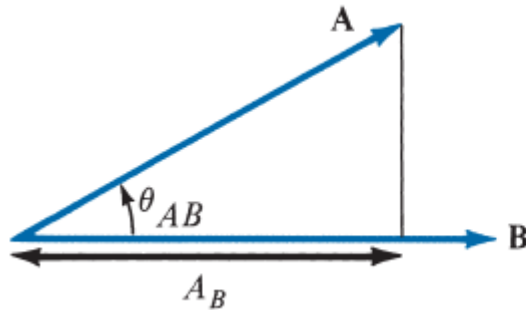
$$\vec{C} = 2\mathbf{a}_x + \mathbf{a}_y - \mathbf{a}_z$$

Vector Algebra – Components of a Vector

- Scalar product can be used to find the projection of **A** along **B**. For this purpose, we need to do dot multiplication between **A** and the unit vector in the direction of **B**, **a_B**.

$$A_B = A \cos \theta_{AB} = |\mathbf{A}| |\mathbf{a}_B| \cos \theta_{AB} = \mathbf{A} \cdot \mathbf{a}_B$$

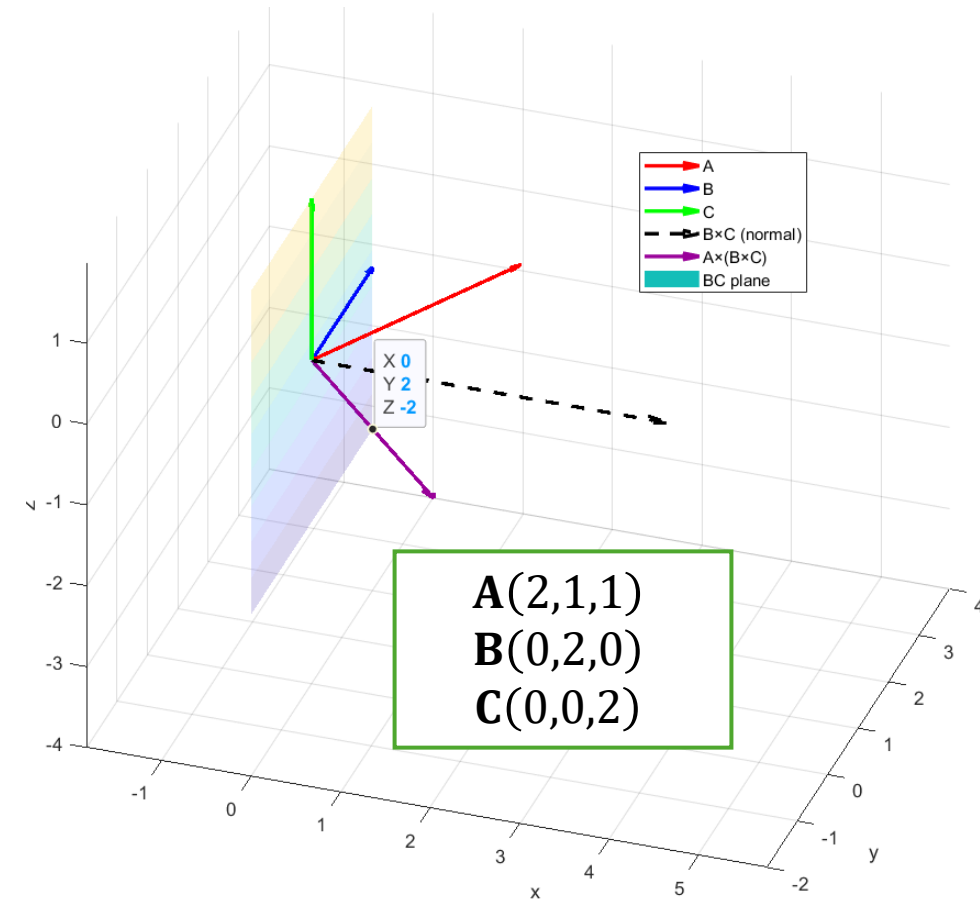
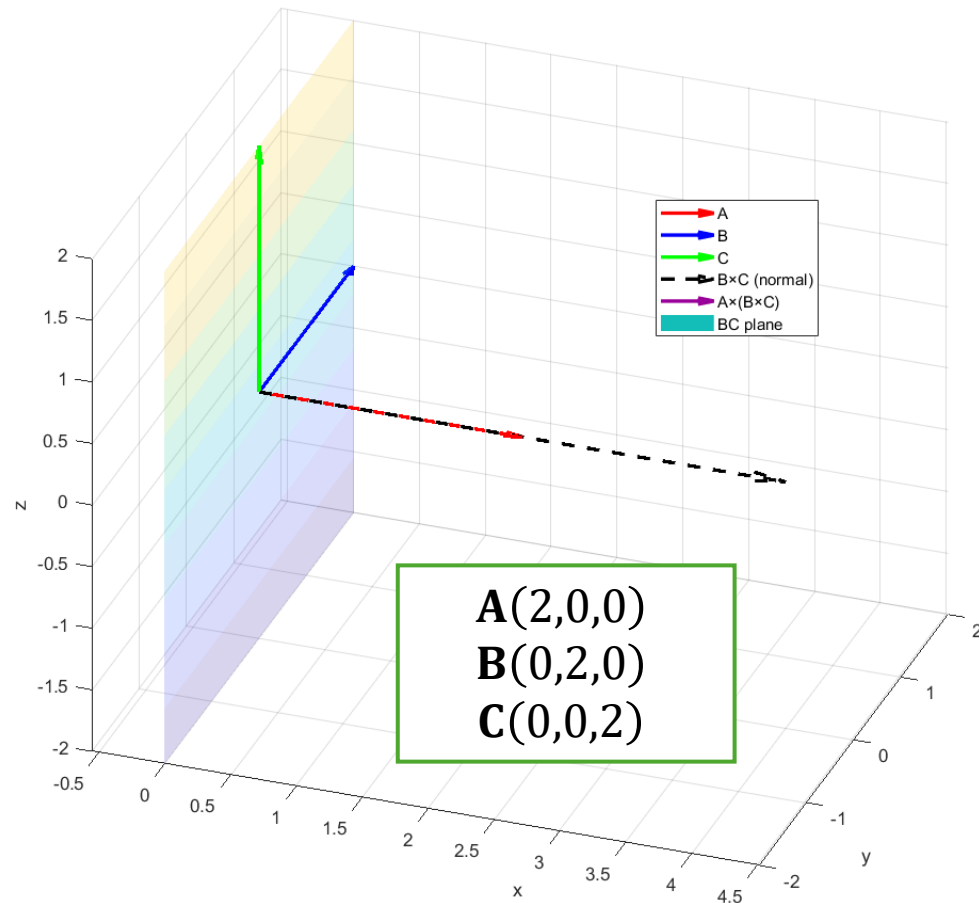
$$\mathbf{A}_B = A_B \mathbf{a}_B = (\mathbf{A} \cdot \mathbf{a}_B) \mathbf{a}_B$$



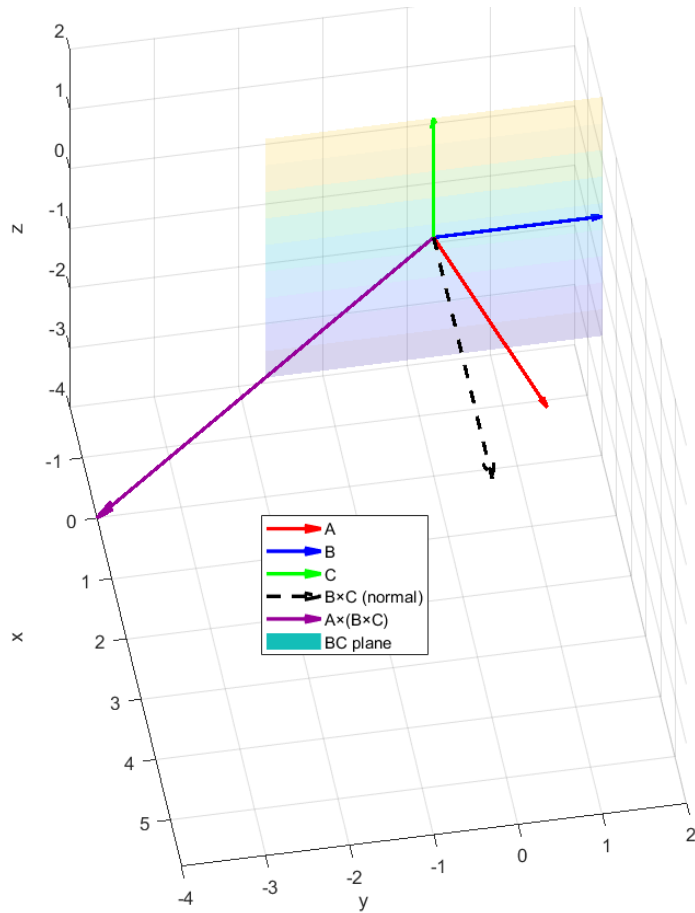
Vector Algebra – Cross Triple Product

$$\mathbf{A} \times (\mathbf{B} \times \mathbf{C}) = \mathbf{B}(\mathbf{A} \cdot \mathbf{C}) - \mathbf{C}(\mathbf{A} \cdot \mathbf{B})$$

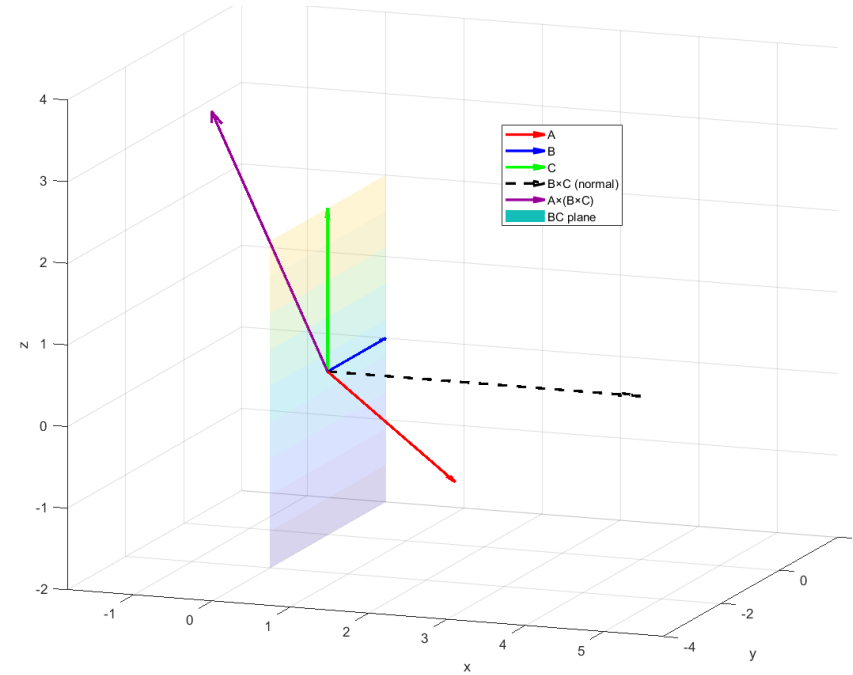
$$= \mathbf{0}$$



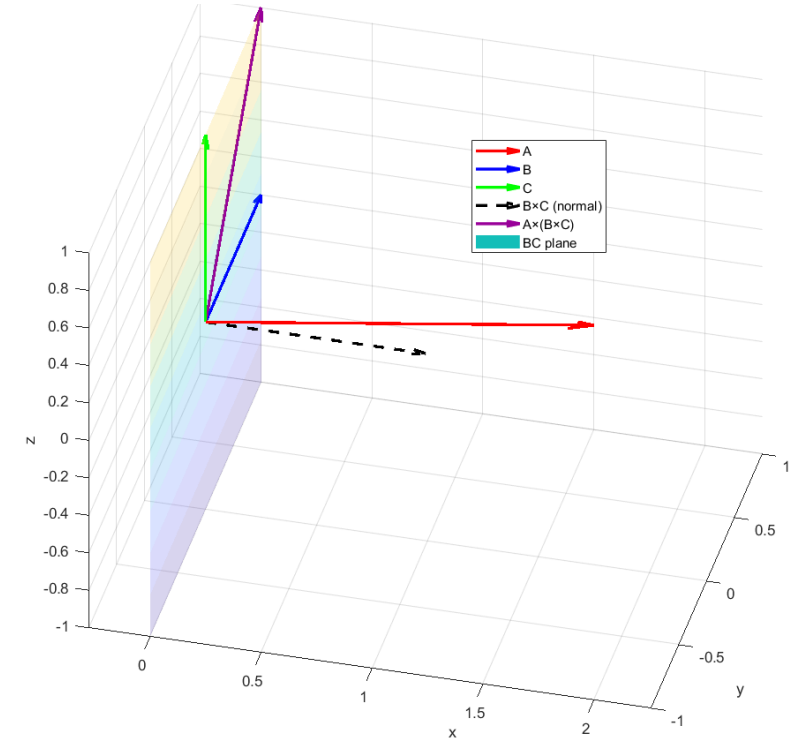
Vector Algebra – Cross Triple Product



$A(2, 1, -1)$
 $B(0, 1, 0)$
 $C(0, 0, 1)$



$A(2, -1, -1)$
 $B(0, 2, 0)$
 $C(0, 0, 2)$



$A(2, -1, 1)$
 $B(0, 1, 0)$
 $C(0, 0, 1)$

Vector Algebra – Cross Triple Product – Example 1

Give the following vectors verify the BAC-CAB rule,

$$\vec{A} = \mathbf{a}_x + 2\mathbf{a}_y + 3\mathbf{a}_z$$

$$\vec{B} = -\mathbf{a}_x + 2\mathbf{a}_z$$

$$\vec{C} = 2\mathbf{a}_x + \mathbf{a}_y - \mathbf{a}_z$$

LH

$$A \times (B \times C)$$

$$B \times C = \begin{vmatrix} \mathbf{a}_x & \mathbf{a}_y & \mathbf{a}_z \\ -1 & 0 & 2 \\ 2 & 1 & -1 \end{vmatrix}$$

$$\vec{D} = -2\mathbf{a}_x + 3\mathbf{a}_y - \mathbf{a}_z$$

$$\vec{A} \times \vec{D} = \begin{vmatrix} \mathbf{a}_x & \mathbf{a}_y & \mathbf{a}_z \\ 1 & 2 & 3 \\ -2 & 3 & -1 \end{vmatrix}$$

$$A \times B \times C = -11\mathbf{a}_x - 5\mathbf{a}_y + 7\mathbf{a}_z$$

= RH

$$= B(A \cdot C) - C(A \cdot B)$$

$$\begin{aligned} A \cdot C &= 1 \cdot 2 + 2 \cdot 1 + 3 \cdot (-1) \\ &= 2 + 2 - 3 \\ &= 1 \end{aligned}$$

$$\begin{aligned} A \cdot B &= 1 \cdot (-1) + 2 \cdot 0 + 3 \cdot 2 \\ &= -1 + 0 + 6 \\ &= 5 \end{aligned}$$

$$= 1B + 5C$$

$$= -\mathbf{a}_x + 2\mathbf{a}_z + 10\mathbf{a}_x + 5\mathbf{a}_y - 5\mathbf{a}_z$$

$$= 9\mathbf{a}_x + 5\mathbf{a}_y - 3\mathbf{a}_z$$

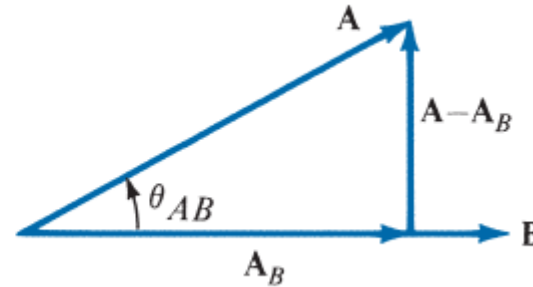
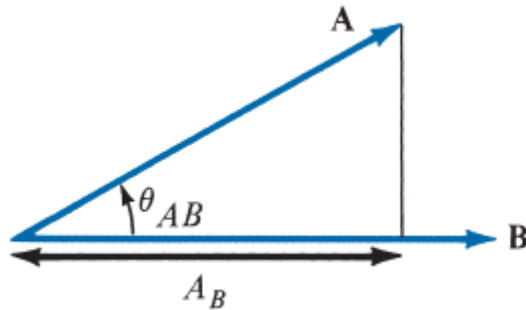
oops

Vector Algebra – Components of a Vector

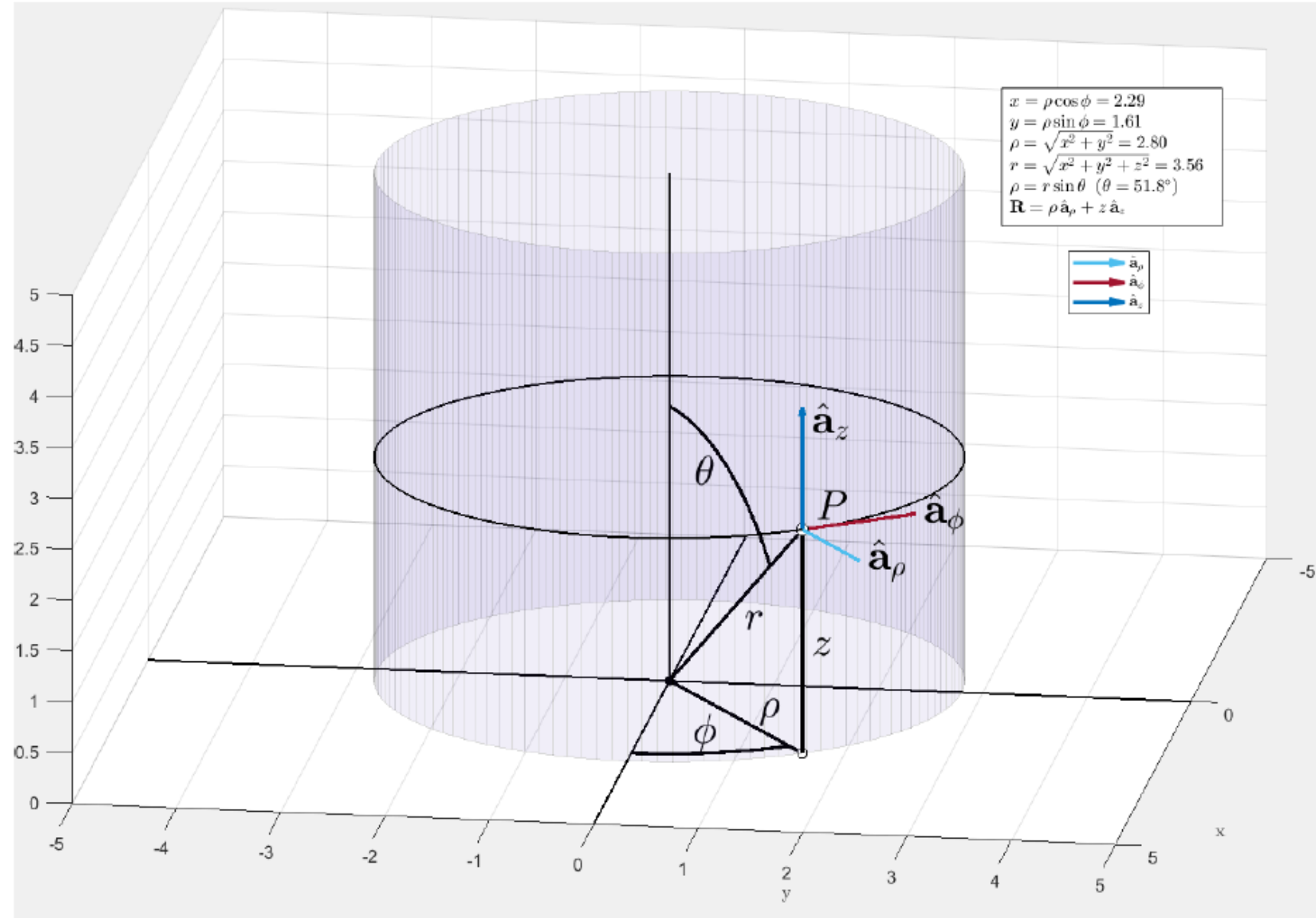
- Scalar product can be used to find the projection of **A** along **B**. For this purpose, we need to do dot multiplication between **A** and the unit vector in the direction of **B**, **a_B**.

$$A_B = A \cos \theta_{AB} = |\mathbf{A}| |\mathbf{a}_B| \cos \theta_{AB} = \mathbf{A} \cdot \mathbf{a}_B$$

$$\mathbf{A}_B = A_B \mathbf{a}_B = (\mathbf{A} \cdot \mathbf{a}_B) \mathbf{a}_B$$

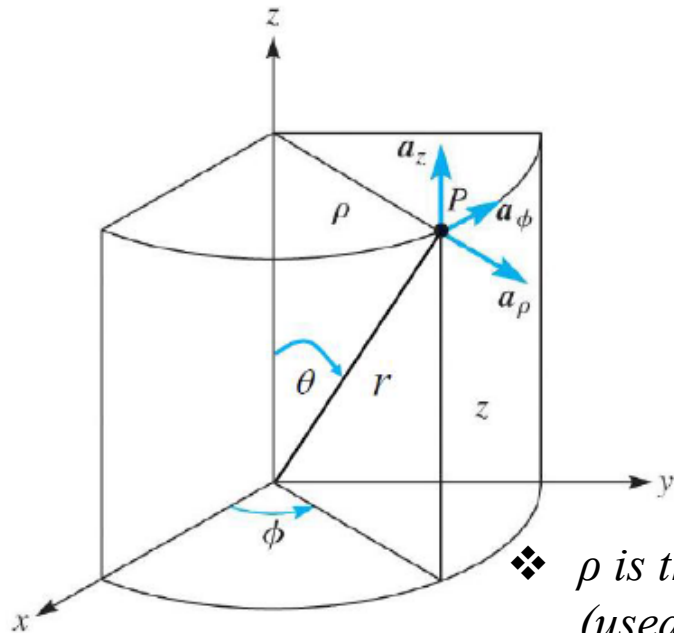


Coordinate Systems – Cylindrical Coordinates



Coordinate Systems – Cylindrical Coordinates (Cont'd)

- Cylindrical coordinates are best suited for rotational symmetry around the z-axis
- A point P is specified by:
 - 1) ρ : distance from the z-axis
 - 2) ϕ : angle in the x — y plane, measured from $+x$
 - 3) z : height along the z-axis



$$x = \rho \cos \phi, \quad y = \rho \sin \phi$$

$$\rho = \sqrt{x^2 + y^2}, \quad r = \|\mathbf{R}\| = \sqrt{x^2 + y^2 + z^2}$$

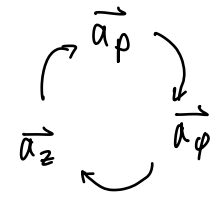
$$\rho = r \sin \theta$$

❖ ρ is the cylindrical radial distance, while $r = \|\mathbf{R}\|$ is the magnitude of the position vector (used in spherical coordinates).

Cylindrical Coordinates – Position Vector and Unit Vectors

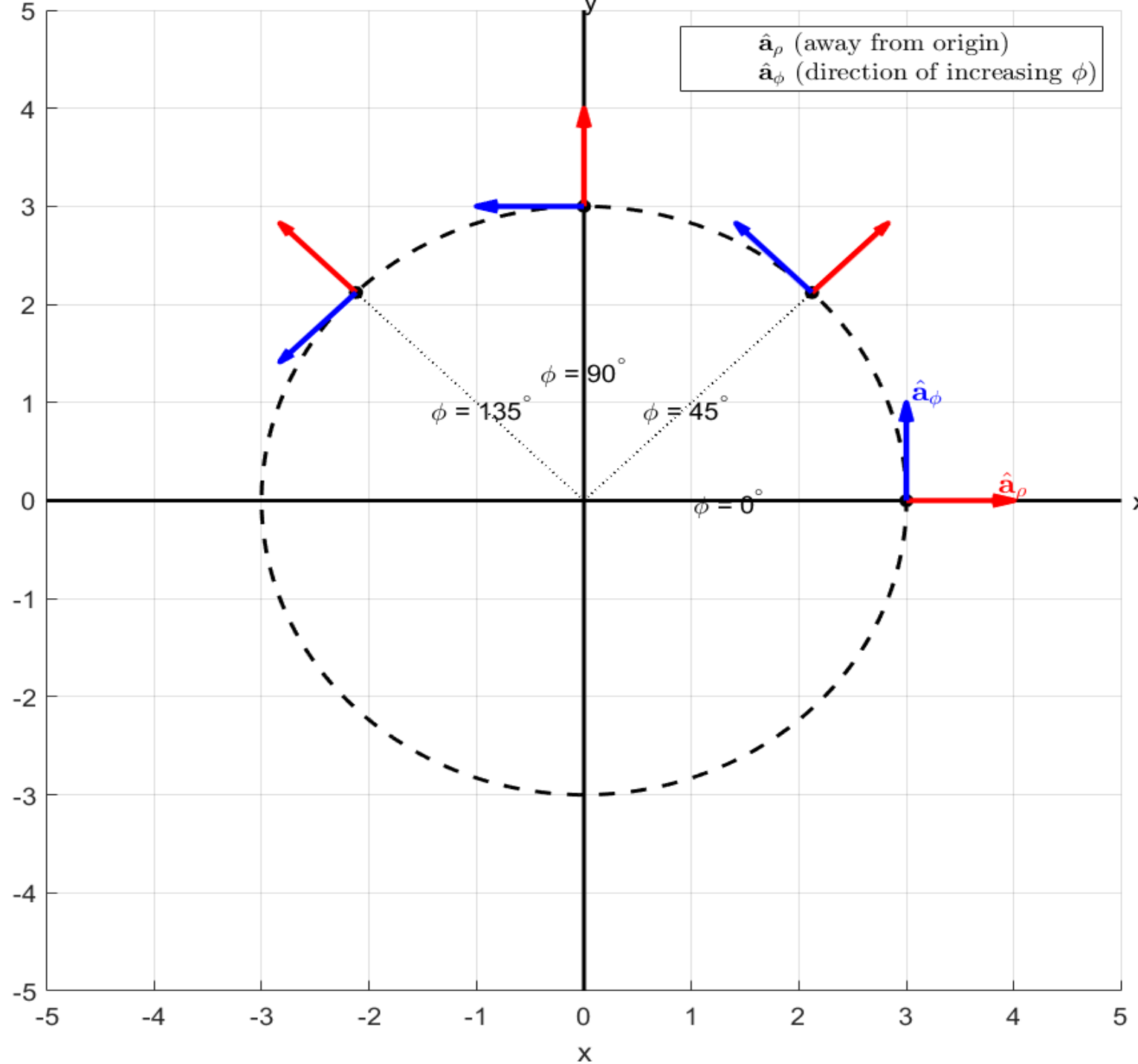
$$\mathbf{R} = \rho \mathbf{a}_\rho + z \mathbf{a}_z$$

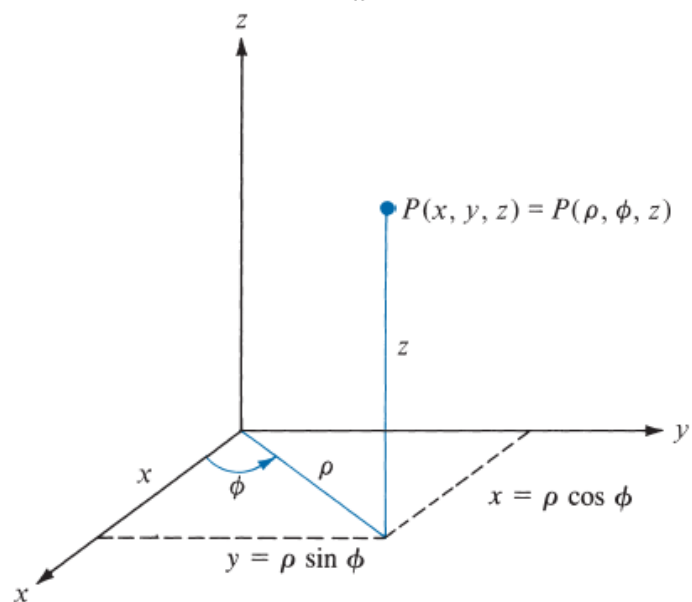
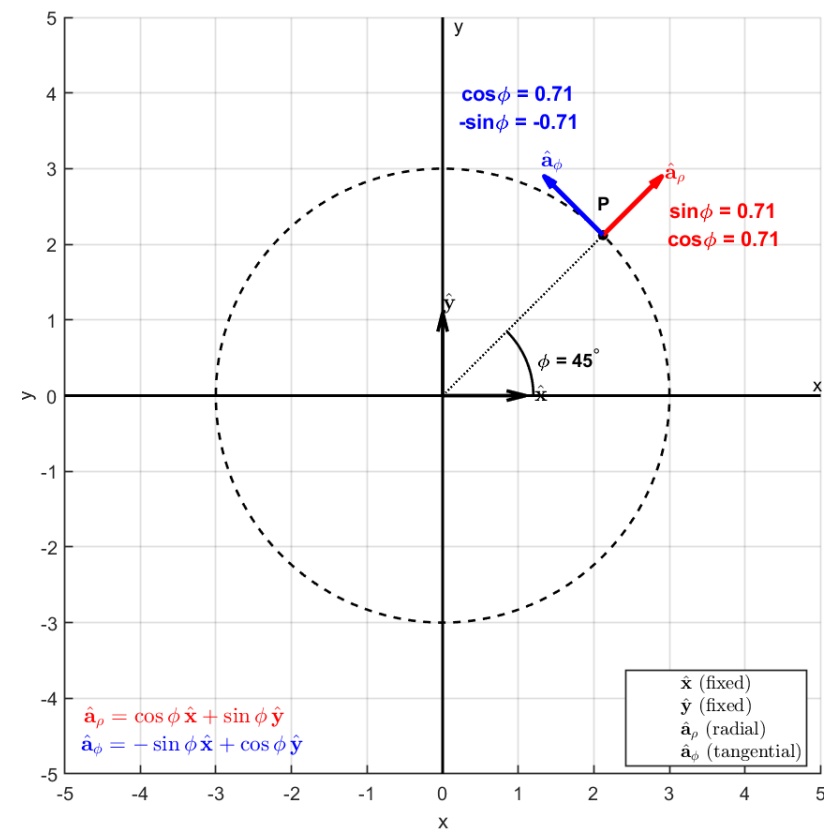
$$\vec{a}_\rho \times \vec{a}_\phi = \vec{a}_z$$



- The position vector always points from the origin to the point
- There is no $\mathbf{a}_\phi$ term
- The unit vector $\mathbf{a}_\rho$ depends on position, $\mathbf{a}_\rho = \cos\phi \mathbf{a}_x + \sin\phi \mathbf{a}_y$. Hence $\mathbf{a}_\rho$ changes direction as ϕ changes, even if $\rho = \text{constant}$.
- The unit vector $\mathbf{a}_\phi$ is perpendicular to the plane formed by $\mathbf{R}$ and $\mathbf{a}_z$

❖ *There is no $\mathbf{a}_\phi$ term because changing ϕ does not move the point away from the origin.*





$$0 \leq \rho < \infty$$

$$0 \leq \phi < 2\pi$$

$$-\infty < z < \infty$$

$$A_\rho \mathbf{a}_\rho + A_\phi \mathbf{a}_\phi + A_z \mathbf{a}_z$$

$$\mathbf{a}_\rho \cdot \mathbf{a}_\rho = \mathbf{a}_\phi \cdot \mathbf{a}_\phi = \mathbf{a}_z \cdot \mathbf{a}_z = 1$$

$$\mathbf{a}_\rho \cdot \mathbf{a}_\phi = \mathbf{a}_\phi \cdot \mathbf{a}_z = \mathbf{a}_z \cdot \mathbf{a}_\rho = 0$$

$$\mathbf{a}_\rho \times \mathbf{a}_\phi = \mathbf{a}_z$$

$$\mathbf{a}_\phi \times \mathbf{a}_z = \mathbf{a}_\rho$$

$$\mathbf{a}_z \times \mathbf{a}_\rho = \mathbf{a}_\phi$$

➤ Cylindrical to Rectangular

$$x = \rho \cos \phi, \quad y = \rho \sin \phi, \quad z = z$$

➤ Rectangular to Cylindrical

$$\rho = \sqrt{x^2 + y^2}, \quad \phi = \tan^{-1} \frac{y}{x}, \quad z = z$$

❖ form a right-handed coordinate system.

➤ Position vector and magnitude

$$\mathbf{R} = x \mathbf{a}_x + y \mathbf{a}_y + z \mathbf{a}_z$$

$$\mathbf{R} = \rho \mathbf{a}_\rho + z \mathbf{a}_z$$

$$|\mathbf{R}| = \sqrt{x^2 + y^2 + z^2} = \sqrt{\rho^2 + z^2}$$

➤ Cylindrical to Rectangular and vice-versa unit vectors

$$\mathbf{a}_\rho = \cos \phi \mathbf{a}_x + \sin \phi \mathbf{a}_y$$

$$\mathbf{a}_\phi = -\sin \phi \mathbf{a}_x + \cos \phi \mathbf{a}_y$$

$$\mathbf{a}_x = \cos \phi \mathbf{a}_\rho - \sin \phi \mathbf{a}_\phi$$

$$\mathbf{a}_y = \sin \phi \mathbf{a}_\rho + \cos \phi \mathbf{a}_\phi$$

* possible exam question

Coordinate Systems – Rectangular to Cylindrical

➤ Given $\mathbf{A} = A_x \mathbf{a}_x + A_y \mathbf{a}_y + A_z \mathbf{a}_z$

1. Identify the angle $\phi = \tan^{-1} \frac{y}{x}$,

2. Write rectangular unit vectors in cylindrical form

$$\mathbf{a}_x = \cos \phi \mathbf{a}_\rho - \sin \phi \mathbf{a}_\phi$$

$$\mathbf{a}_y = \sin \phi \mathbf{a}_\rho + \cos \phi \mathbf{a}_\phi$$

$$\mathbf{a}_z = \mathbf{a}_z$$

3. Substitute into $\mathbf{A}$, $\mathbf{A} = A_x(\cos \phi \mathbf{a}_\rho - \sin \phi \mathbf{a}_\phi) + A_y(\sin \phi \mathbf{a}_\rho + \cos \phi \mathbf{a}_\phi) + A_z \mathbf{a}_z$

4. Collect terms

Coordinate Systems – Cylindrical to Rectangular

➤ Given

$$\mathbf{A} = A_\rho \hat{\mathbf{a}}_\rho + A_\phi \hat{\mathbf{a}}_\phi + A_z \hat{\mathbf{a}}_z$$

1. Write cylindrical unit vectors in rectangular form

$$\hat{\mathbf{a}}_\rho = \cos \phi \hat{\mathbf{a}}_x + \sin \phi \hat{\mathbf{a}}_y$$

$$\hat{\mathbf{a}}_\phi = -\sin \phi \hat{\mathbf{a}}_x + \cos \phi \hat{\mathbf{a}}_y$$

$$\hat{\mathbf{a}}_z = \hat{\mathbf{a}}_z$$

2. Substitute

$$\mathbf{A} = A_\rho(\cos \phi \hat{\mathbf{a}}_x + \sin \phi \hat{\mathbf{a}}_y) + A_\phi(-\sin \phi \hat{\mathbf{a}}_x + \cos \phi \hat{\mathbf{a}}_y) + A_z \hat{\mathbf{a}}_z$$

3. Collect terms

$$\mathbf{A} = A_x \hat{\mathbf{a}}_x + A_y \hat{\mathbf{a}}_y + A_z \hat{\mathbf{a}}_z$$

$$A_x = A_\rho \cos \phi - A_\phi \sin \phi$$

$$A_y = A_\rho \sin \phi + A_\phi \cos \phi$$

Coordinate Systems – Matrix Form

➤ Rectangular → Cylindrical

$$\begin{bmatrix} A_\rho \\ A_\phi \\ A_z \end{bmatrix} = \begin{bmatrix} \cos \phi & \sin \phi & 0 \\ -\sin \phi & \cos \phi & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} A_x \\ A_y \\ A_z \end{bmatrix}$$

➤ Cylindrical → Rectangular

$$\begin{bmatrix} A_x \\ A_y \\ A_z \end{bmatrix} = \begin{bmatrix} \cos \phi & -\sin \phi & 0 \\ \sin \phi & \cos \phi & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} A_\rho \\ A_\phi \\ A_z \end{bmatrix}$$

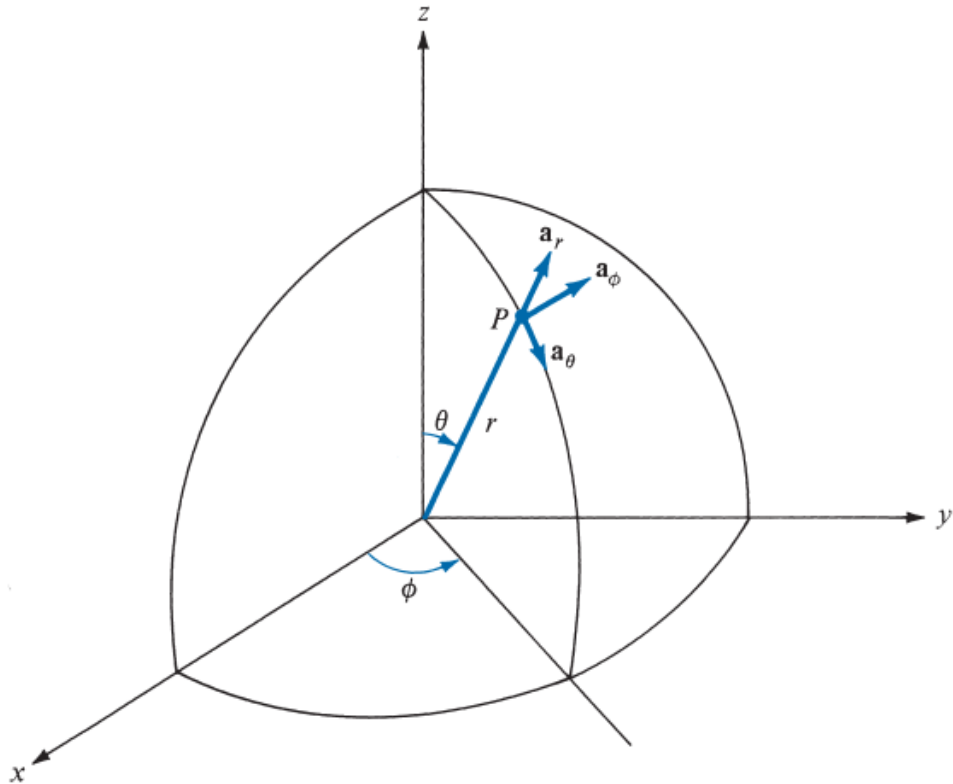
Coordinate Systems – Spherical Coordinates

- A vector can be shown as $A_r \mathbf{a}_r + A_\theta \mathbf{a}_\theta + A_\phi \mathbf{a}_\phi$
- The dot products among the unit vectors: $\mathbf{a}_r \cdot \mathbf{a}_r = \mathbf{a}_\theta \cdot \mathbf{a}_\theta = \mathbf{a}_\phi \cdot \mathbf{a}_\phi = 1$
 $\mathbf{a}_r \cdot \mathbf{a}_\theta = \mathbf{a}_\theta \cdot \mathbf{a}_\phi = \mathbf{a}_\phi \cdot \mathbf{a}_r = 0$
- For the cross products we have:

$$\mathbf{a}_r \times \mathbf{a}_\theta = \mathbf{a}_\phi$$

$$\mathbf{a}_\theta \times \mathbf{a}_\phi = \mathbf{a}_r$$

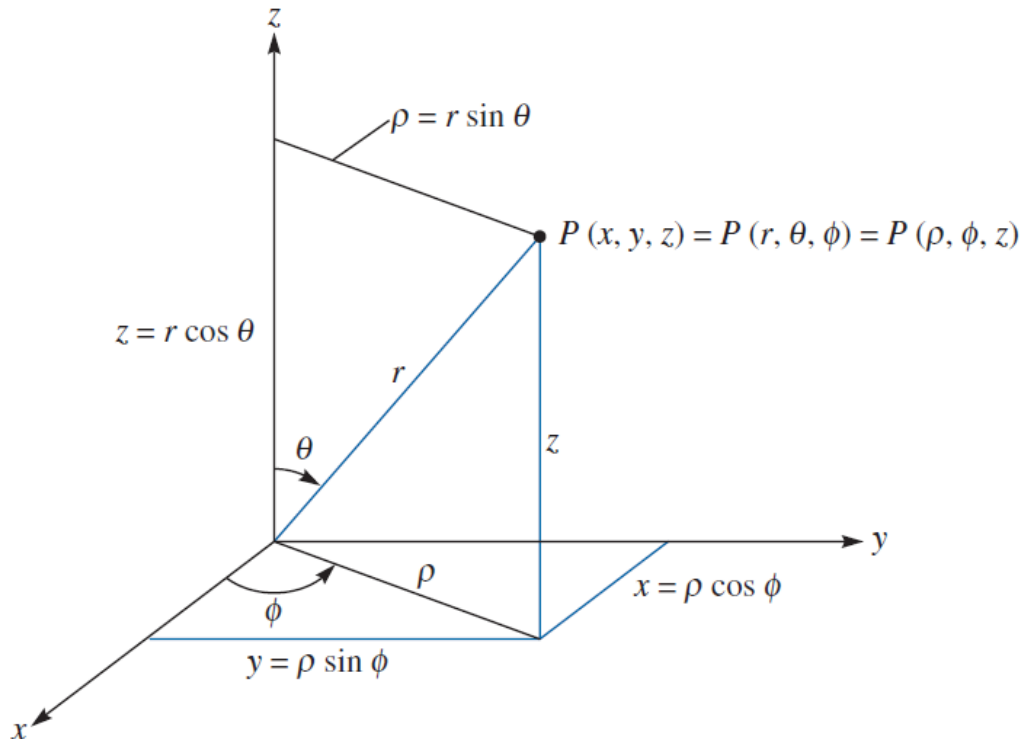
$$\mathbf{a}_\phi \times \mathbf{a}_r = \mathbf{a}_\theta$$



Coordinate Systems – Spherical Coordinates

$$r = \sqrt{x^2 + y^2 + z^2}, \quad \theta = \tan^{-1} \frac{\sqrt{x^2 + y^2}}{z}, \quad \phi = \tan^{-1} \frac{y}{x}$$

$$x = r \sin \theta \cos \phi, \quad y = r \sin \theta \sin \phi, \quad z = r \cos \theta$$

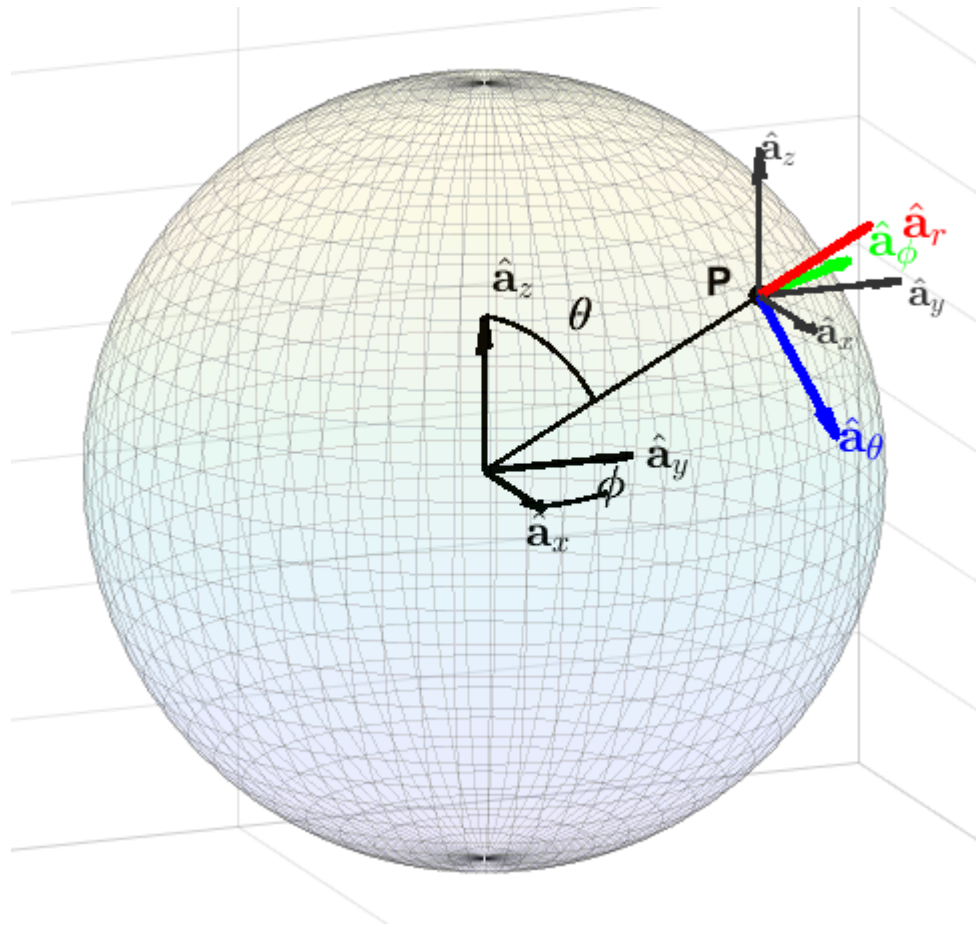


$$\mathbf{a}_x = \sin \theta \cos \phi \mathbf{a}_r + \cos \theta \cos \phi \mathbf{a}_\theta - \sin \phi \mathbf{a}_\phi$$

$$\mathbf{a}_y = \sin \theta \sin \phi \mathbf{a}_r + \cos \theta \sin \phi \mathbf{a}_\theta + \cos \phi \mathbf{a}_\phi$$

$$\mathbf{a}_z = \cos \theta \mathbf{a}_r - \sin \theta \mathbf{a}_\theta$$

Coordinate Systems – Spherical Coordinates, Unit Vectors



$$\mathbf{a}_x = \sin \theta \cos \phi \mathbf{a}_r + \cos \theta \cos \phi \mathbf{a}_\theta - \sin \phi \mathbf{a}_\phi$$

$$\mathbf{a}_y = \sin \theta \sin \phi \mathbf{a}_r + \cos \theta \sin \phi \mathbf{a}_\theta + \cos \phi \mathbf{a}_\phi$$

$$\mathbf{a}_z = \cos \theta \mathbf{a}_r - \sin \theta \mathbf{a}_\theta$$

$$\mathbf{a}_r = \sin \theta \cos \phi \mathbf{a}_x + \sin \theta \sin \phi \mathbf{a}_y + \cos \theta \mathbf{a}_z$$

$$\mathbf{a}_\theta = \cos \theta \cos \phi \mathbf{a}_x + \cos \theta \sin \phi \mathbf{a}_y - \sin \theta \mathbf{a}_z$$

$$\mathbf{a}_\phi = -\sin \phi \mathbf{a}_x + \cos \phi \mathbf{a}_y$$

Coordinate Systems – Matrix Form

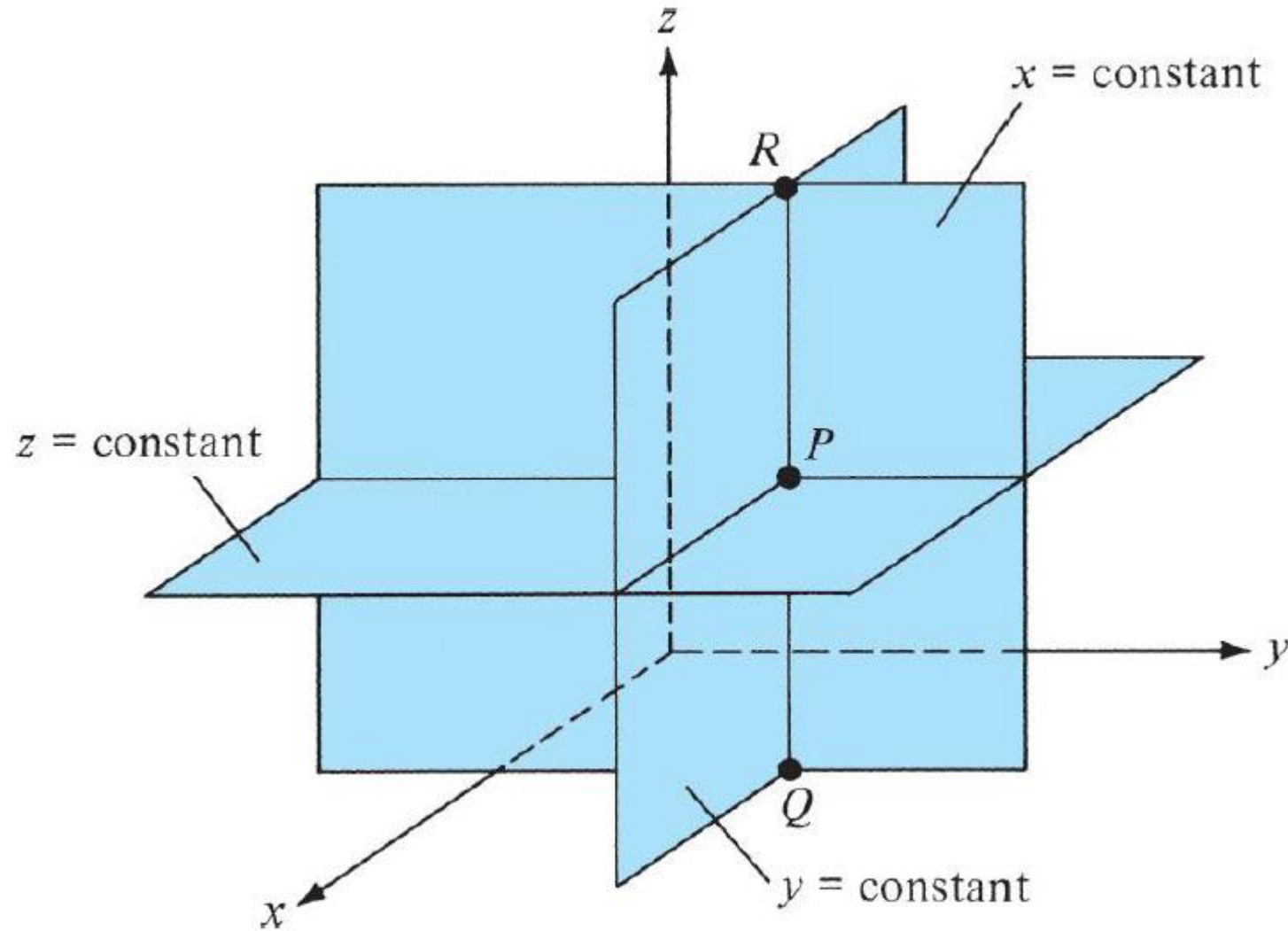
➤ Rectangular → Spherical

$$\begin{bmatrix} A_r \\ A_\theta \\ A_\phi \end{bmatrix} = \begin{bmatrix} \sin \theta \cos \phi & \sin \theta \sin \phi & \cos \theta \\ \cos \theta \cos \phi & \cos \theta \sin \phi & -\sin \theta \\ -\sin \phi & \cos \phi & 0 \end{bmatrix} \begin{bmatrix} A_x \\ A_y \\ A_z \end{bmatrix}$$

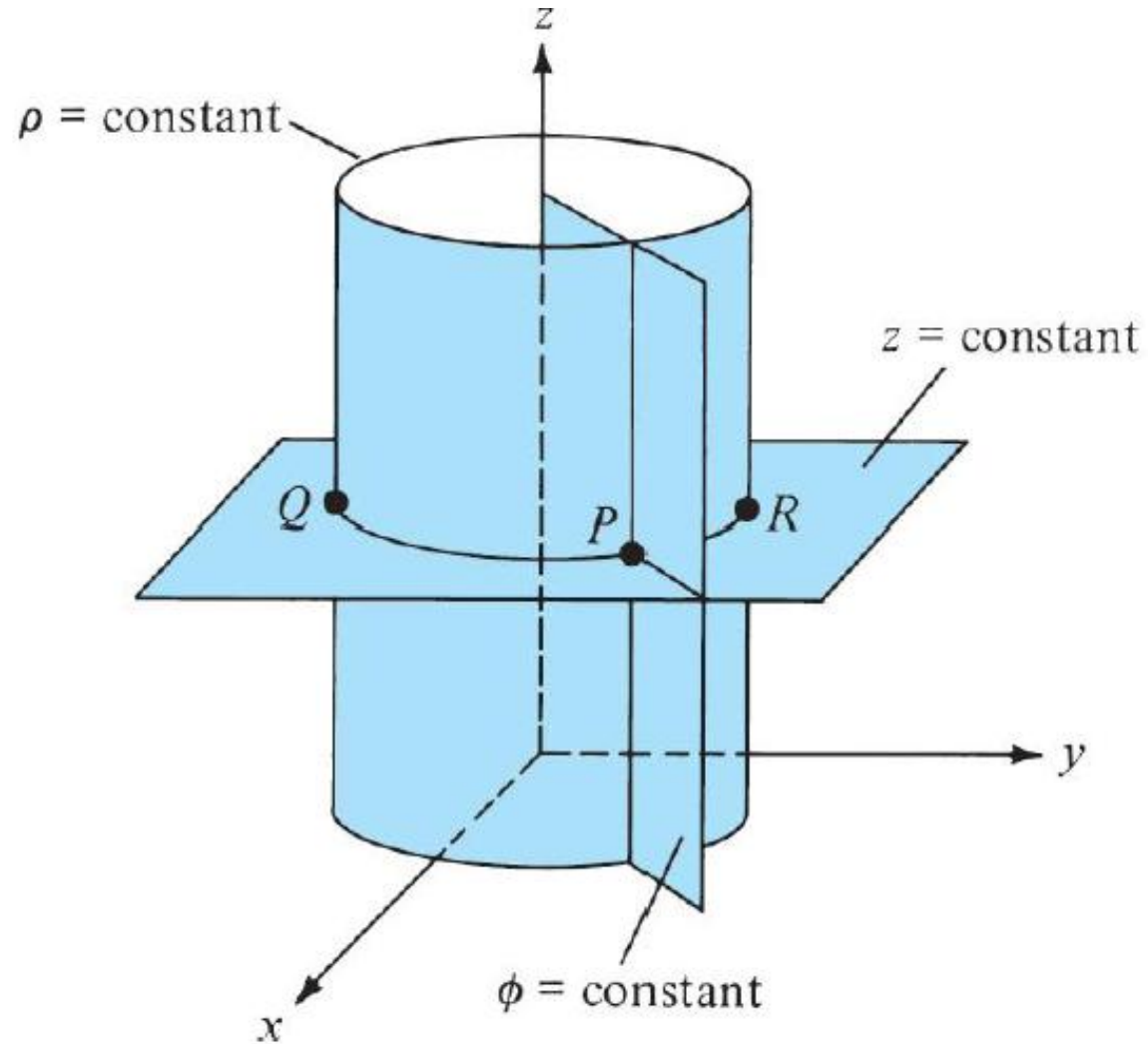
➤ Spherical → Rectangular

$$\begin{bmatrix} A_x \\ A_y \\ A_z \end{bmatrix} = \begin{bmatrix} \sin \theta \cos \phi & \cos \theta \cos \phi & -\sin \phi \\ \sin \theta \sin \phi & \cos \theta \sin \phi & \cos \phi \\ \cos \theta & -\sin \theta & 0 \end{bmatrix} \begin{bmatrix} A_r \\ A_\theta \\ A_\phi \end{bmatrix}$$

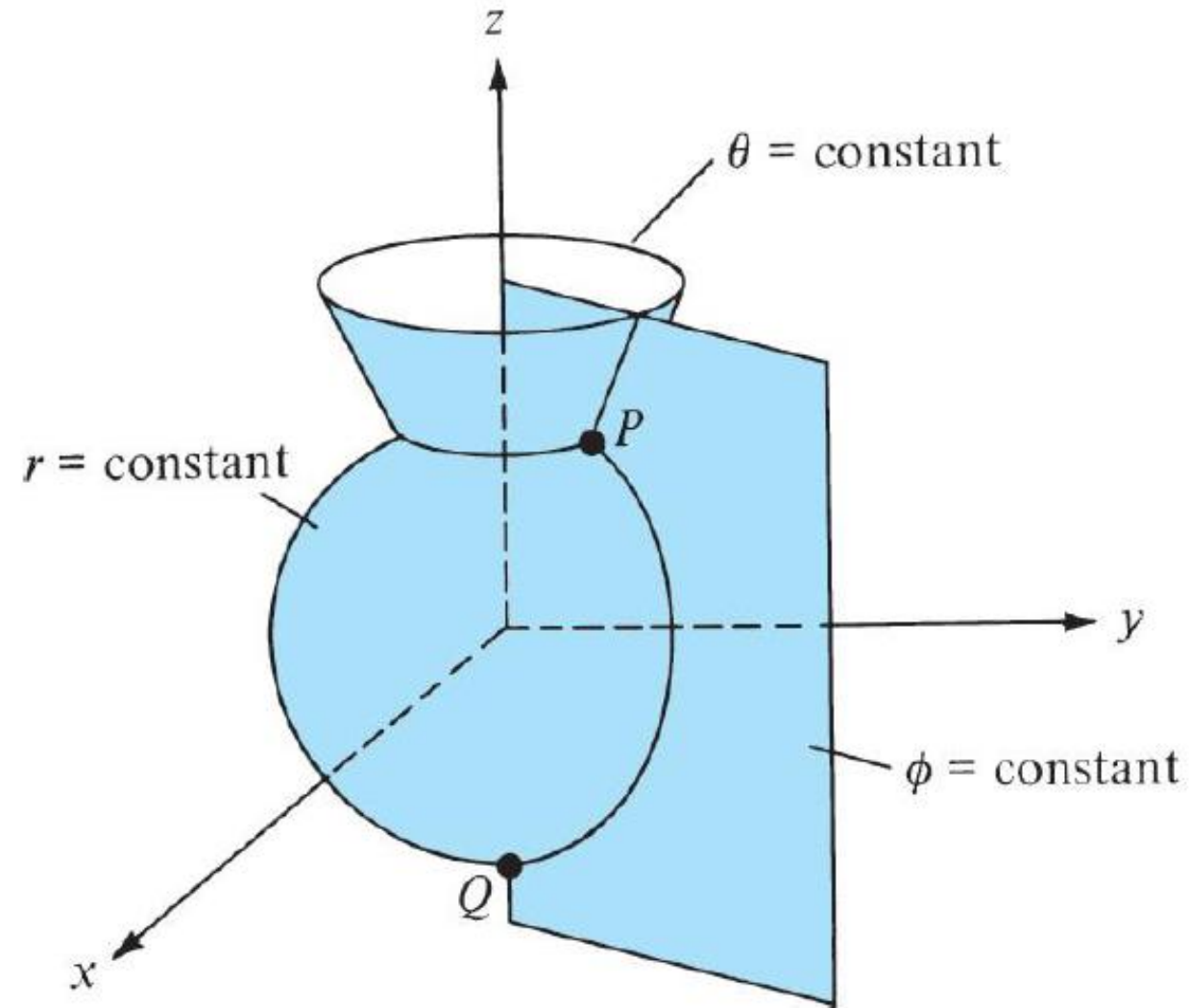
Coordinate Systems – Constant Surfaces in Rectangular



Coordinate Systems – Constant Surfaces in Cylindrical



Coordinate Systems – Constant Surfaces in Spherical



Vector Calculus – Rectangular Coordinate

- Differential displacement or length element,

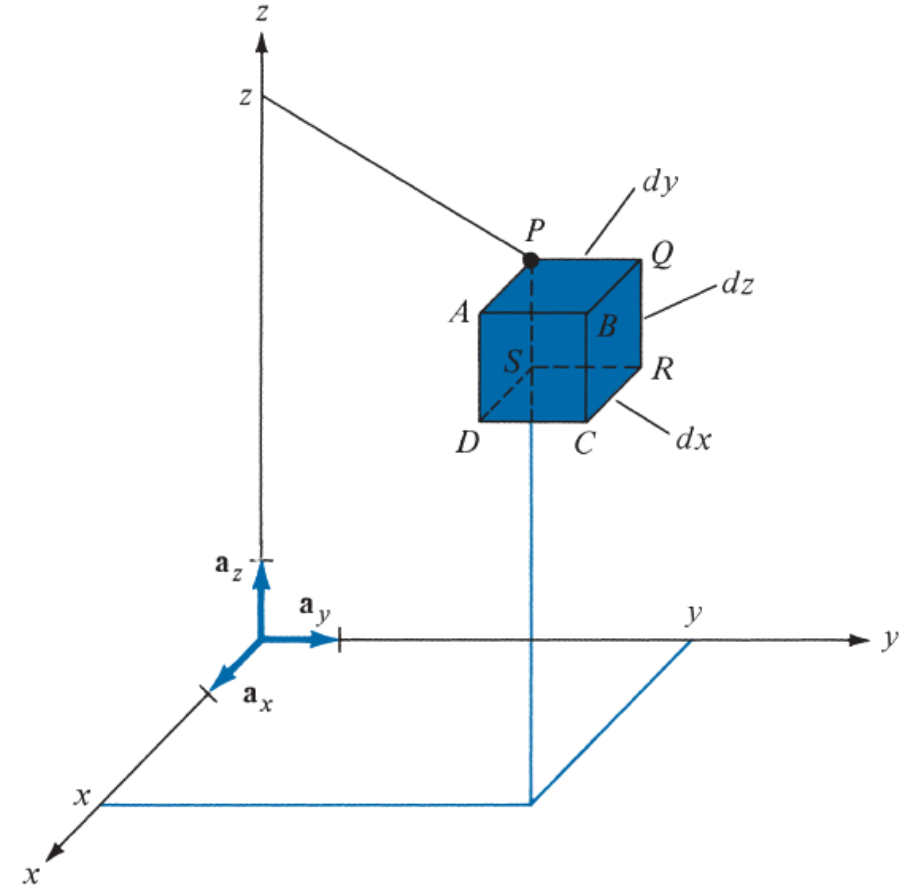
$$d\mathbf{l} = dx \mathbf{a}_x + dy \mathbf{a}_y + dz \mathbf{a}_z$$

- Differential normal surface area,

$$d\mathbf{S} = \mathbf{n} dS$$

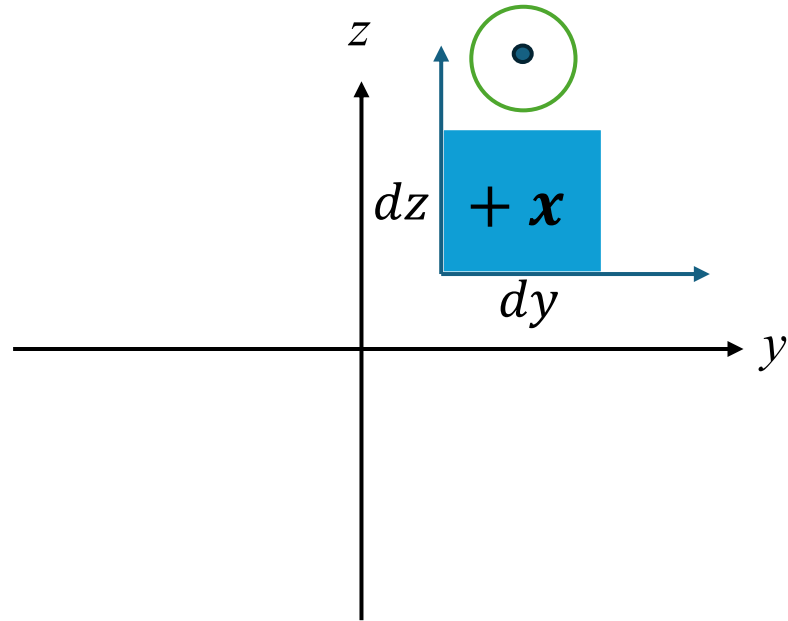
- Differential volume,

$$dv = dx dy dz$$

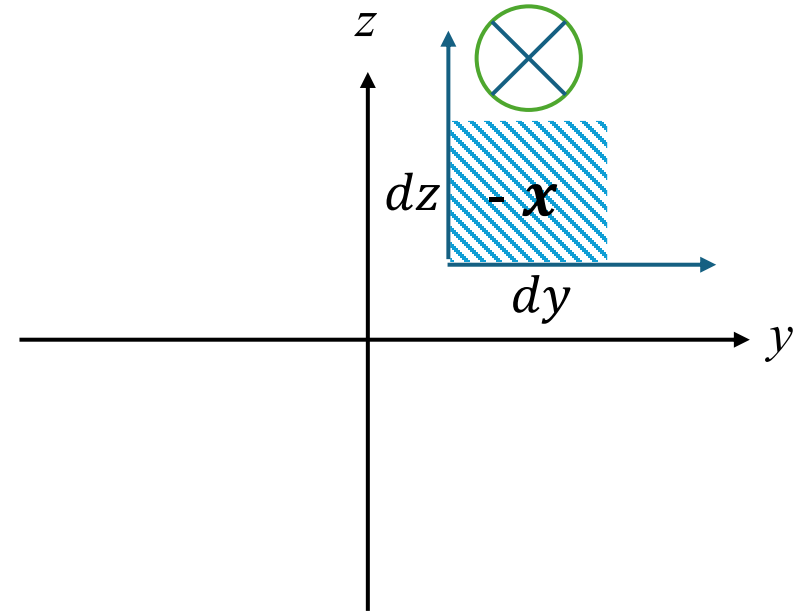


Vector Calculus – Rectangular Coordinate

➤ Differential displacement or length element,



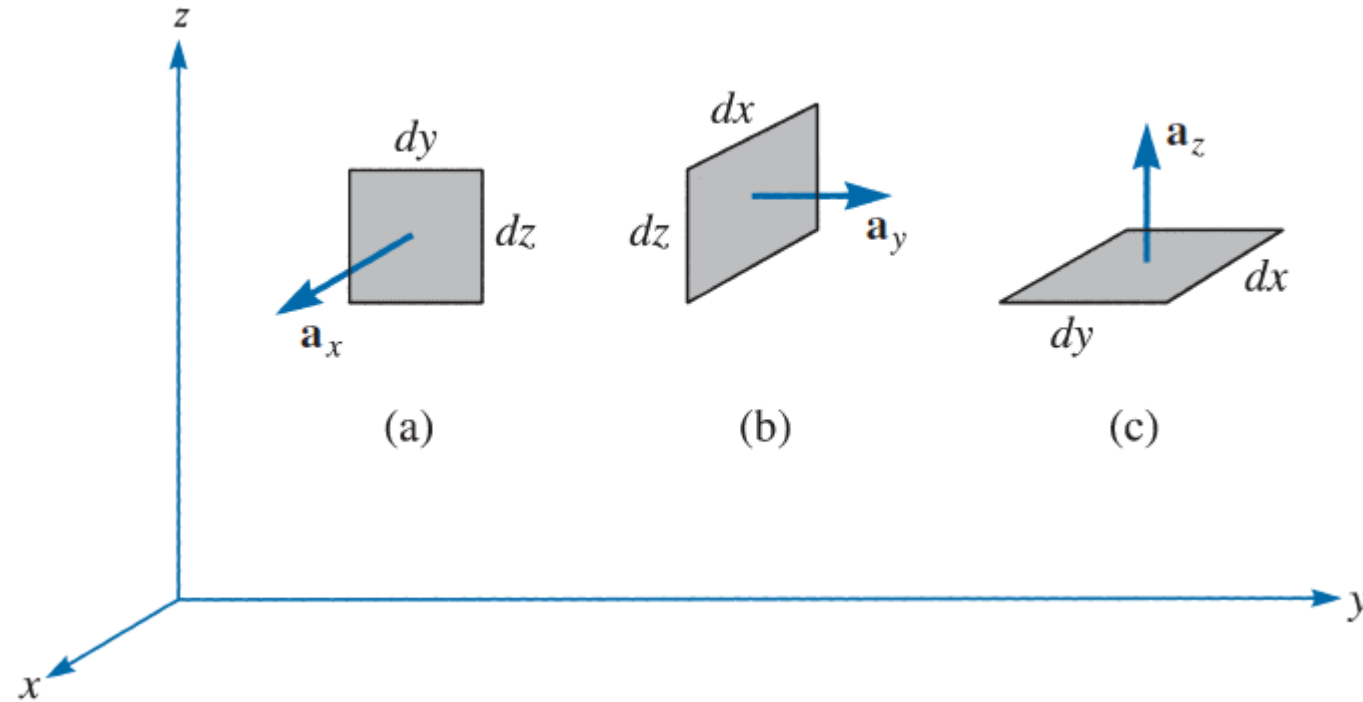
$$d\mathbf{S} = dy\mathbf{a}_y \times dz\mathbf{a}_z$$



$$d\mathbf{S} = dz\mathbf{a}_z \times dy\mathbf{a}_y$$

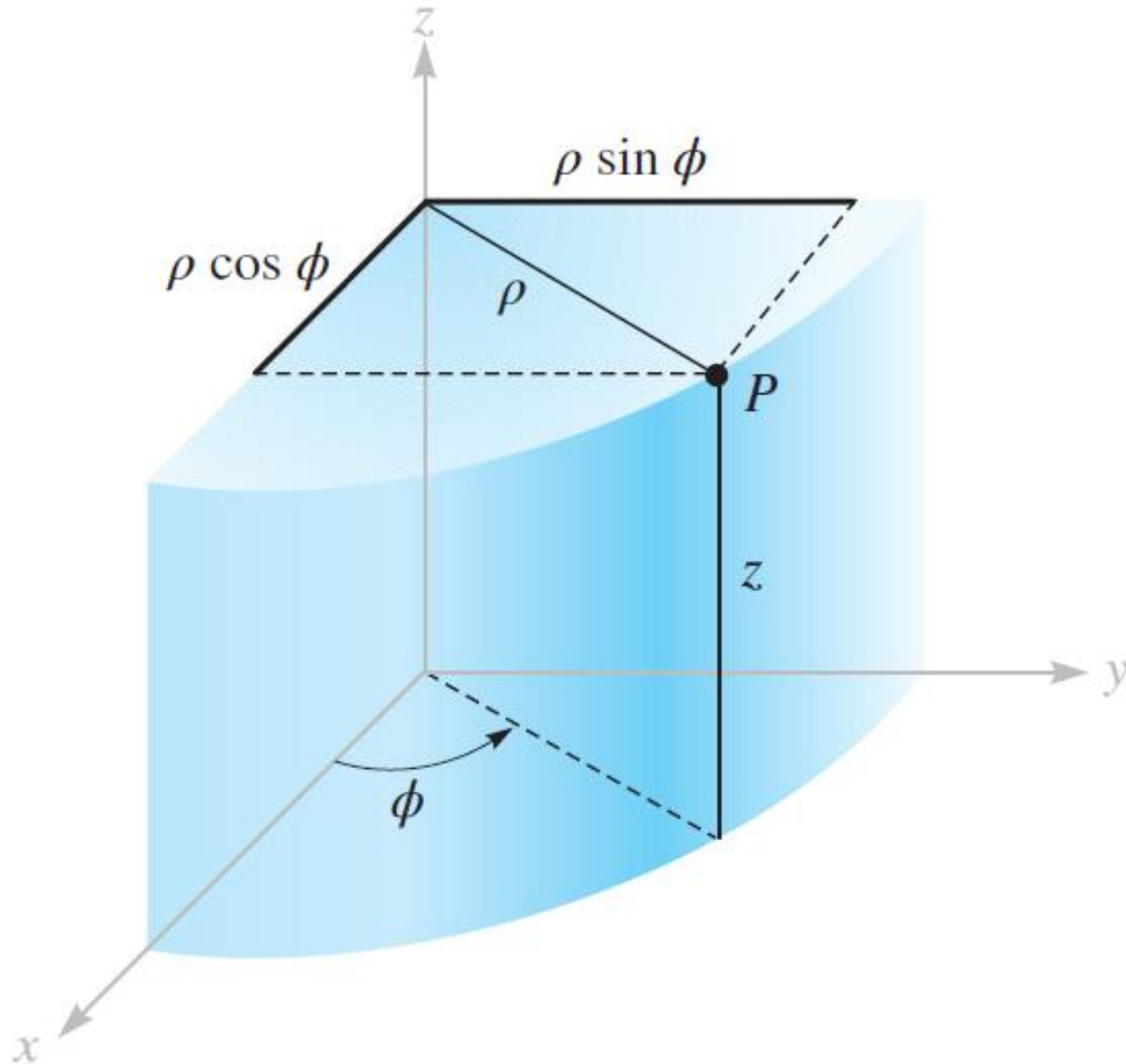
Vector Calculus – Differential Normal Surface Areas in Rectangular Coordinates

- Differential displacement or length element,



(a) $d\mathbf{S} = dy dz \mathbf{a}_x$, (b) $d\mathbf{S} = dx dz \mathbf{a}_y$, (c) $d\mathbf{S} = dx dy \mathbf{a}_z$.

Vector Calculus

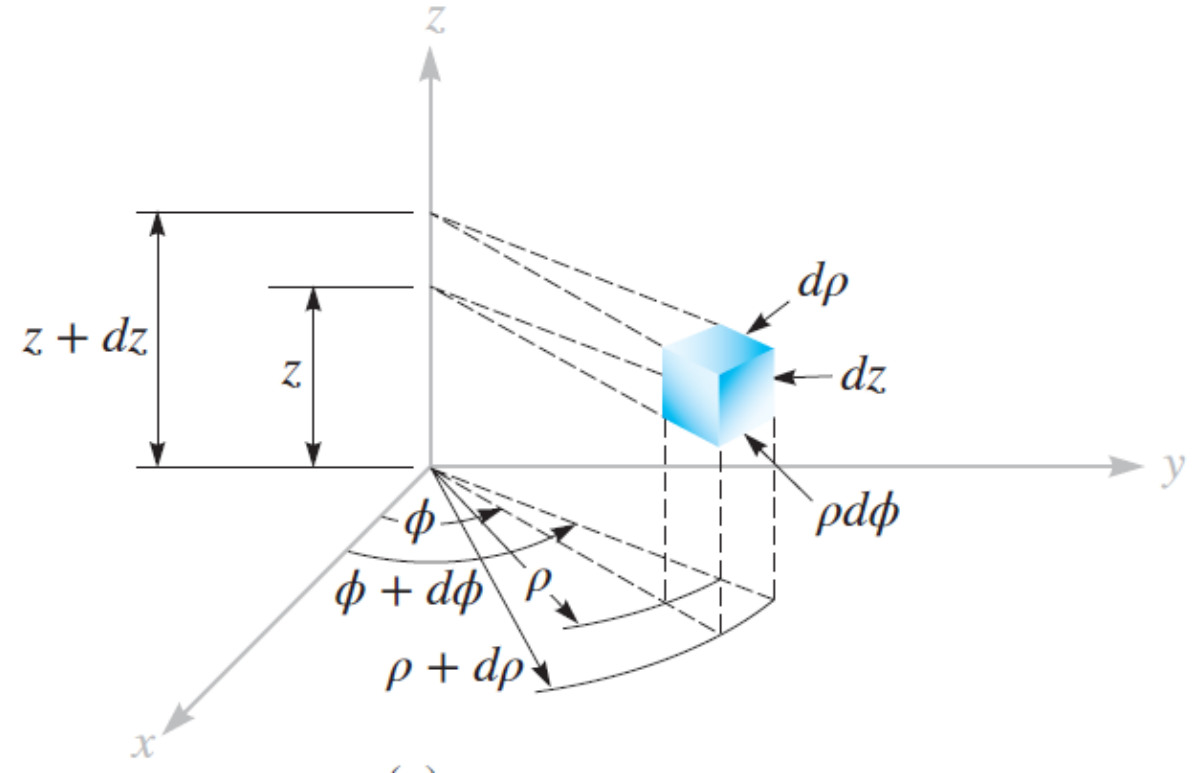
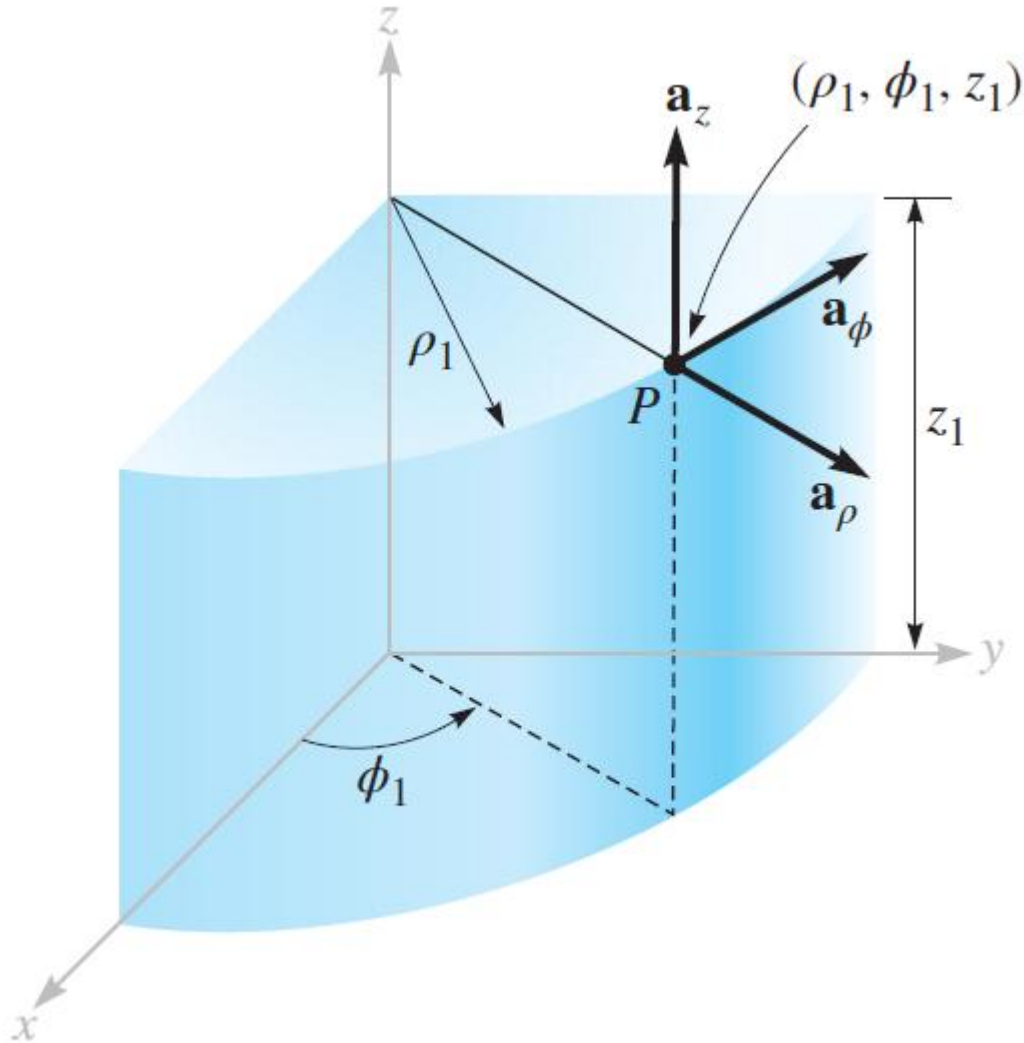


$$\rho = \sqrt{x^2 + y^2}$$

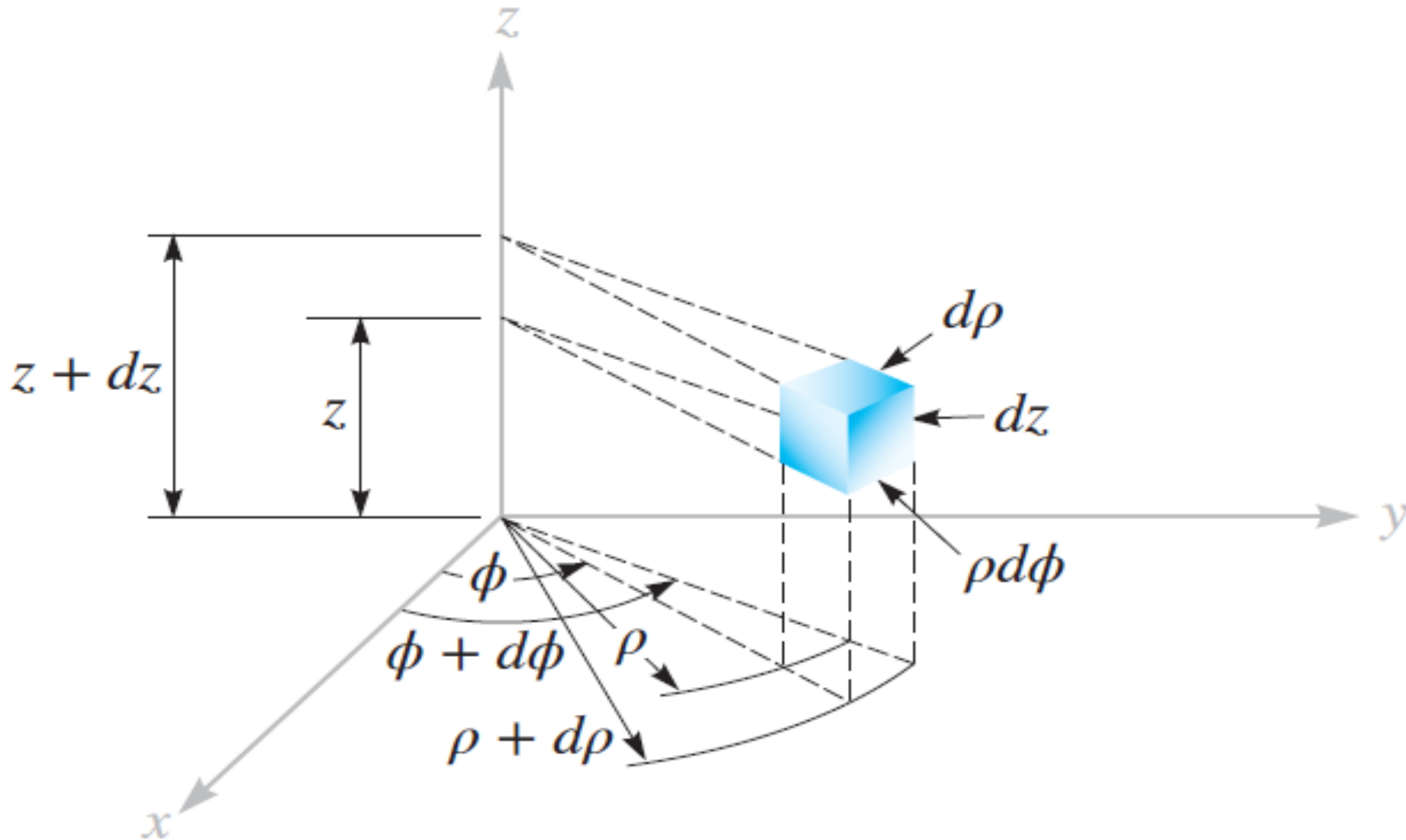
$$\phi = \tan^{-1} \frac{y}{x}$$

$$z = z$$

Vector Calculus – Cylindrical Coordinate



Vector Calculus – Cylindrical Coordinate



Vector Calculus – Cylindrical Coordinate

- Differential displacement or length element,

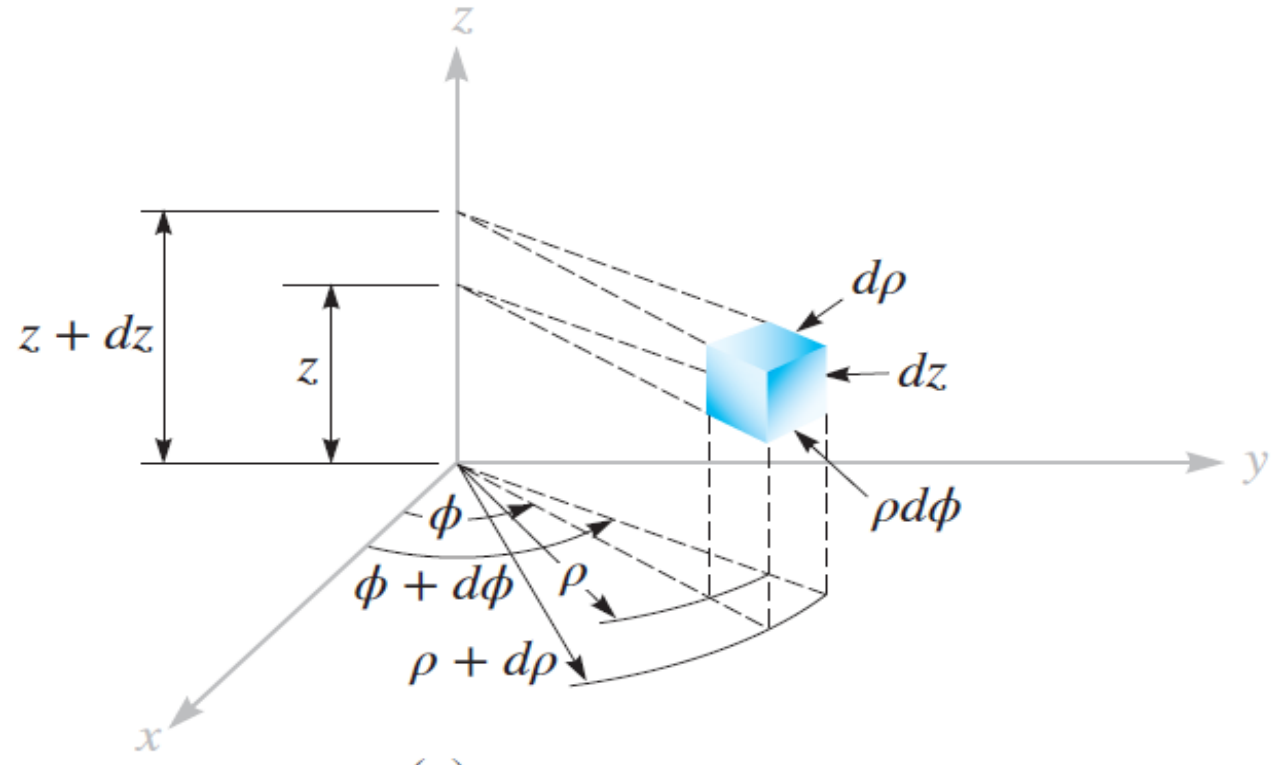
$$d\mathbf{l} = d\rho \mathbf{a}_\rho + \rho d\phi \mathbf{a}_\phi + dz \mathbf{a}_z$$

- Differential normal surface area,

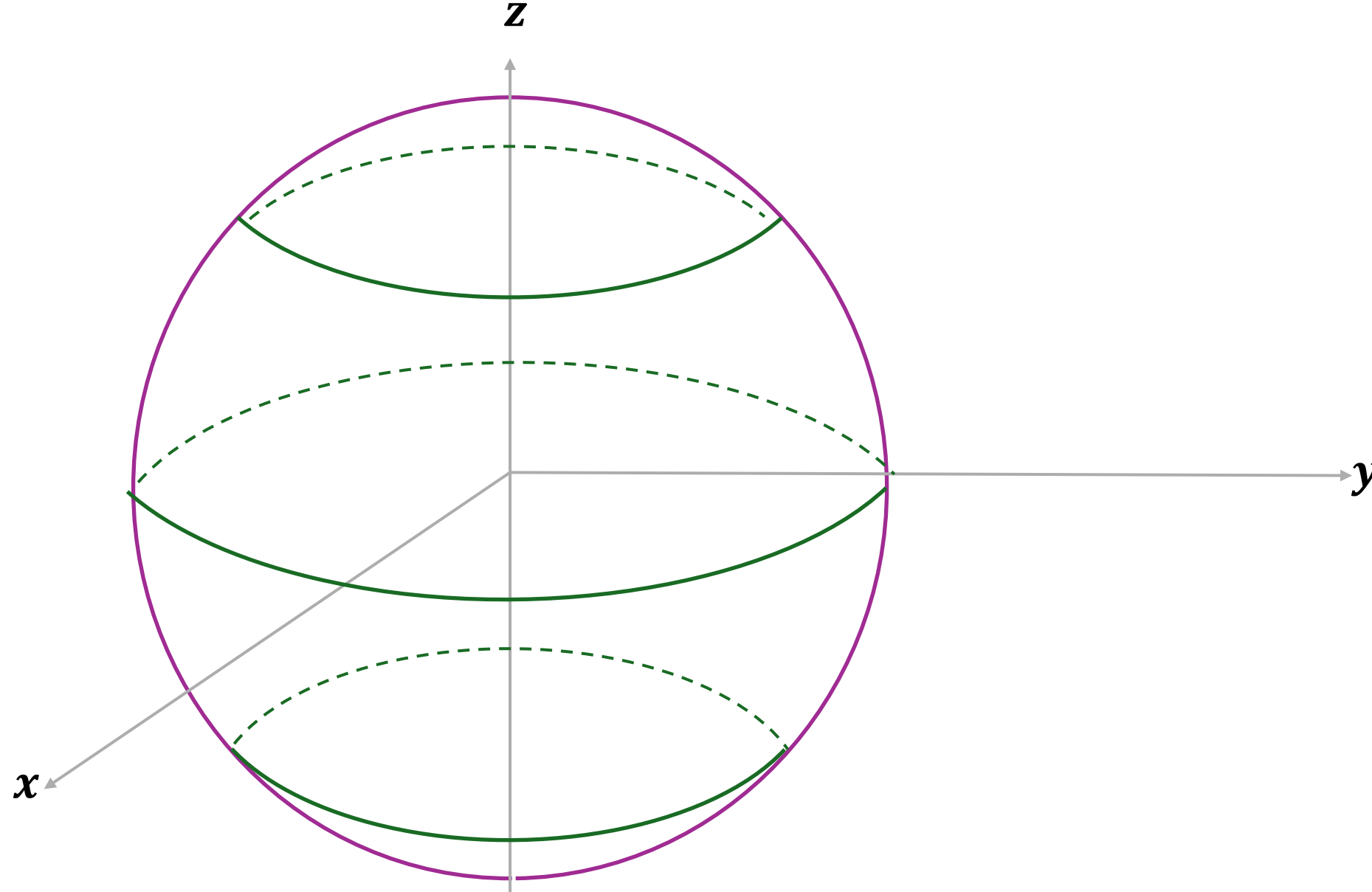
$$d\mathbf{S} = \mathbf{n} dS$$

- Differential volume element,

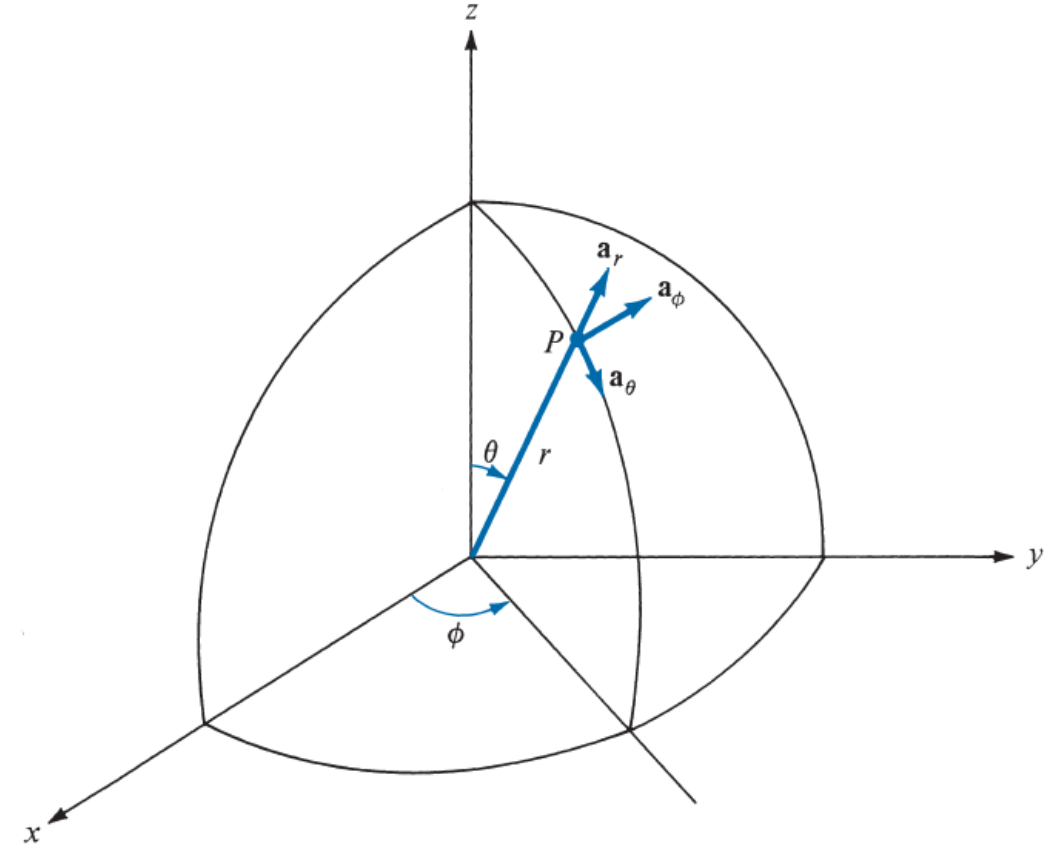
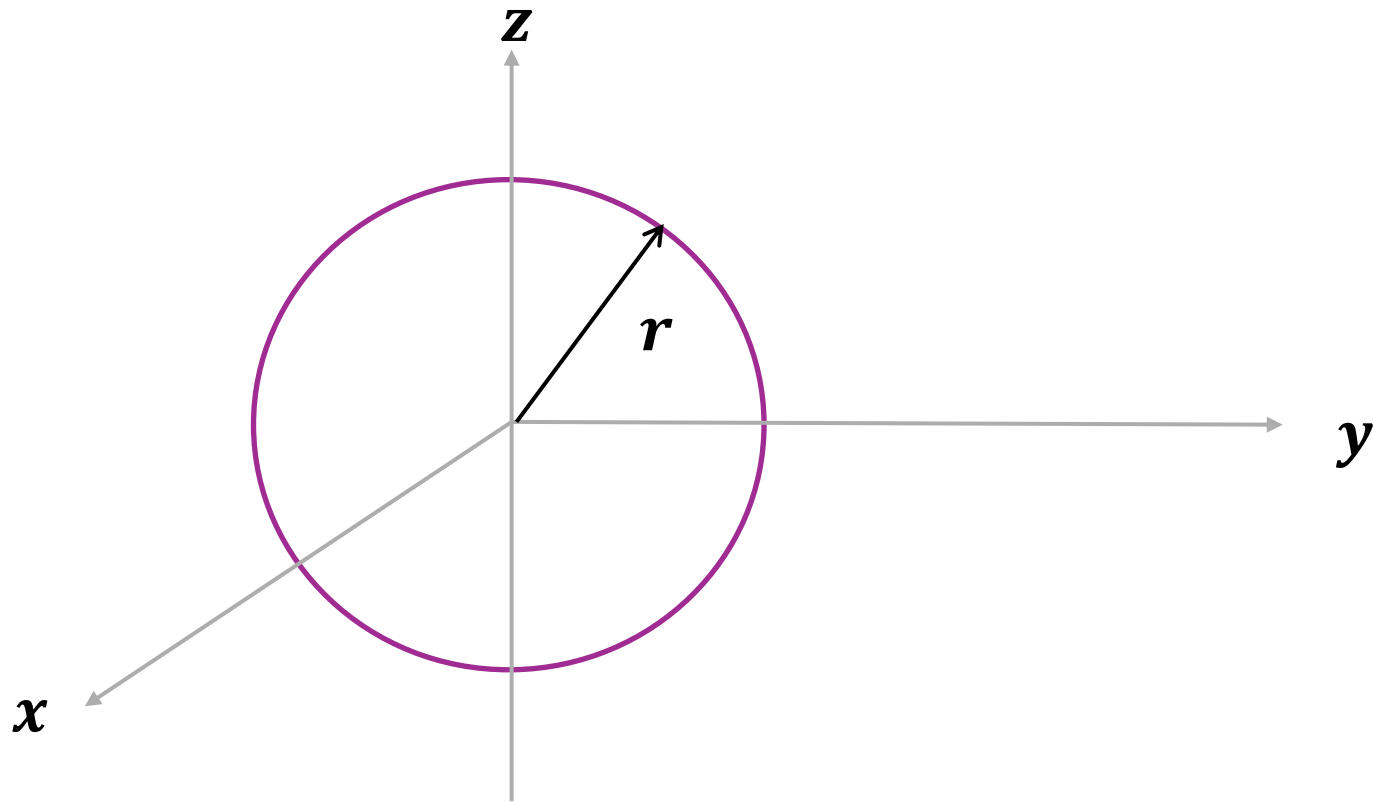
$$dv = \rho d\rho d\phi dz$$



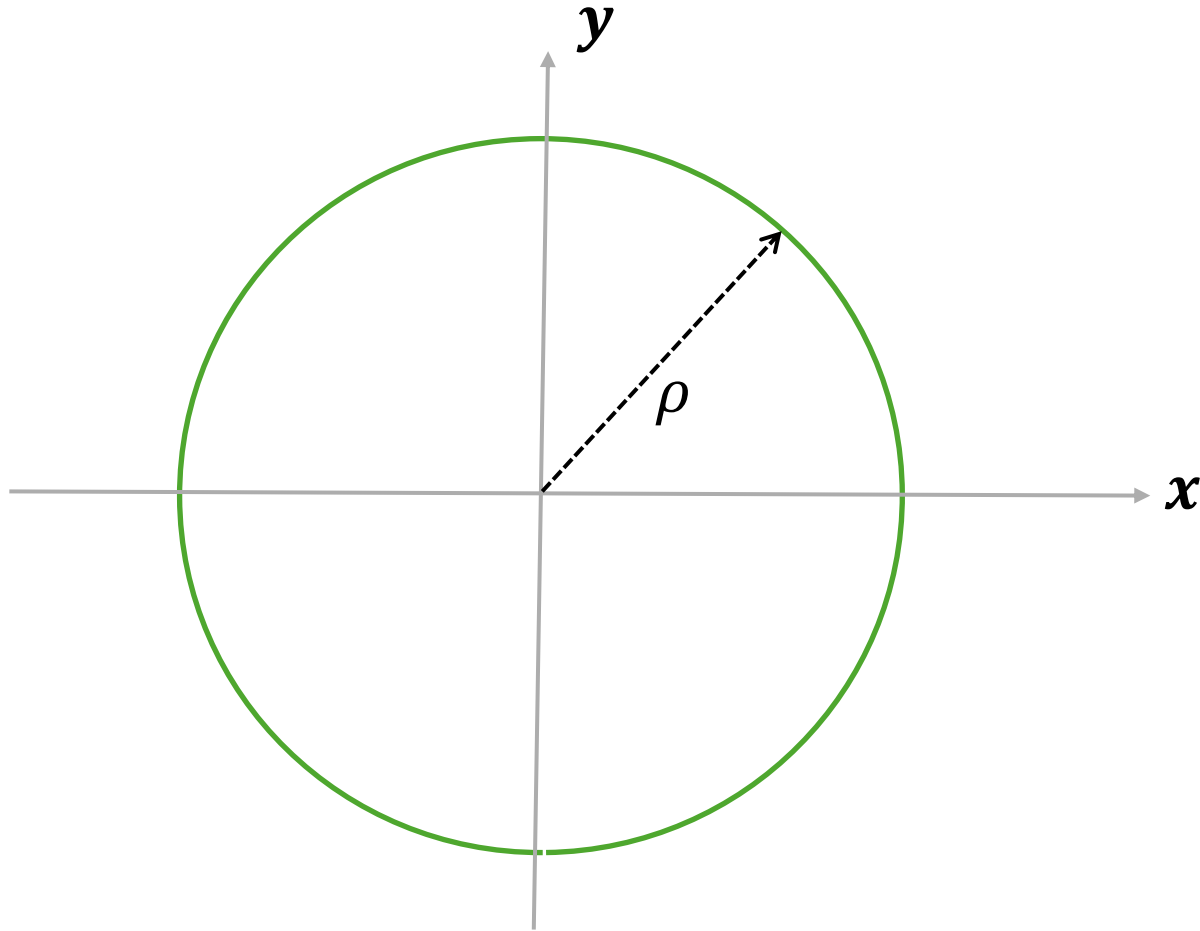
Vector Calculus – Spherical Coordinate Systems



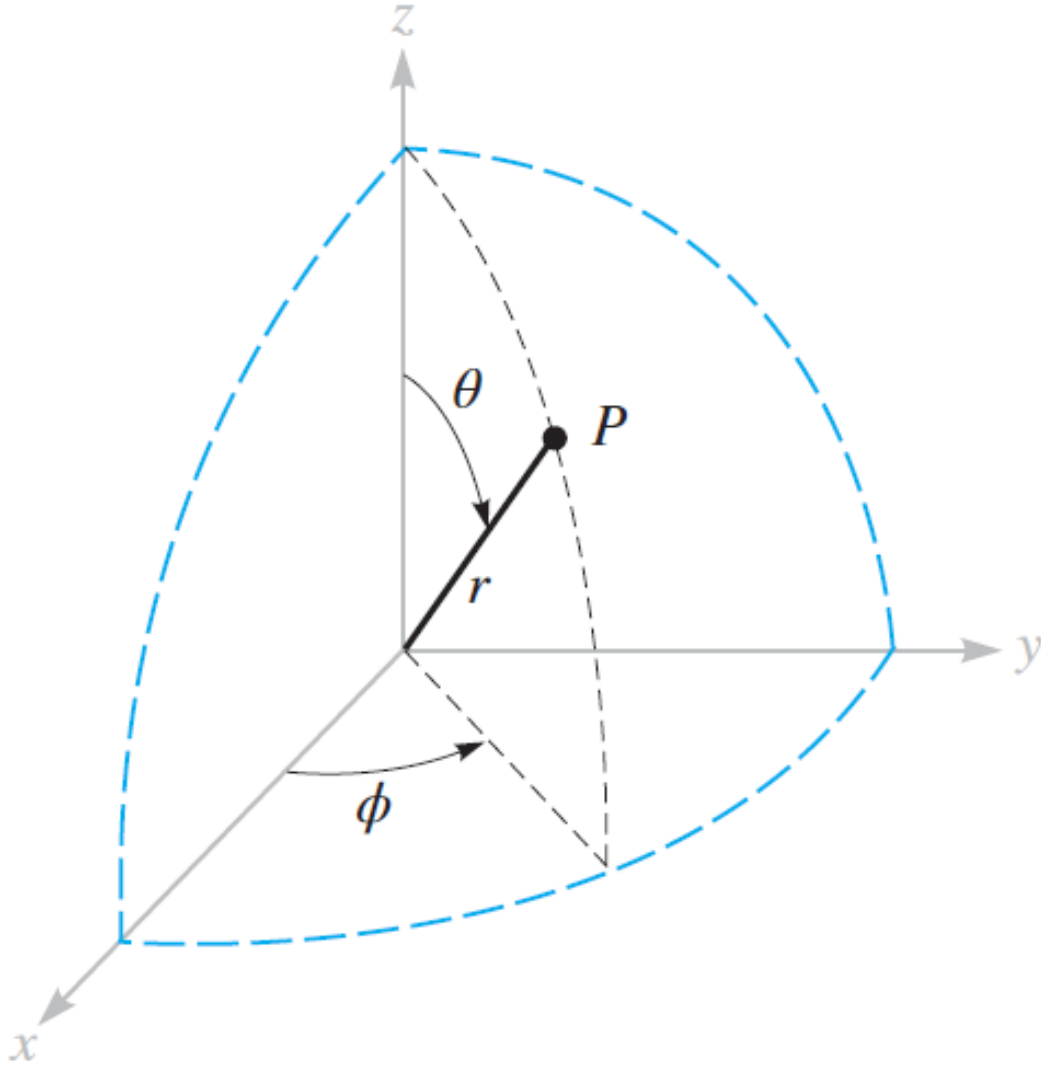
Vector Calculus – Spherical Coordinate Systems



Vector Calculus – Spherical Coordinate Systems

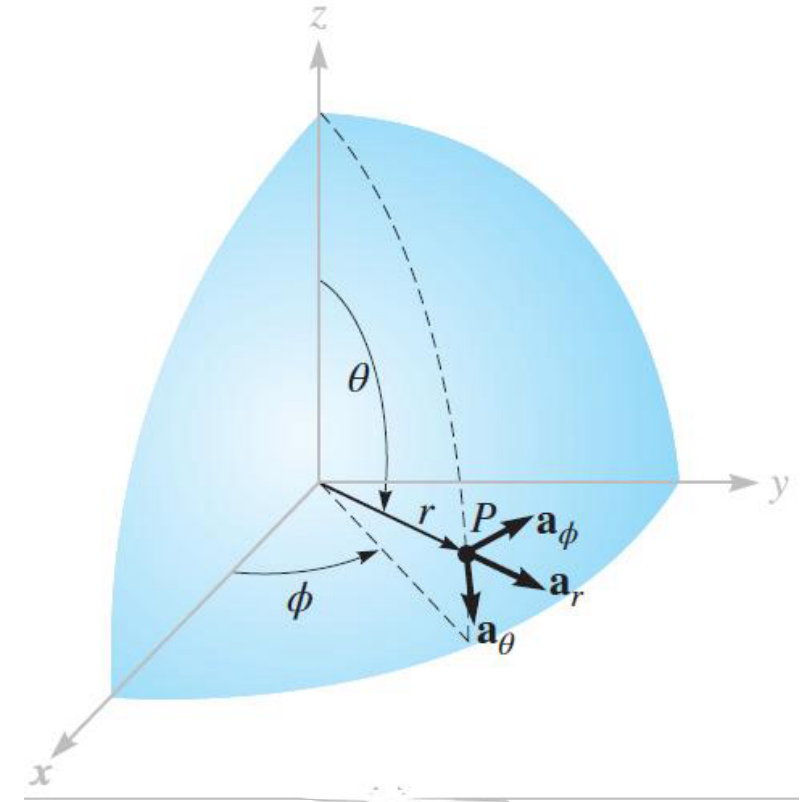
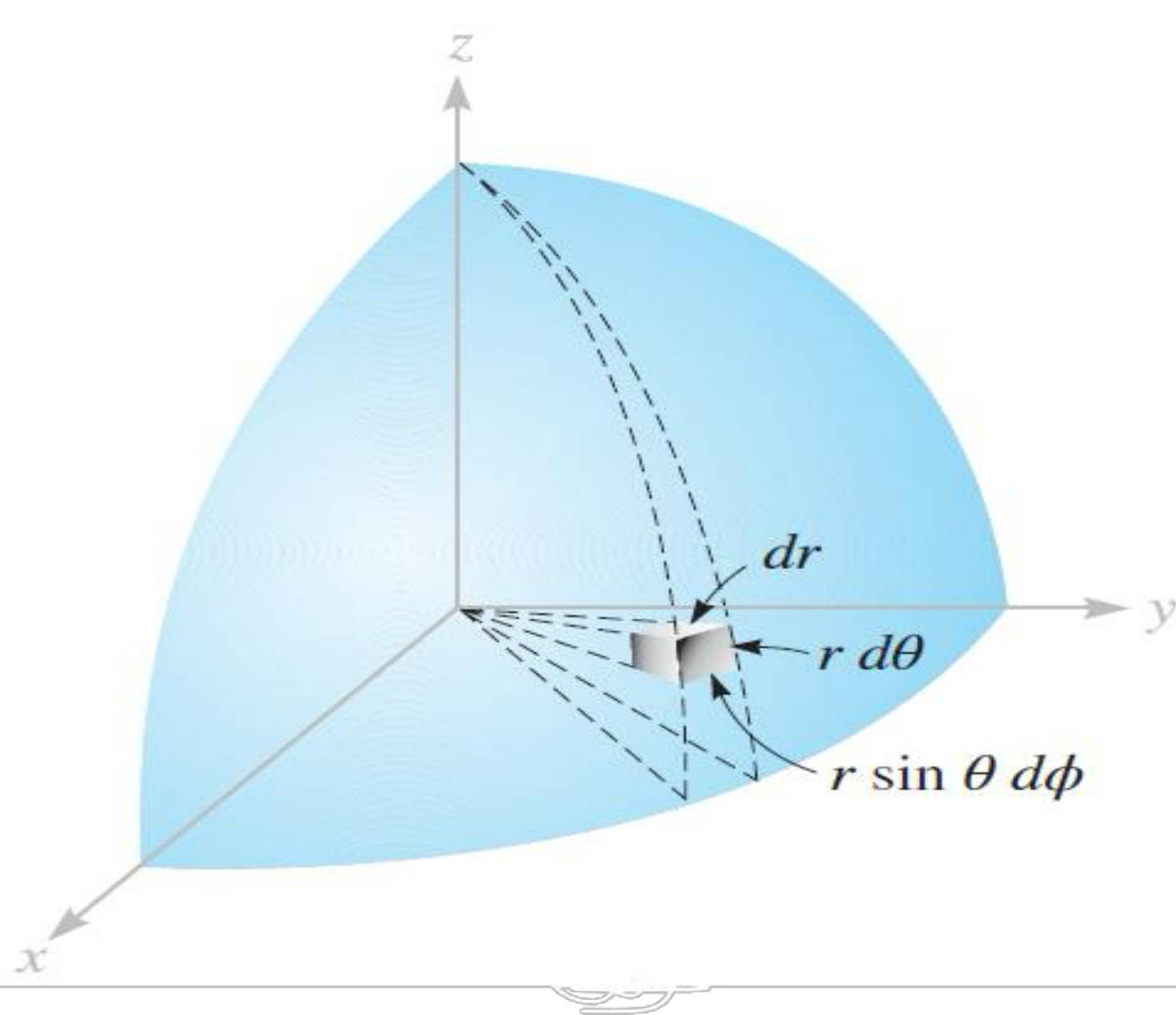


Vector Calculus – Spherical Coordinate Systems



$$r = \sqrt{x^2 + y^2 + z^2}$$
$$\theta = \cos^{-1} \frac{z}{\sqrt{x^2 + y^2 + z^2}}$$
$$\phi = \tan^{-1} \frac{y}{x}$$

Vector Calculus – Spherical Coordinate Systems



Vector Calculus – Spherical Coordinate Systems

- Differential displacement or length element,

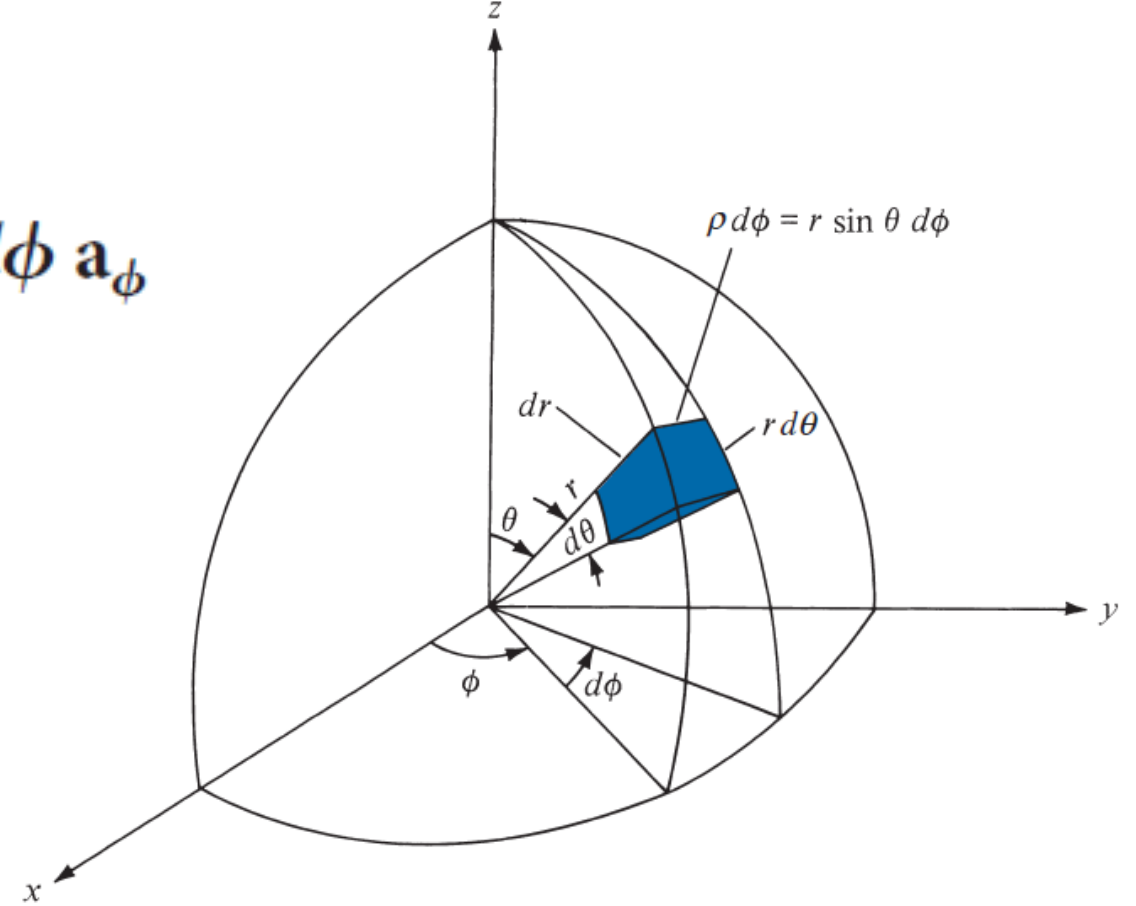
$$d\mathbf{l} = dr \mathbf{a}_r + r d\theta \mathbf{a}_\theta + r \sin \theta d\phi \mathbf{a}_\phi$$

- Differential normal surface area,

$$d\mathbf{S} = \mathbf{n} dS$$

- Differential volume element,

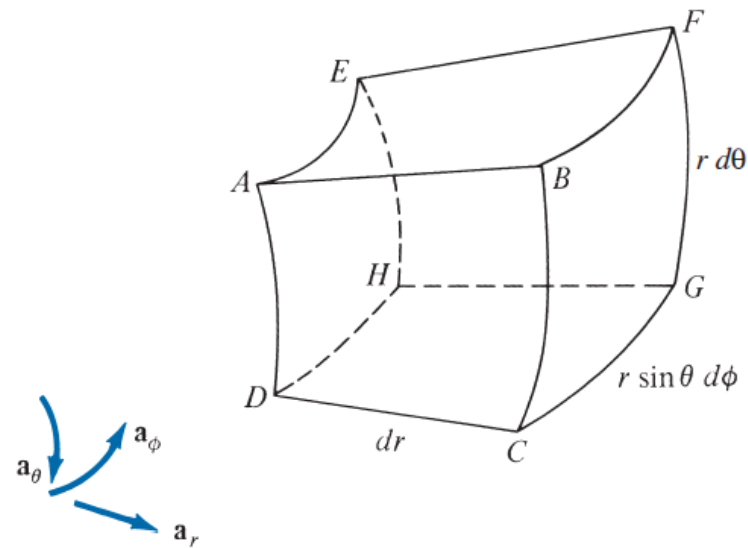
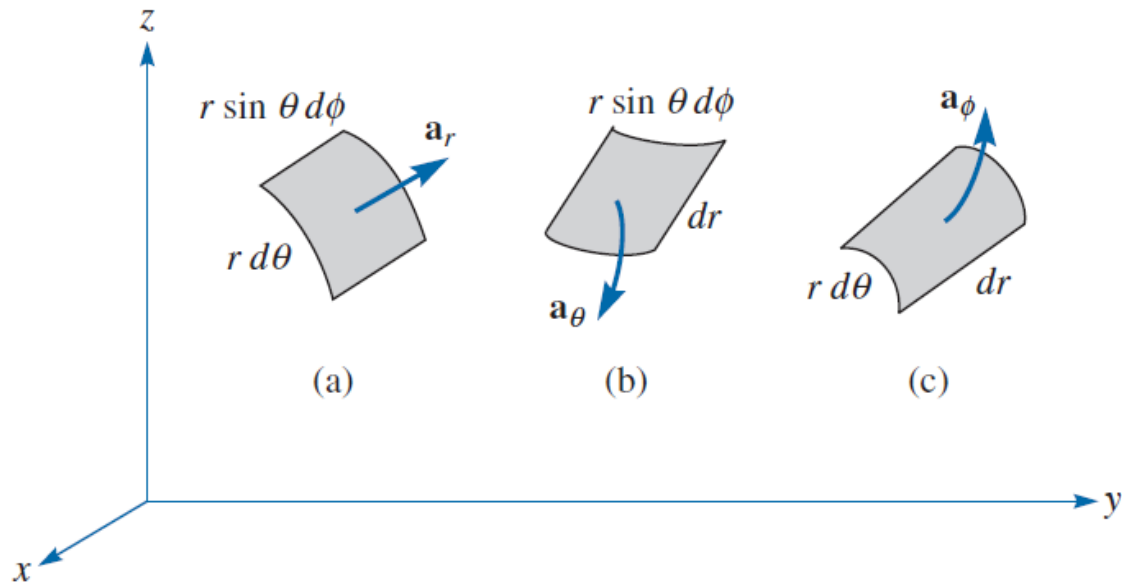
$$dv = r^2 \sin \theta dr d\theta d\phi$$



Vector Calculus – Spherical Coordinate Systems

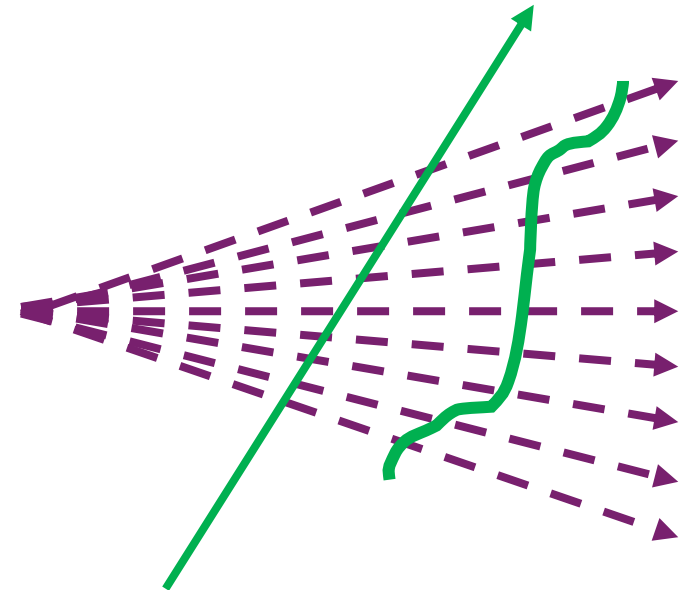
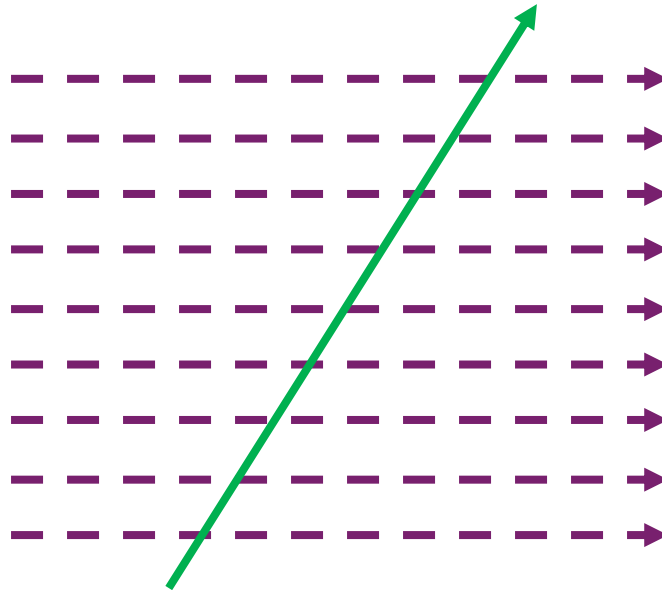
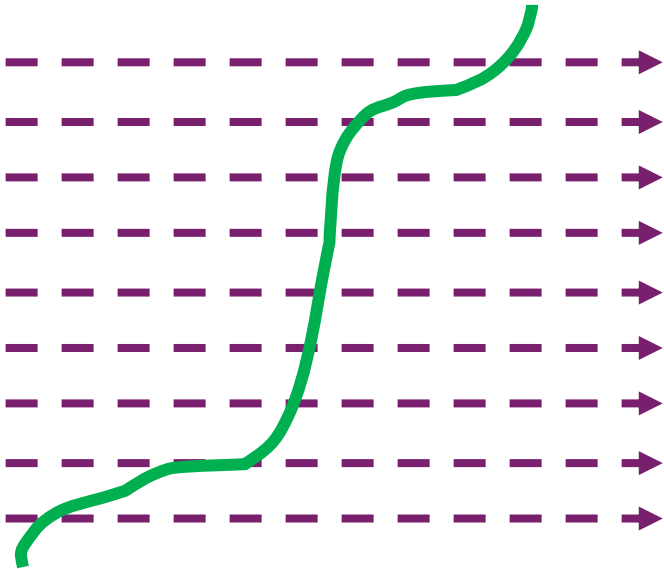
- Differential normal surface area,

$$d\mathbf{S} = \mathbf{n}dS$$



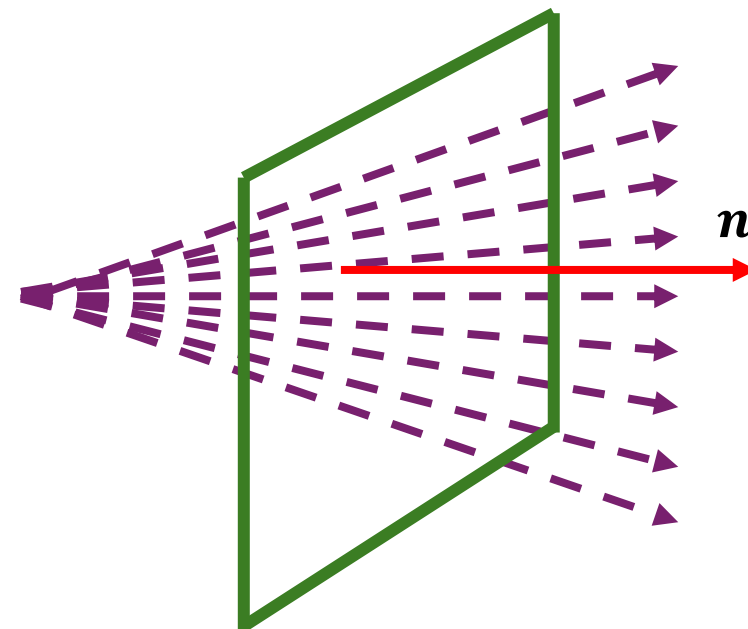
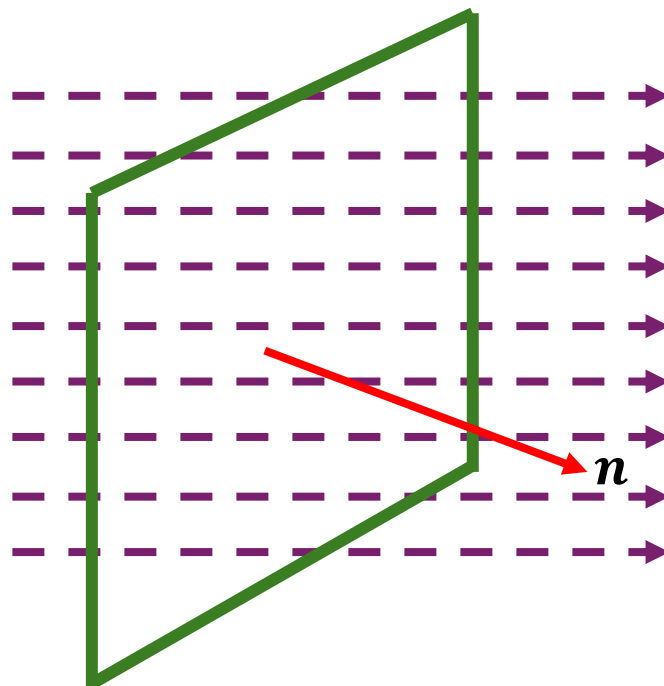
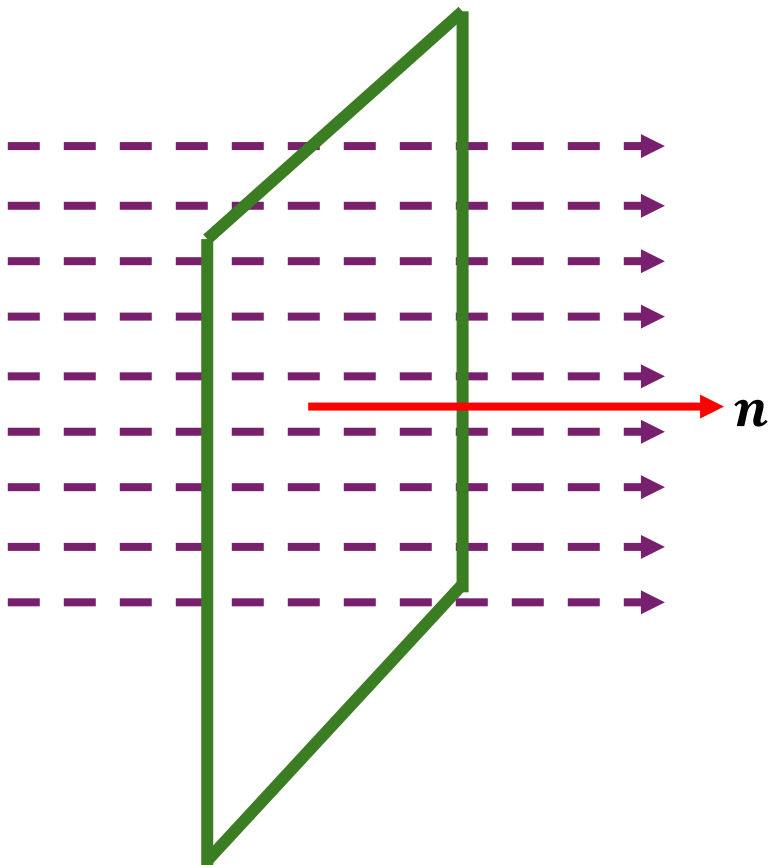
Vector Calculus – Line Integral

$$\oint_L \mathbf{A} \cdot d\mathbf{l}$$



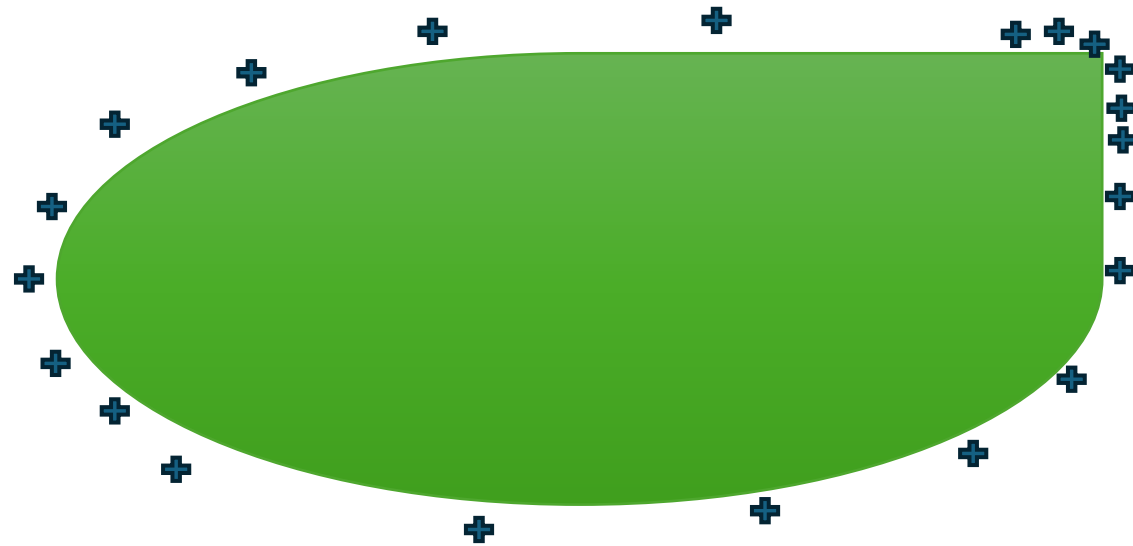
Vector Calculus – Surface Integral

$$\Psi = \int_S \mathbf{A} \cdot d\mathbf{S}$$



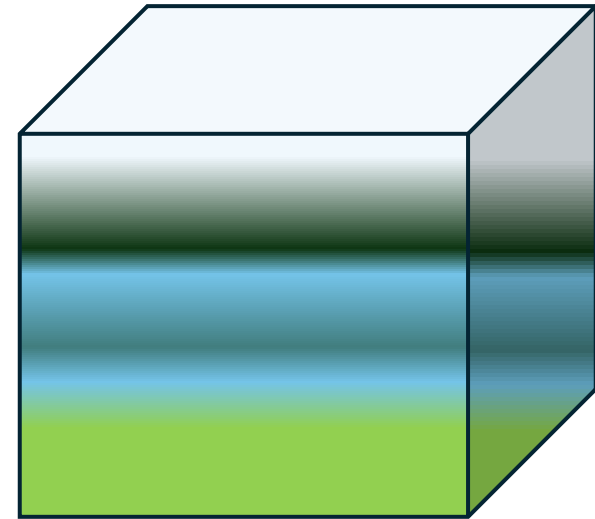
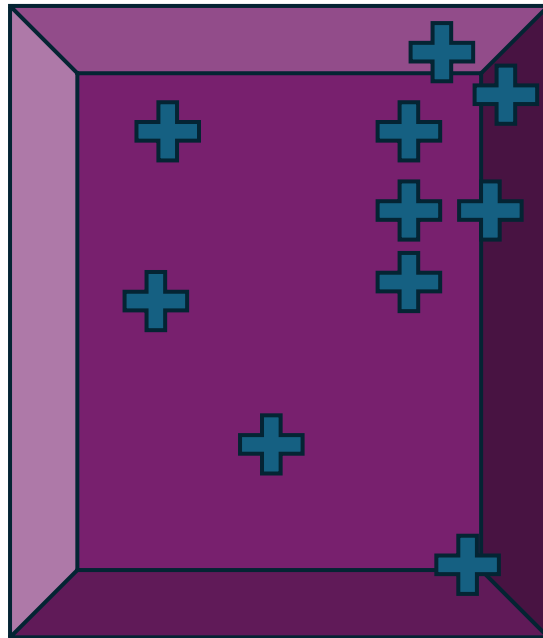
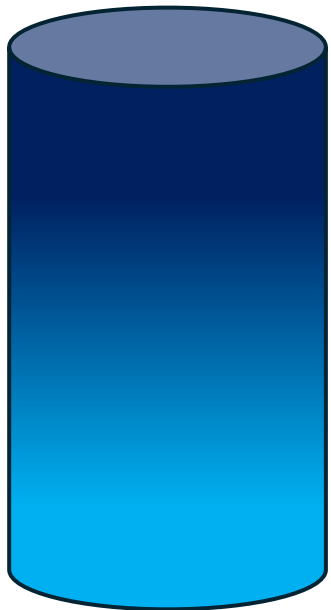
Vector Calculus – Surface Integral

$$\Psi = \oint_S \mathbf{A} \cdot d\mathbf{S}$$



Vector Calculus – Volume Integral

$$\int_v \rho_v dv$$



Field

- Quantities are not confined to a single point.
- They exist everywhere in space.
- At each location, the quantity maybe different.
- We need a way to describe how a quantity depends on position.

Field

- ✓ What a field is?
 - A **field** is a **physical quantity** defined at every point in **space**.
scalar → lower case

$$\underbrace{\phi(x, y, z), \quad \underline{f}(x, y, z), \quad V(x, y, z)}$$

$$\mathbf{A}(x, y, z), \quad \mathbf{E}(x, y, z)$$

$$\mathbf{A}(x, y, z) = A_x(x, y, z) \hat{a}_x + A_y(x, y, z) \hat{a}_y + A_z(x, y, z) \hat{a}_z$$

Scalar Field

- A scalar field describes how a quantity is distributed throughout space.
- At each point, the quantity has magnitude only, with no direction.
- The field is fully described by how this value varies with position.
- ❖ *A scalar field tells us how much of something exists at every point in space.*

Scalar Field

✓ What a scalar field is?

➤ A **scalar field** is a **physical quantity** defined at every point in space, where each point is assigned a **single scalar value**.

1. Temperature $T = T(x, y, z)$

2. Electric Potential $V = V(x, y, z)$

3. Charge Density $\rho = \rho(x, y, z)$

$$\frac{\partial}{\partial x} \quad \frac{\partial}{\partial y} \quad \frac{\partial}{\partial z}$$

Del Operator

- ✓ The del operator (∇), also called Nabla, is a vector calculus operator.
- ✓ The del operator has vector-like form, but its components are differential operators, not physical vector components.
- ✓ Like other differential operators, the del operator is used to describe change.
- ✓ When applied to scalar or vector fields, it tells us how those fields vary throughout space.

Del Operator in Different Coordinate Systems

➤ Rectangular Coordinate System

$$\nabla = \frac{\partial}{\partial x} \mathbf{a}_x + \frac{\partial}{\partial y} \mathbf{a}_y + \frac{\partial}{\partial z} \mathbf{a}_z$$

➤ Cylindrical Coordinate System

$$\nabla = \mathbf{a}_\rho \frac{\partial}{\partial \rho} + \mathbf{a}_\phi \frac{1}{\rho} \frac{\partial}{\partial \phi} + \mathbf{a}_z \frac{\partial}{\partial z}$$

to remove ρ component from ϕ .
only want angle change, & nothing else

➤ Spherical Coordinate System

$$\nabla = \mathbf{a}_r \frac{\partial}{\partial r} + \mathbf{a}_\theta \frac{1}{r} \frac{\partial}{\partial \theta} + \mathbf{a}_\phi \frac{1}{r \sin \theta} \frac{\partial}{\partial \phi}$$

remove unwanted components

Different Operations using Del – Gradient

- For this purpose, the del operator acts **on a scalar field**.
↳ gives direction for scalar
- The result is a vector field whose magnitude represents the rate of change of the scalar field, and whose direction indicates the direction of the greatest increase of the field.
- The gradient determines the rate of change of the scalar field along a given direction.
* gradient is result of Δ operator on scalar field.
◦ gives direction of max change / ROC
↳ ie where change is the strongest

Gradient in different Coordinate Systems

➤ Rectangular Coordinate System

$$\nabla V = \frac{\partial V}{\partial x} \mathbf{a}_x + \frac{\partial V}{\partial y} \mathbf{a}_y + \frac{\partial V}{\partial z} \mathbf{a}_z$$

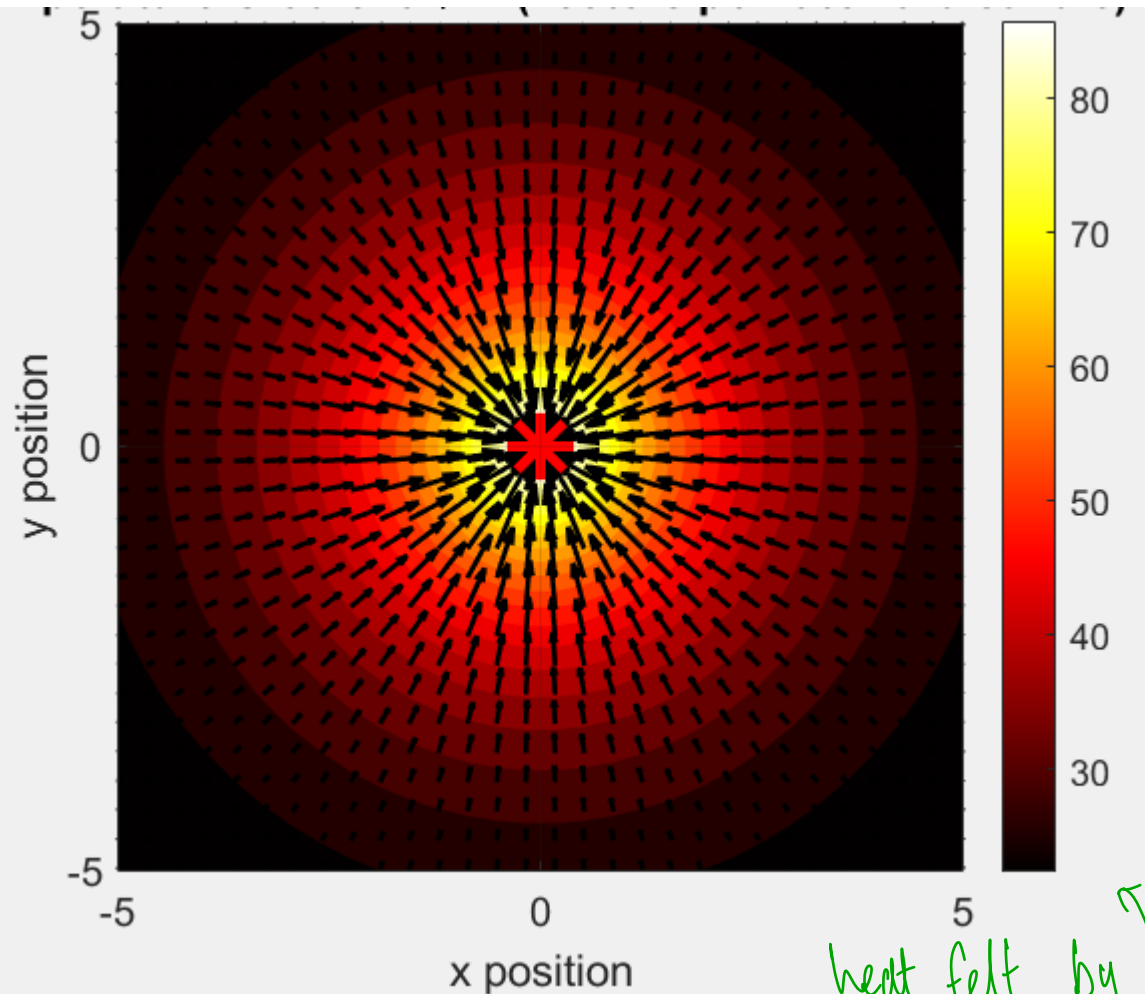
➤ Cylindrical Coordinate System

$$\nabla V = \frac{\partial V}{\partial \rho} \mathbf{a}_\rho + \frac{1}{\rho} \frac{\partial V}{\partial \phi} \mathbf{a}_\phi + \frac{\partial V}{\partial z} \mathbf{a}_z$$

➤ Spherical Coordinate System

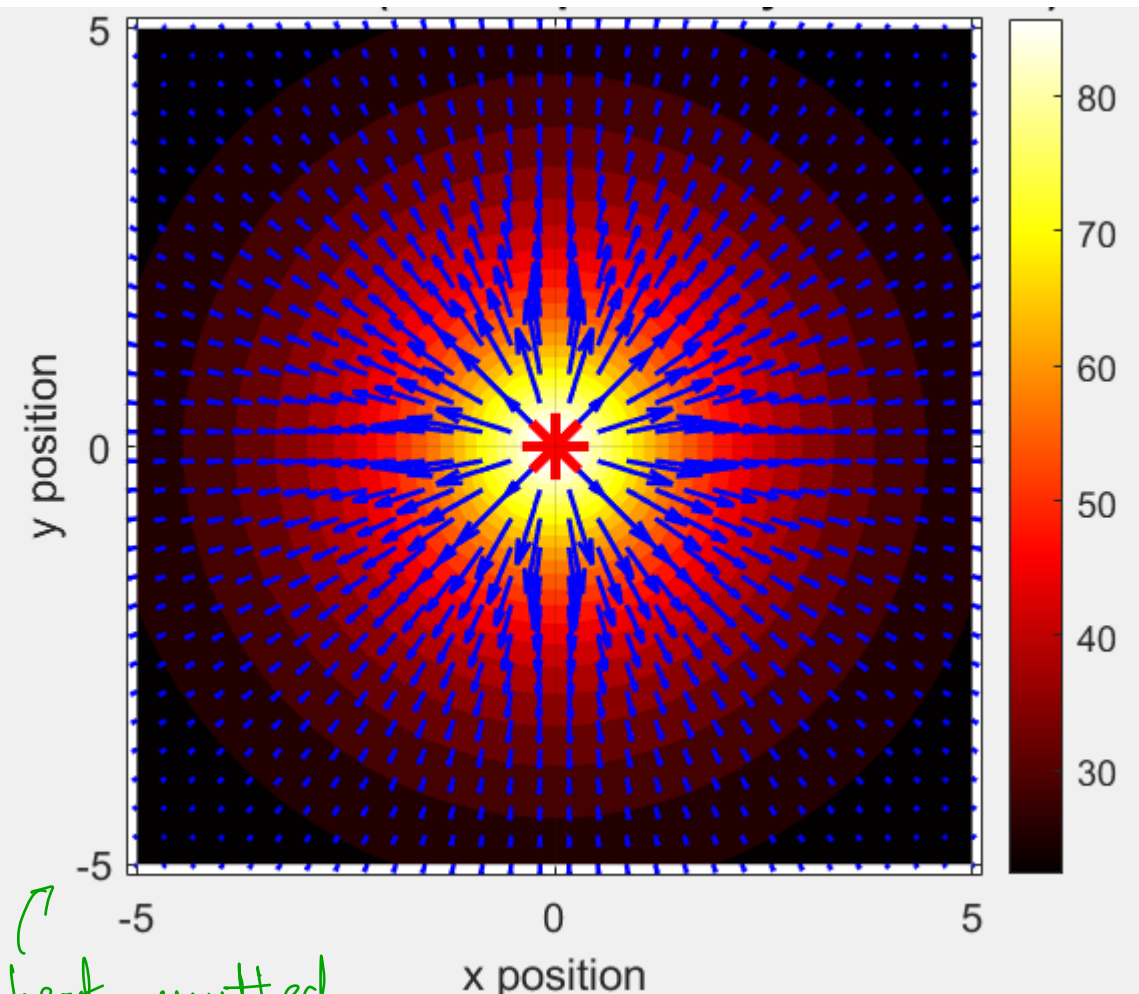
$$\nabla V = \frac{\partial V}{\partial r} \mathbf{a}_r + \frac{1}{r} \frac{\partial V}{\partial \theta} \mathbf{a}_\theta + \frac{1}{r \sin \theta} \frac{\partial V}{\partial \phi} \mathbf{a}_\phi$$

Gradient of Temperature – Single source



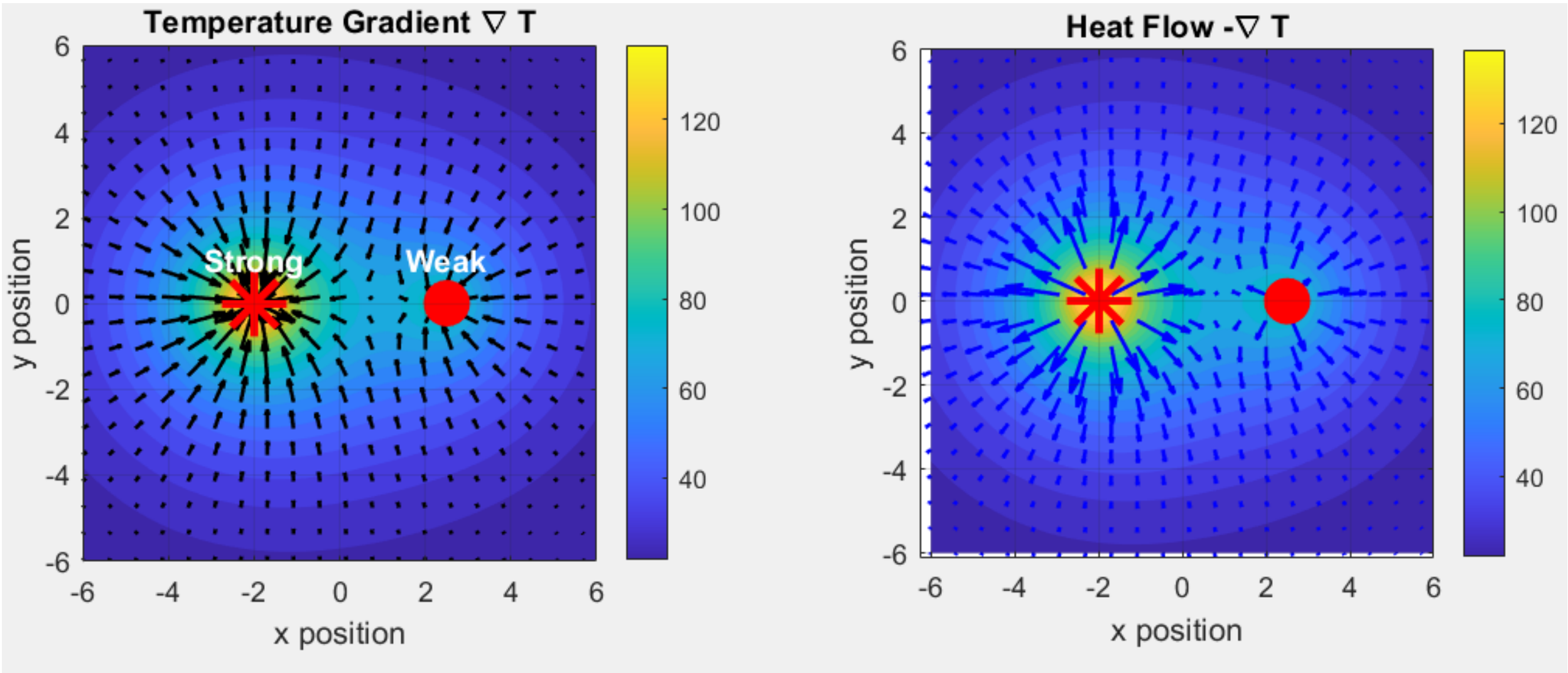
heat felt by
hand around
candle

↳ cylindrical coord system



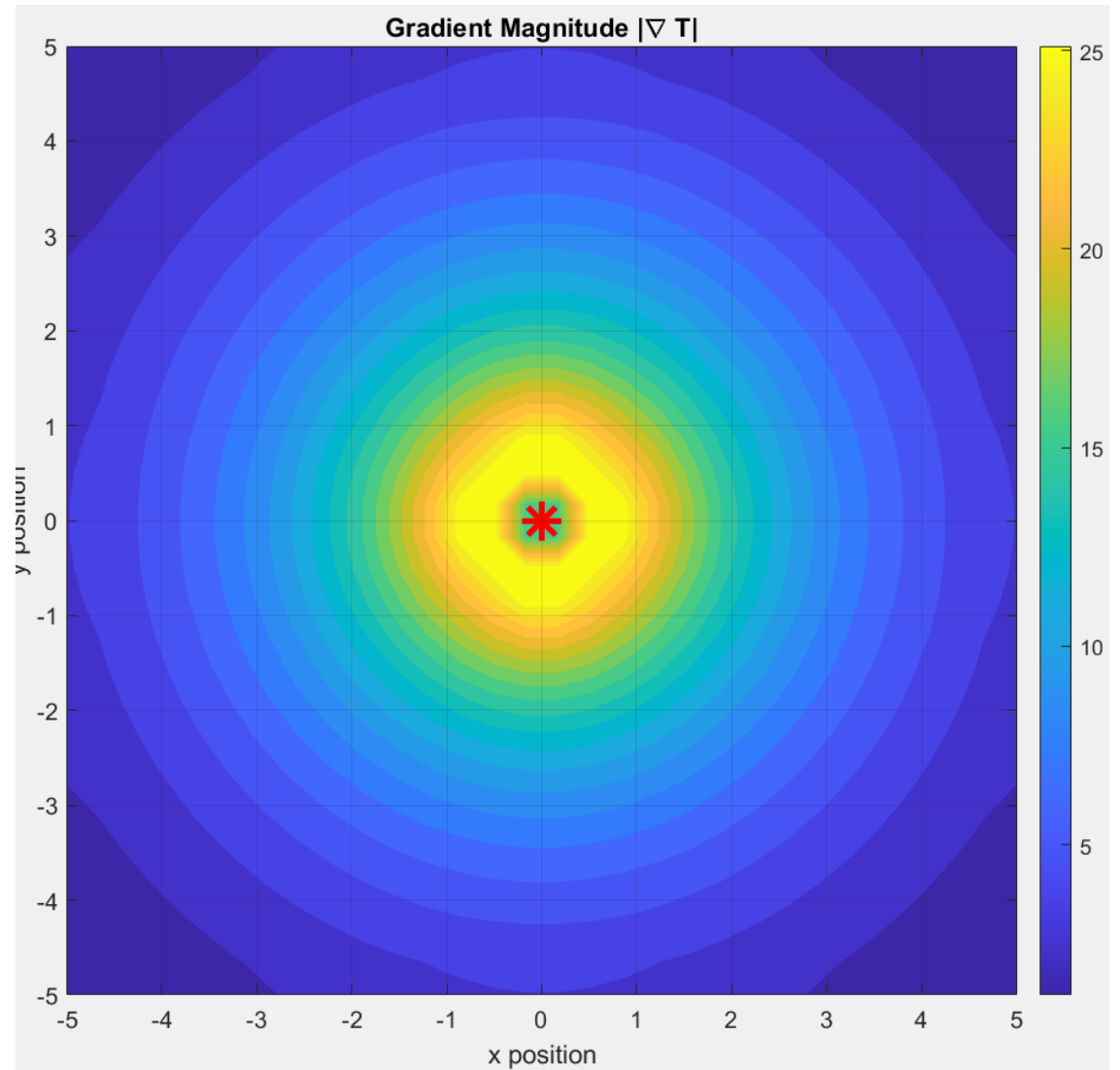
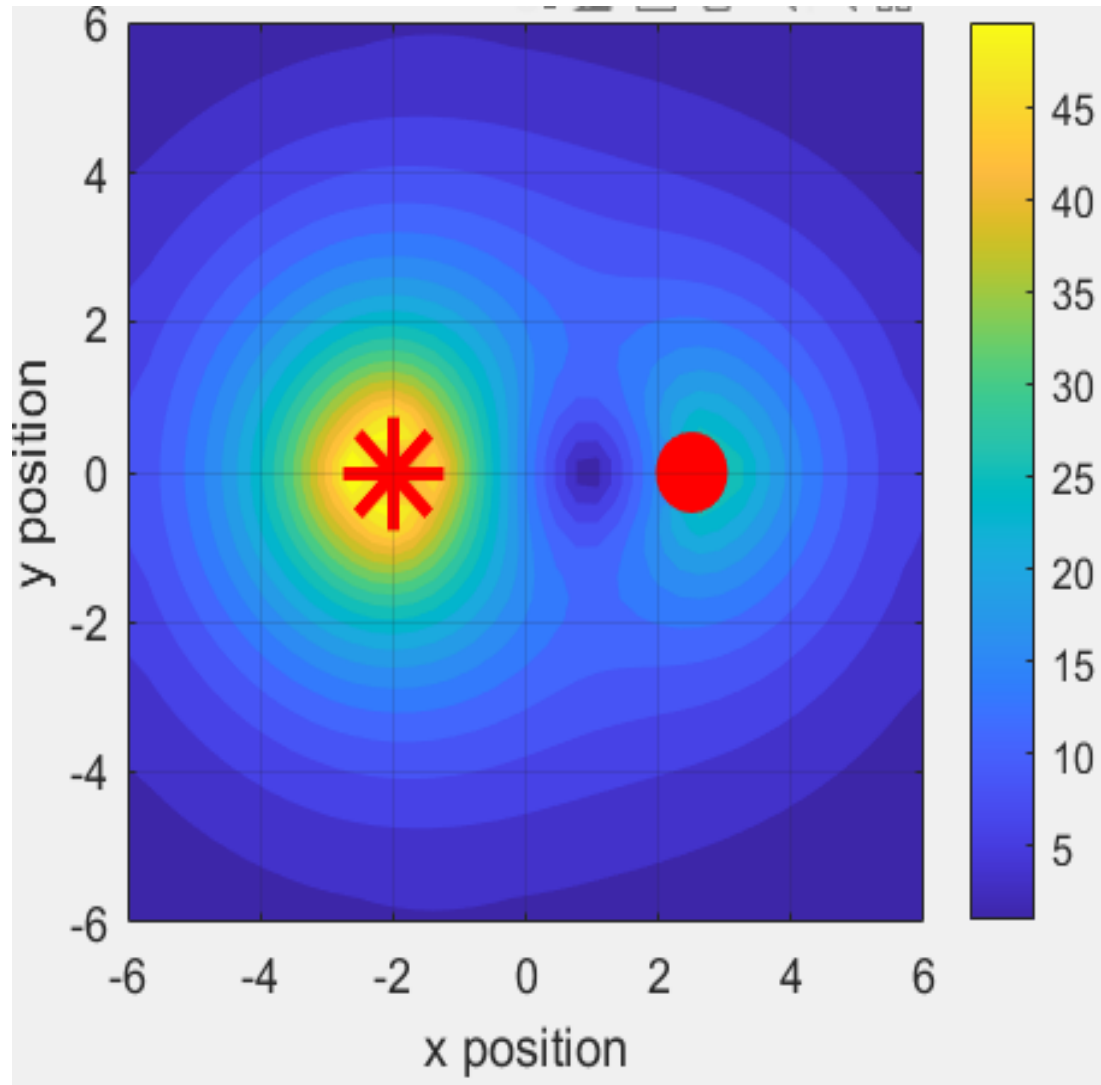
heat emitted
by candle

Gradient of Temperature – Two sources



* points to candle w/ stronger force

Gradient of Temperature – Magnitude



magnitude: $|\nabla T|$

Different Operations using Del – Divergence

- When the del operator takes the divergence of a vector field, it results in a scalar field that tells us how much the field spreads out or converges at each point in space.

$\text{divergence: } \nabla \cdot \vec{A}$

$\leftarrow$ dot product
 $\leftarrow$ answer = scalar
- The divergence is positive where field lines spread out and negative where they converge.
- In physical fields, divergence is often related to the presence of sources or sinks.
- Divergence is a local quantity. It depends only on the field behavior in an infinitesimal region around a point.

Divergence in different Coordinate Systems

➤ Rectangular Coordinate System

$$\nabla \cdot \mathbf{A} = \frac{\partial A_x}{\partial x} + \frac{\partial A_y}{\partial y} + \frac{\partial A_z}{\partial z}$$

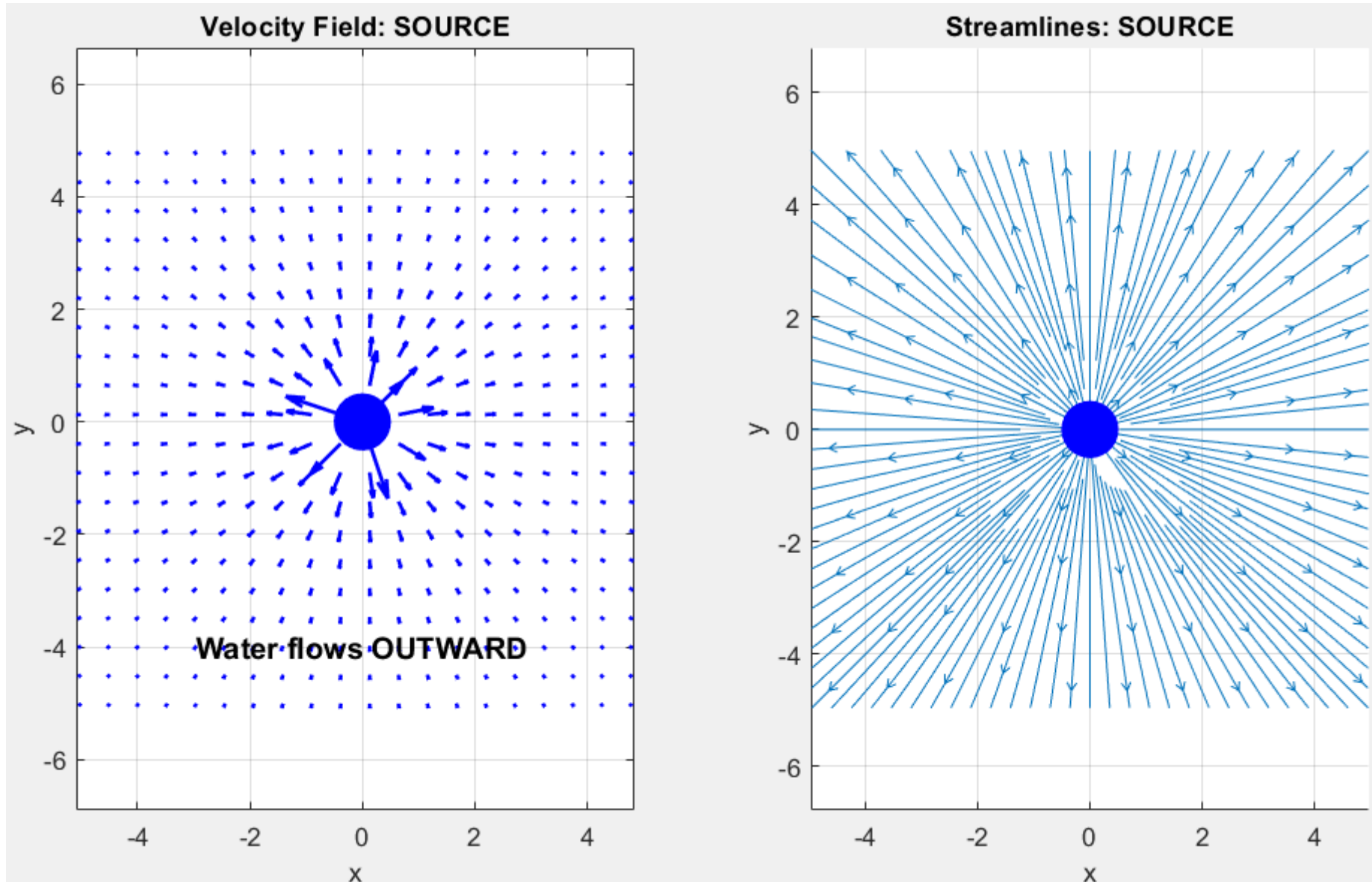
➤ Cylindrical Coordinate System

$$\nabla \cdot \mathbf{A} = \frac{1}{\rho} \frac{\partial}{\partial \rho} (\rho A_\rho) + \frac{1}{\rho} \frac{\partial A_\phi}{\partial \phi} + \frac{\partial A_z}{\partial z}$$

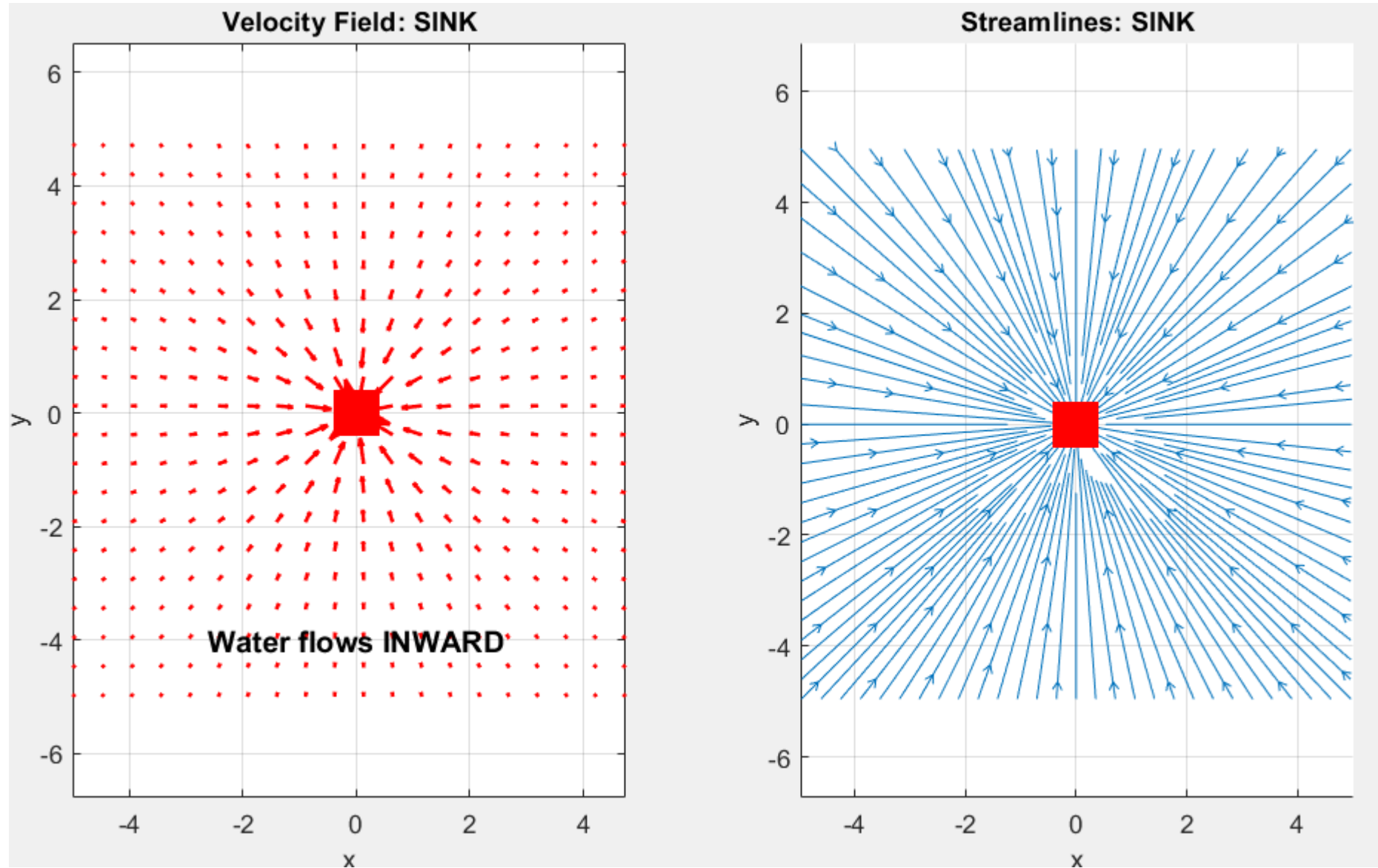
➤ Spherical Coordinate System

$$\nabla \cdot \mathbf{A} = \frac{1}{r^2} \frac{\partial}{\partial r} (r^2 A_r) + \frac{1}{r \sin \theta} \frac{\partial}{\partial \theta} (A_\theta \sin \theta) + \frac{1}{r \sin \theta} \frac{\partial A_\phi}{\partial \phi}$$

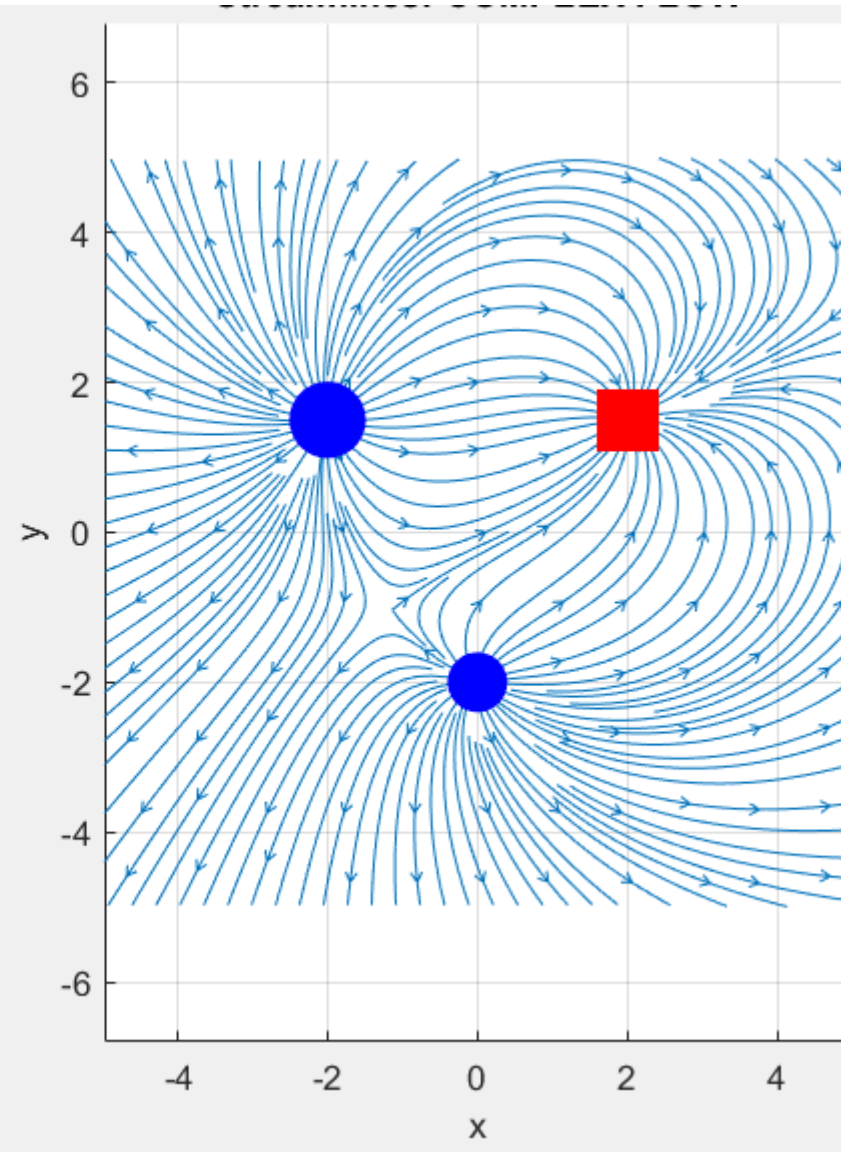
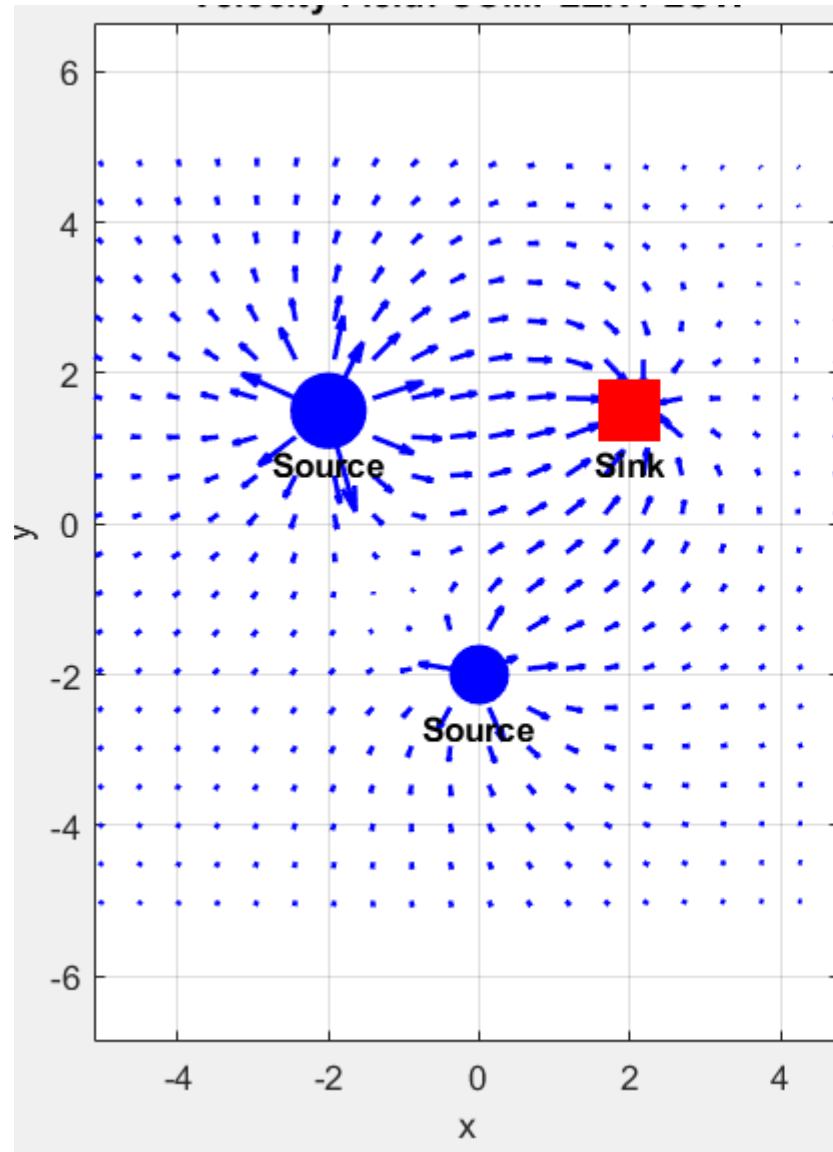
Divergence of a Water Spring



Divergence of a Water Drain



Divergence of two Sources and a sink



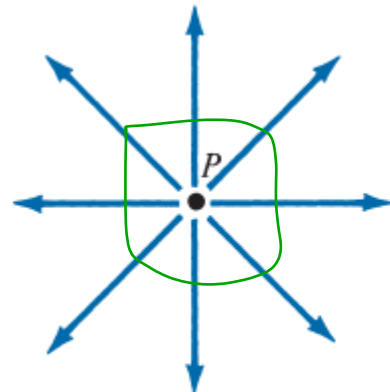
$$\text{source} = + \vec{a}$$
$$\text{sink} = - \vec{a}$$

Divergence Theorem

- The divergence of an arbitrary vector field ($\mathbf{A}$) at point P is the net outward flux of $\mathbf{A}$ through a small closed surface surrounding P , divided by the volume enclosed, as that volume shrinks to zero.

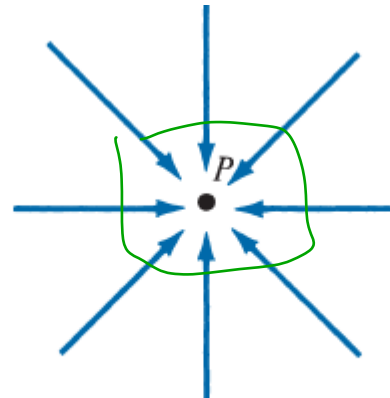
$$\text{div } \mathbf{A} = \nabla \cdot \mathbf{A} = \lim_{\Delta v \rightarrow 0} \frac{\oint_S \mathbf{A} \cdot d\mathbf{S}}{\Delta v}$$

$$\nabla \cdot \mathbf{A} > 0$$

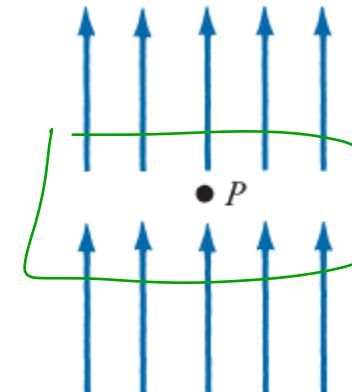


(a)

$$\nabla \cdot \mathbf{A} < 0$$



(b)

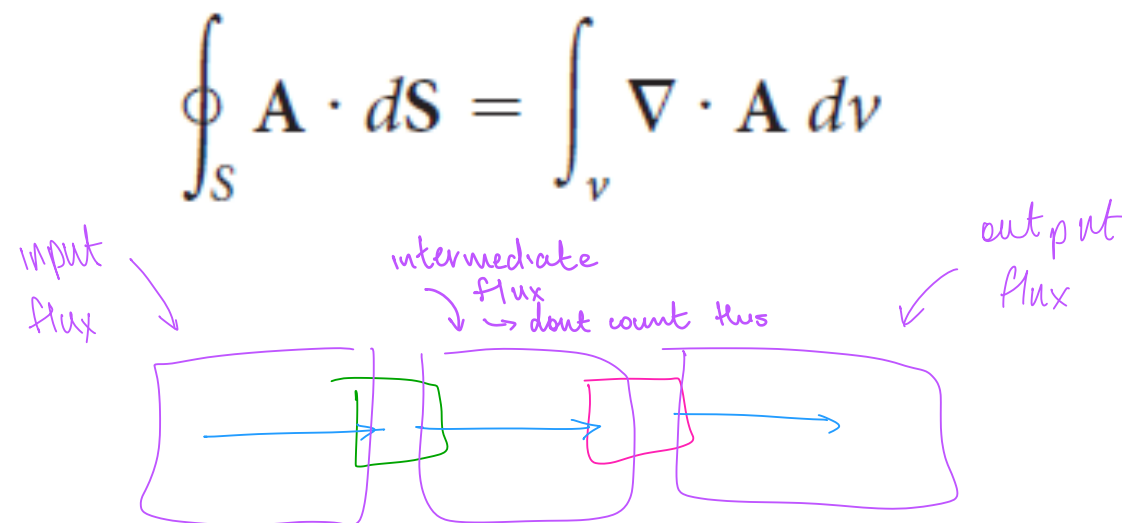


(c)

$$\nabla \cdot \mathbf{A} = 0$$

Divergence Theorem

- The divergence theorem states that the total outward flux of a vector field $\mathbf{A}$ through a closed surface S is equal to the sum of the divergences of $\mathbf{A}$ throughout the volume enclosed by that surface.
- ❖ *What flows out through the boundary of a region is exactly equal to what is generated inside the region.*



Divergence

- One physical meaning, two mathematical expressions; **Divergence**
- **Point Form**, what is happening **at a single point**

$$\nabla \cdot \mathbf{A}$$

- Use this when
 1. You want the divergence at **a point**
 2. You are given $A(x, y, z)$ explicitly
 3. You are analyzing sources and sinks locally
- Applications
 1. Charge density in electromagnetics
 2. Fluid flow, compression or expansion at a point
 3. Heat flow or diffusion problems

Divergence

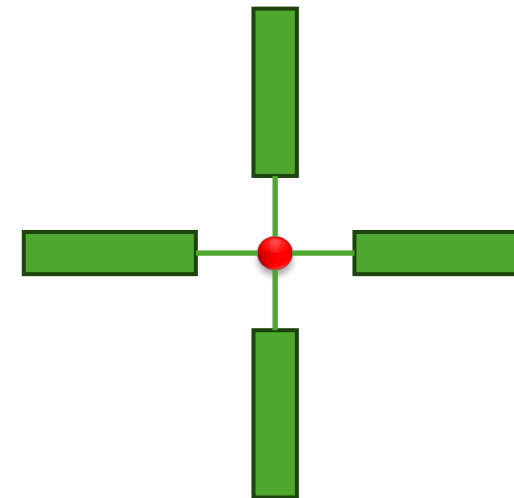
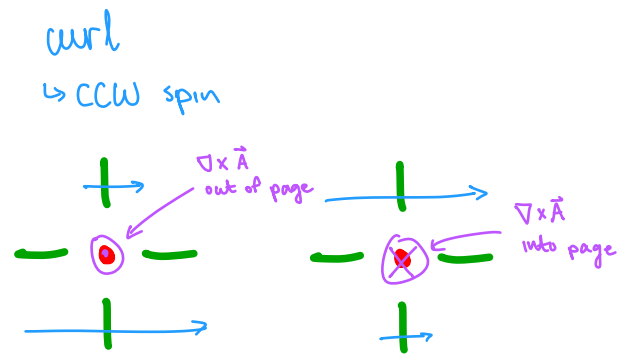
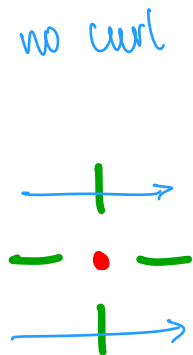
- One physical meaning, two mathematical expression; **Divergence**
- Integral or flux form, what is happening over a finite region

$$\oint_S \mathbf{A} \cdot d\mathbf{S} = \int_V (\nabla \cdot \mathbf{A}) dV$$

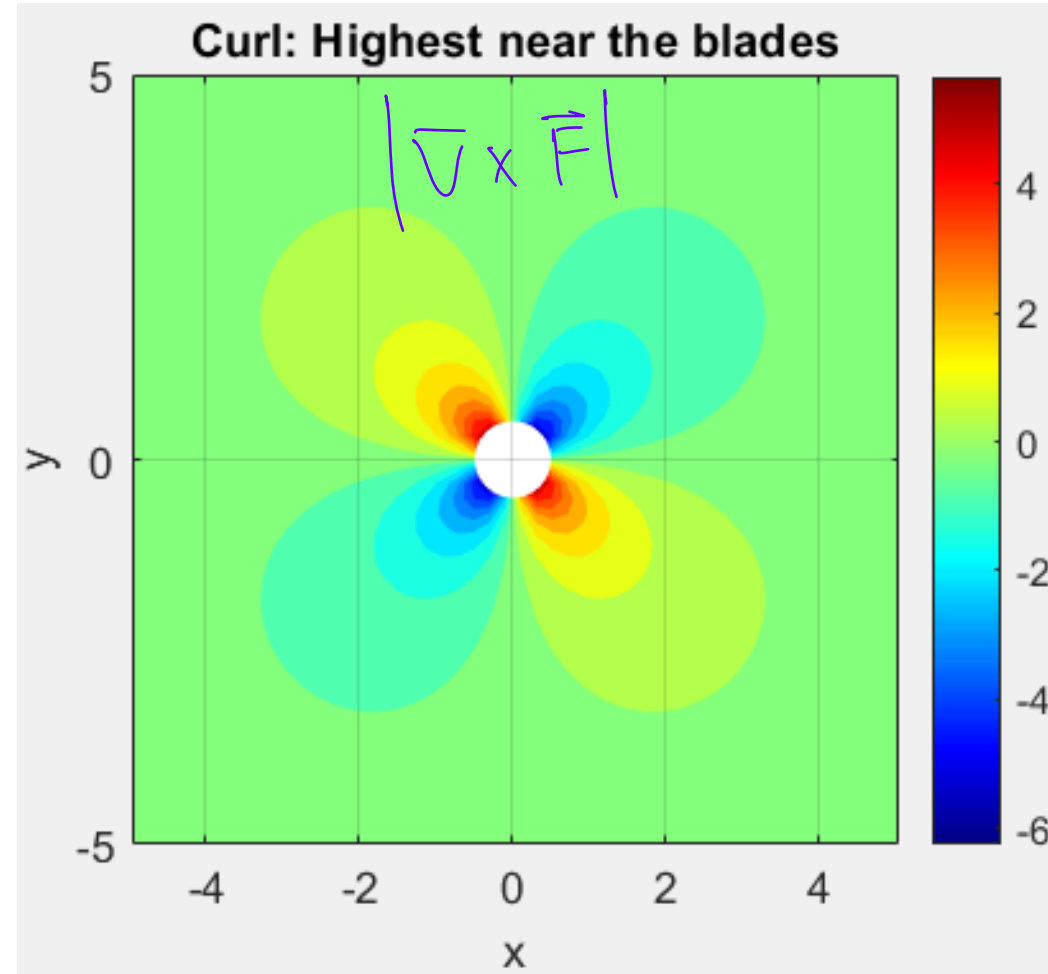
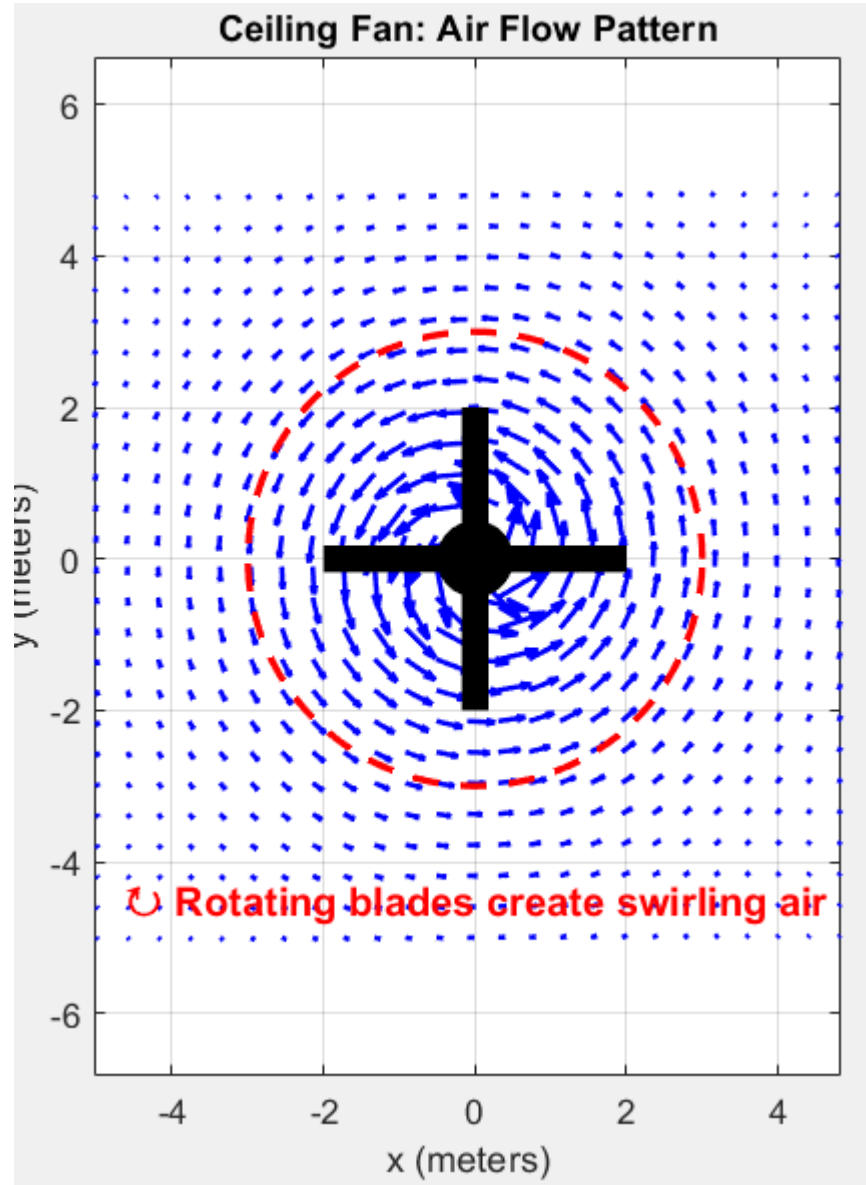
- Use this when
 1. You want the total flux through a surface
 2. The surface has symmetry
 3. Direct surface integration is hard, but volume behavior is simple
- Applications
 1. Gauss's law in electromagnetics
 2. Total charge enclosed
 3. Verifying conservation laws

Curl

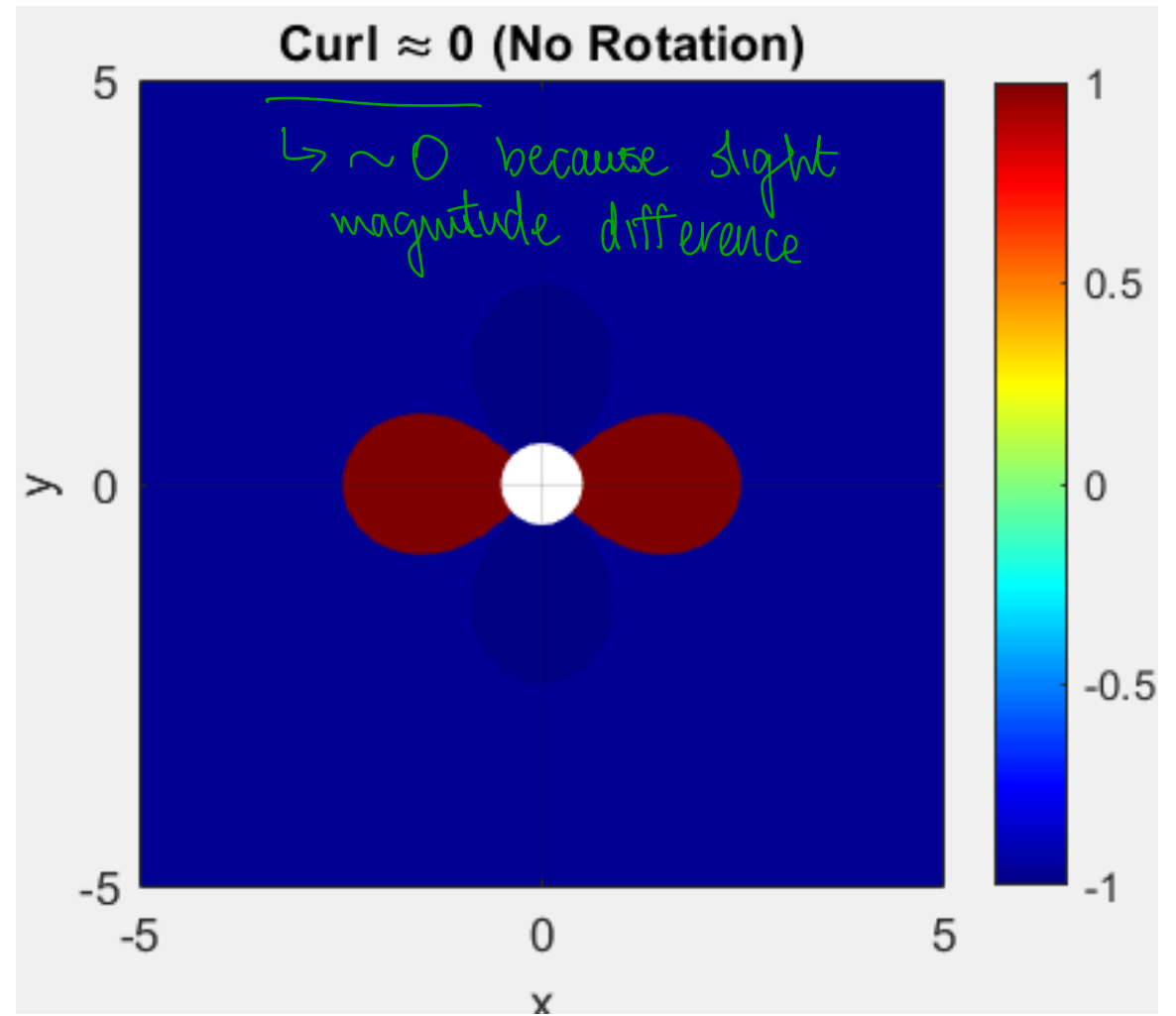
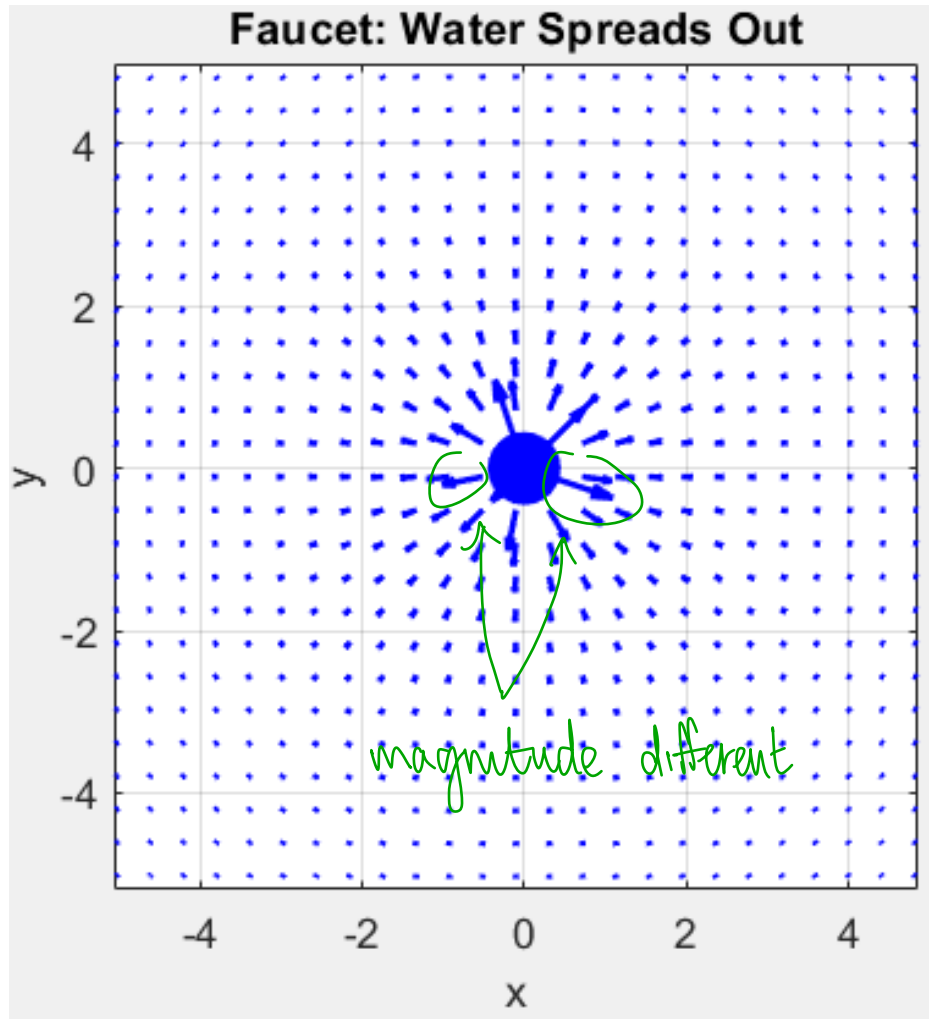
- If I put a tiny paddle wheel at this point, will it spin?
- Direction of curl = axis of rotation
- **Zero curl** does NOT mean **zero field**
- Curl of a vector field at a point measures how much and in which direction the field wants to rotate about that point.



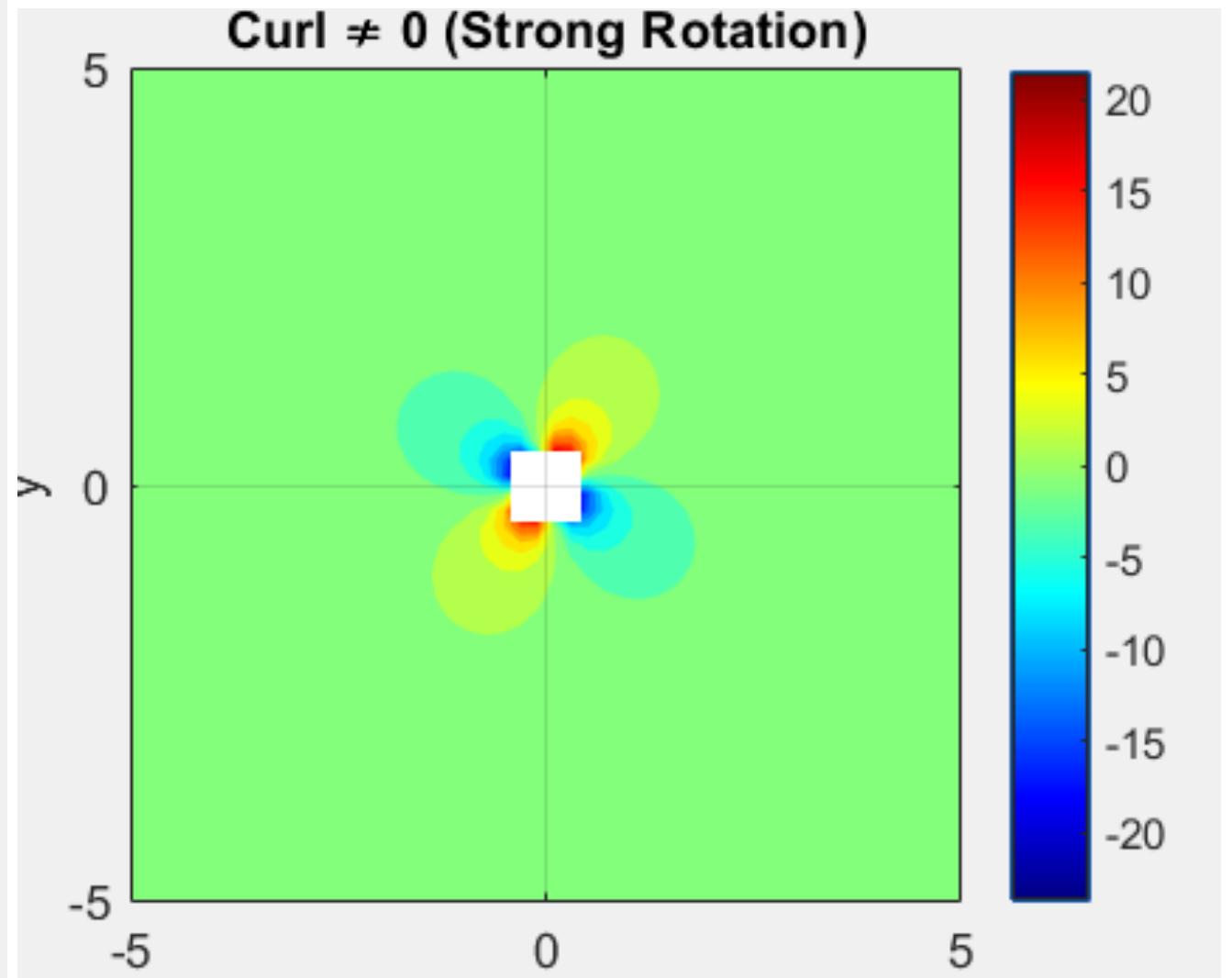
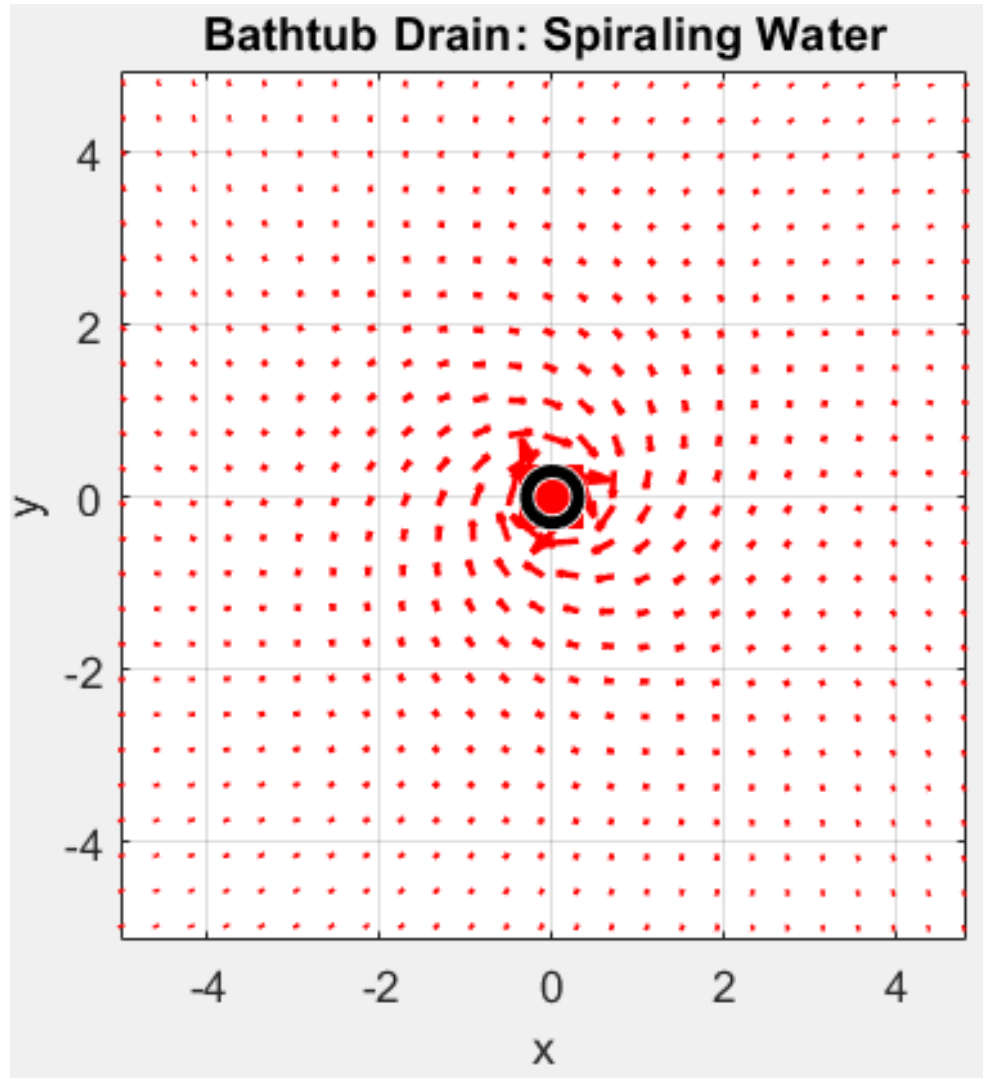
Curl



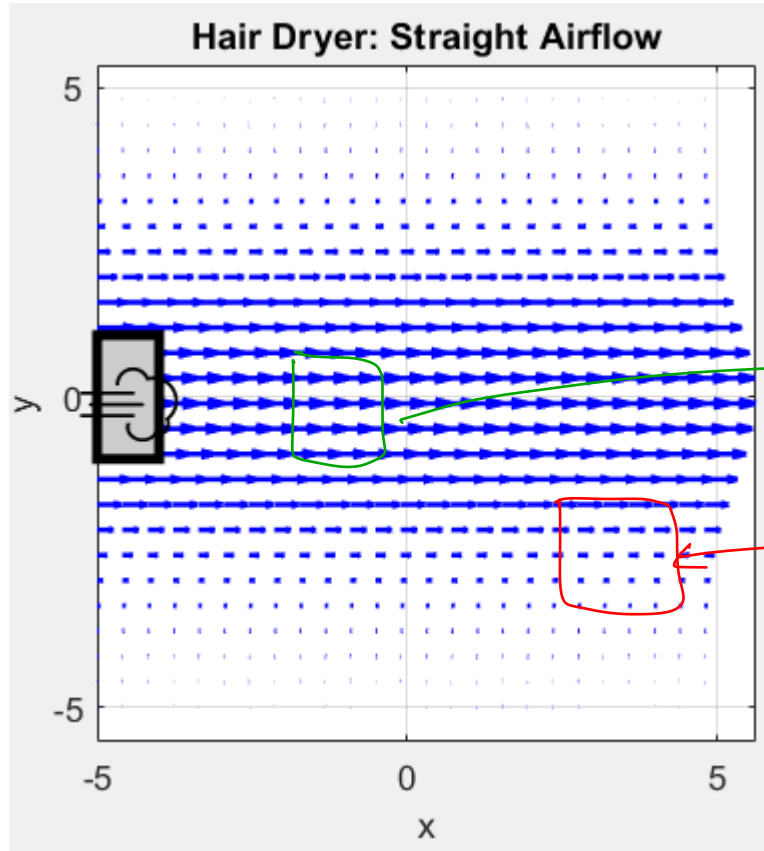
Curl



Curl

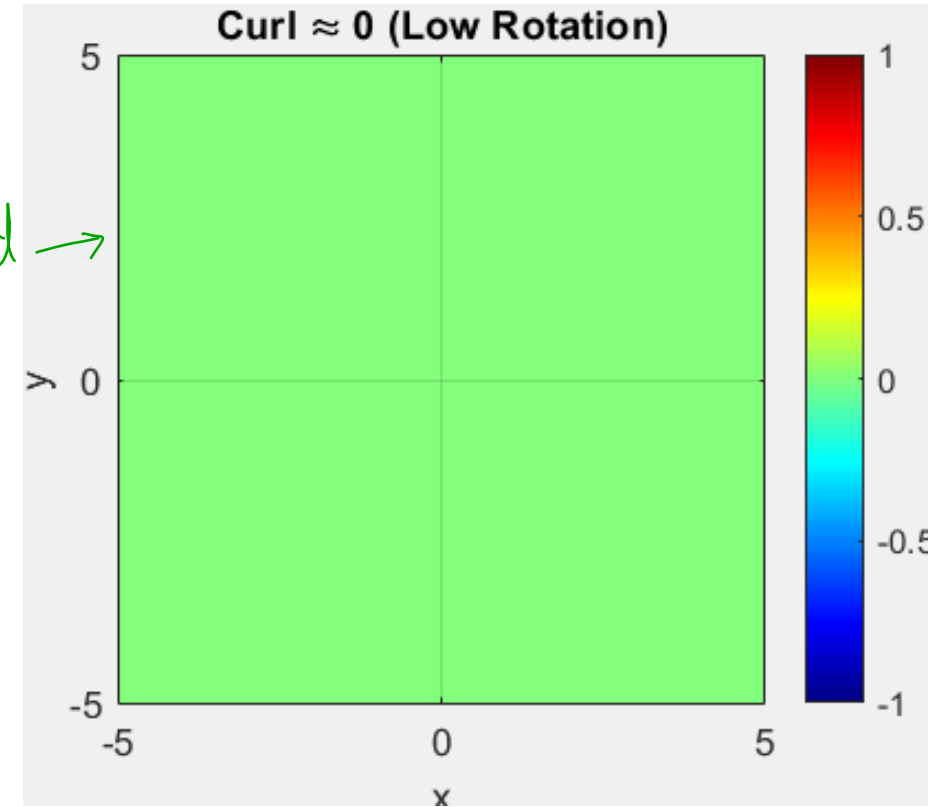


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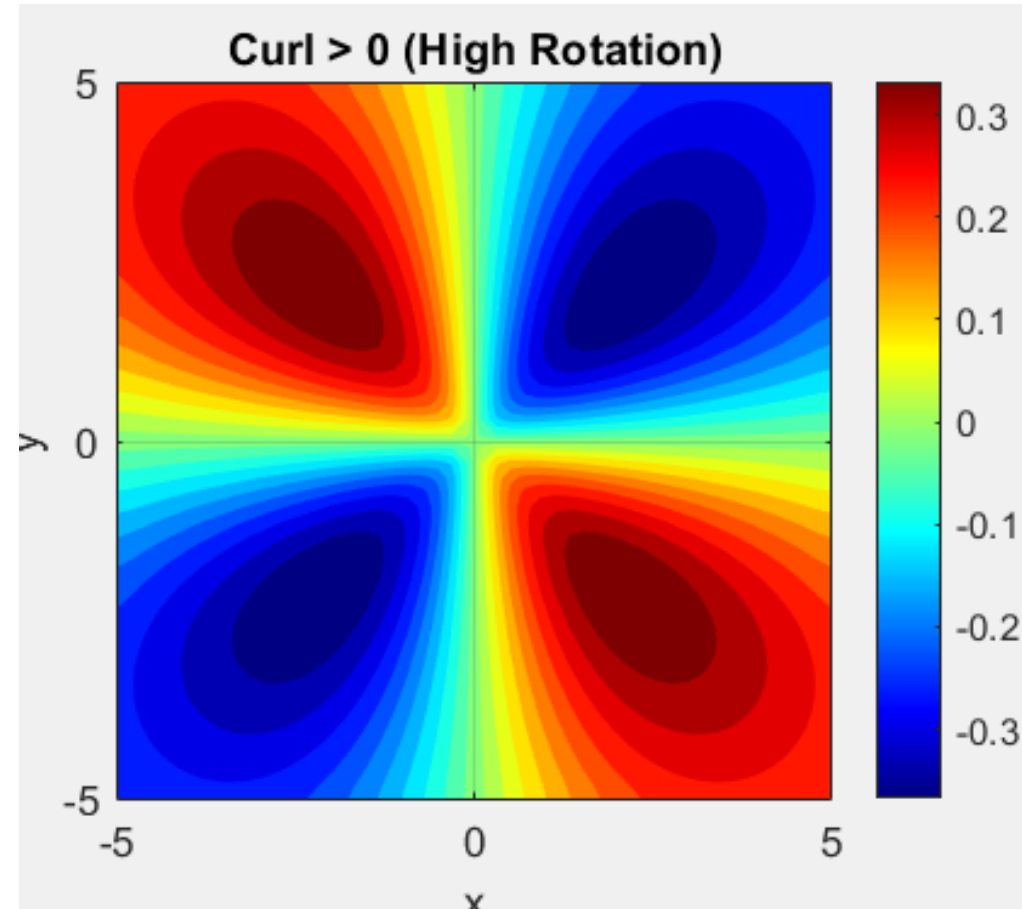
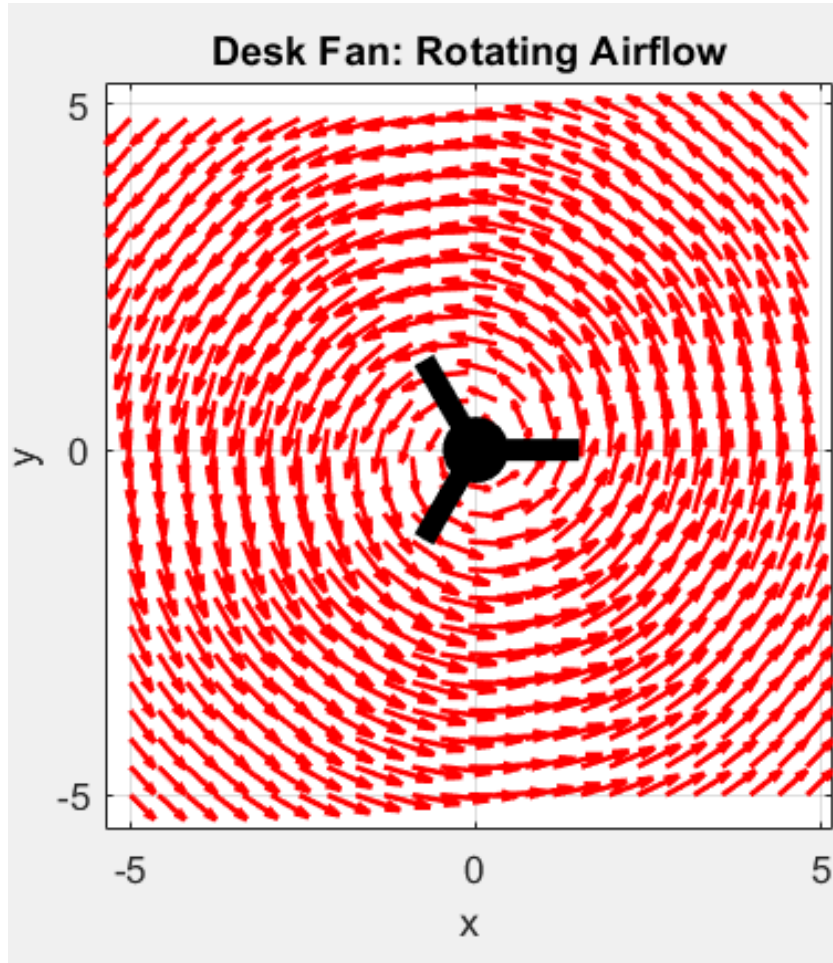


uniform represented →

← not uniform



Curl



Curl

➤ Rectangular Coordinate System

$$\nabla \times \mathbf{A} = \begin{vmatrix} \mathbf{a}_x & \mathbf{a}_y & \mathbf{a}_z \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ A_x & A_y & A_z \end{vmatrix}$$

➤ Cylindrical Coordinate System

$$\nabla \times \mathbf{A} = \frac{1}{\rho} \begin{vmatrix} \mathbf{a}_\rho & \rho \mathbf{a}_\phi & \mathbf{a}_z \\ \frac{\partial}{\partial \rho} & \frac{\partial}{\partial \phi} & \frac{\partial}{\partial z} \\ A_\rho & \rho A_\phi & A_z \end{vmatrix}$$

➤ Spherical Coordinate System

$$\nabla \times \mathbf{A} = \frac{1}{r^2 \sin \theta} \begin{vmatrix} \mathbf{a}_r & r \mathbf{a}_\theta & r \sin \theta \mathbf{a}_\phi \\ \frac{\partial}{\partial r} & \frac{\partial}{\partial \theta} & \frac{\partial}{\partial \phi} \\ A_r & r A_\theta & r \sin \theta A_\phi \end{vmatrix}$$

Curl – Cylindrical Coordinate System

➤ Cylindrical Coordinate System $\mathbf{a}_\rho = \cos \phi \mathbf{a}_x + \sin \phi \mathbf{a}_y$ $\mathbf{a}_\phi = -\sin \phi \mathbf{a}_x + \cos \phi \mathbf{a}_y$

$$\nabla \times \mathbf{A} = \frac{1}{\rho} \begin{vmatrix} \mathbf{a}_\rho & \rho \mathbf{a}_\phi & \mathbf{a}_z \\ \frac{\partial}{\partial \rho} & \frac{\partial}{\partial \phi} & \frac{\partial}{\partial z} \\ A_\rho & \rho A_\phi & A_z \end{vmatrix}$$

$$\vec{A} = \vec{A}_\rho \vec{a}_\rho + \vec{A}_\phi \vec{a}_\phi + \vec{A}_z \vec{a}_z$$

$$\vec{\nabla} = \vec{a}_\rho \frac{\partial}{\partial \rho} + \frac{1}{\rho} \vec{a}_\phi \frac{\partial}{\partial \phi} + \vec{a}_z \frac{\partial}{\partial z}$$

$$\nabla \times \mathbf{A} = \left(\vec{a}_\rho \frac{\partial}{\partial \rho} + \frac{1}{\rho} \vec{a}_\phi \frac{\partial}{\partial \phi} + \vec{a}_z \frac{\partial}{\partial z} \right) \times \vec{A}$$

$$\vec{a}_\rho \times \vec{a}_\rho = 0$$

$$\vec{a}_\phi \times \vec{a}_\rho = -\vec{a}_z$$

$$\vec{a}_z \times \vec{a}_\rho = \vec{a}_\phi$$

first term :

Curl

- ✓ **Curl tells us how much the field wants to spin, and which way it spins.**

$$\text{curl } \mathbf{A} = \nabla \times \mathbf{A} = \left(\lim_{\Delta S \rightarrow 0} \frac{\oint_L \mathbf{A} \cdot d\mathbf{l}}{\Delta S} \right)_{\max} \mathbf{a}_n$$

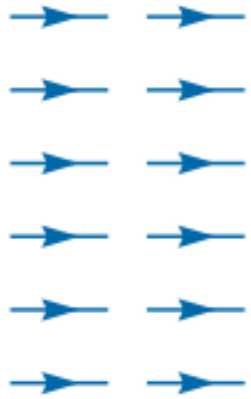
- The curl of a vector field tells us how much the field tends to rotate around a point, and the direction of the curl shows the axis of that rotation.
- The curl of $\mathbf{A}$ measures how strongly the field circulates around a point. Its direction shows the axis about which the field rotates, and its size tells how intense that rotation is.
- The curl of $\mathbf{A}$ is a vector that represents the local rotational behavior of the field. Its magnitude equals the circulation per unit area as the area becomes very small, and its direction is perpendicular to the surface where this circulation is greatest.

Stokes's theorem

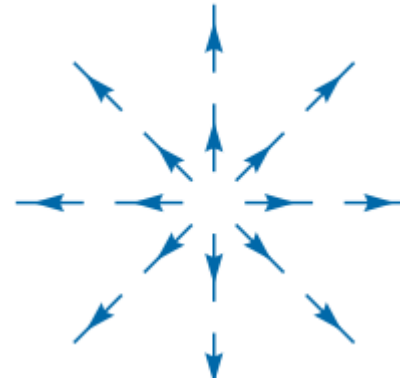
- ✓ **Stokes's theorem:** circulation of a vector field $\mathbf{A}$ around a closed path C is equal to the surface integral of the curl of $\mathbf{A}$ over any surface S bounded by C .
- The direction of C and the surface normal $d\mathbf{S}$ must satisfy the right-hand rule.

$$\oint_{Line} \mathbf{A} \cdot d\mathbf{l} = \int_{Surface} (\nabla \times \mathbf{A}) \cdot d\mathbf{S}$$

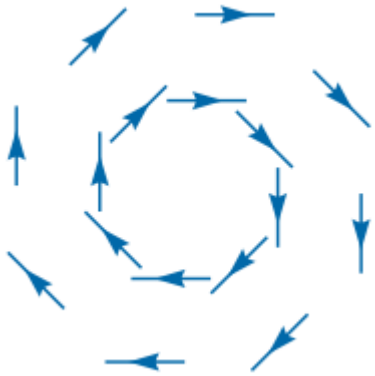
Classification of Vector Fields



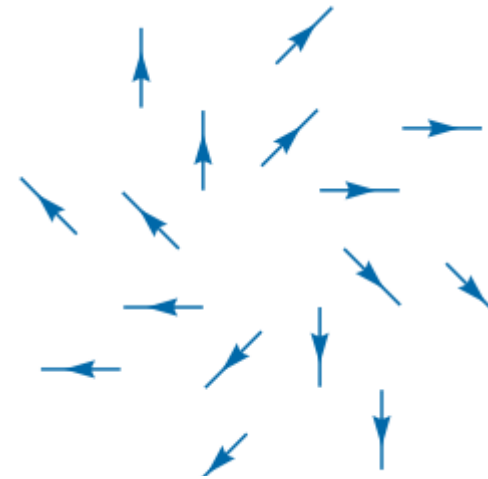
$$\nabla \cdot \mathbf{A} = 0, \nabla \times \mathbf{A} = \mathbf{0}$$



$$\nabla \cdot \mathbf{A} \neq 0, \nabla \times \mathbf{A} = \mathbf{0}$$



$$\nabla \cdot \mathbf{A} = 0, \nabla \times \mathbf{A} \neq \mathbf{0}$$



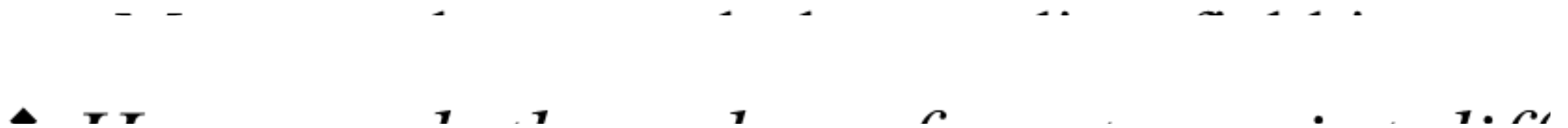
$$\nabla \cdot \mathbf{A} \neq 0, \nabla \times \mathbf{A} \neq \mathbf{0}$$

Laplacian

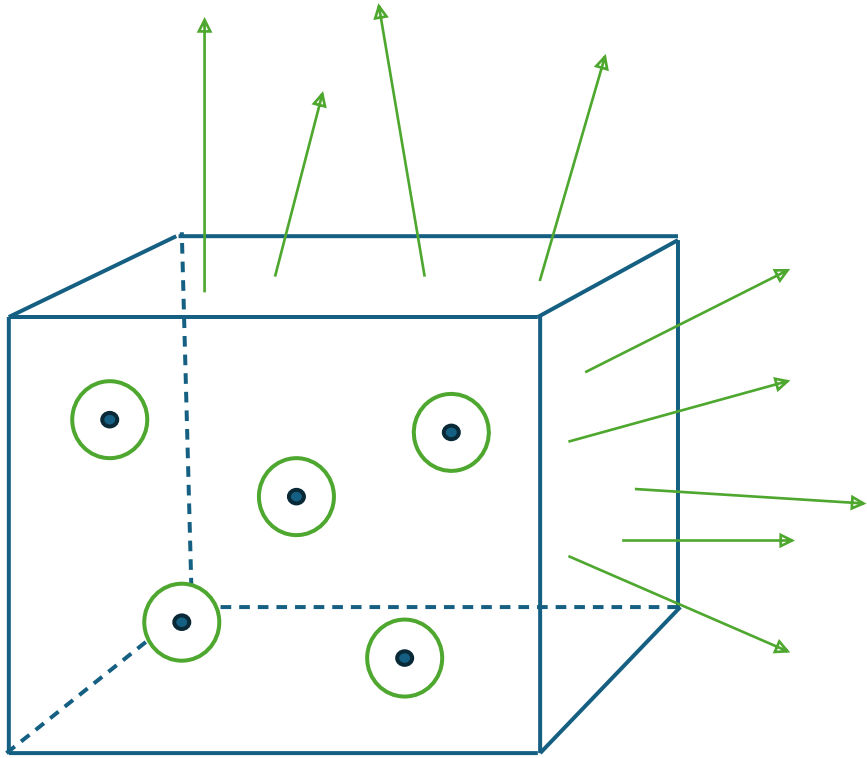
✓ Laplacian of a scalar field is the divergence of

Take gradient of a scalar field (φ) $\nabla\varphi$

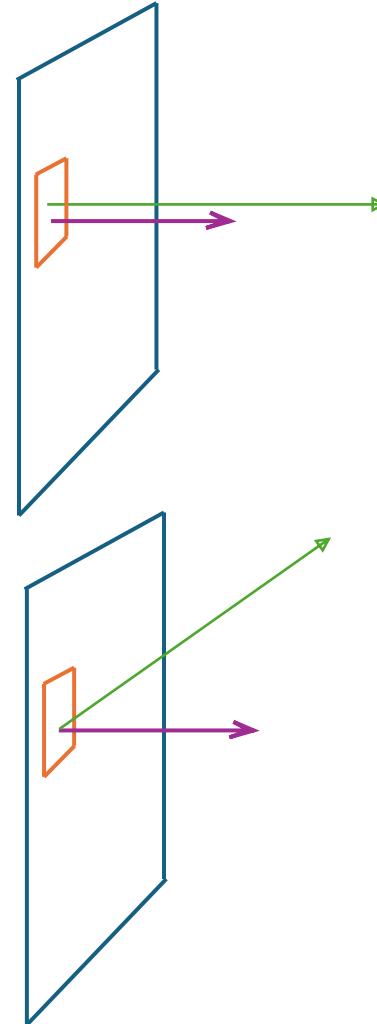
➤ What does the divergence of a vector field tell



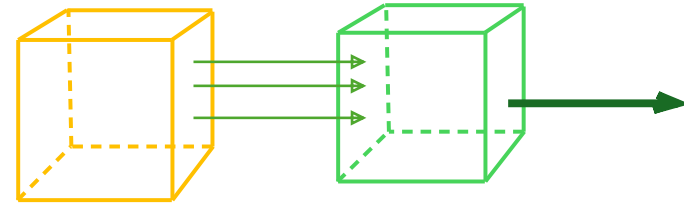
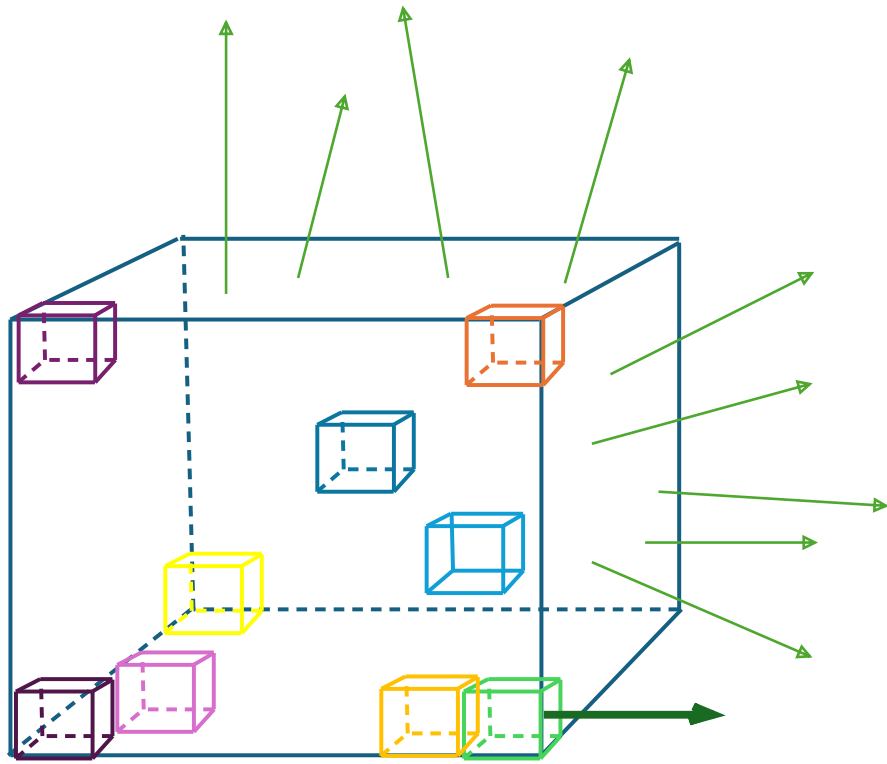
Divergence and Divergence Theorem



$$\oint_S \mathbf{A} \cdot d\mathbf{S}$$

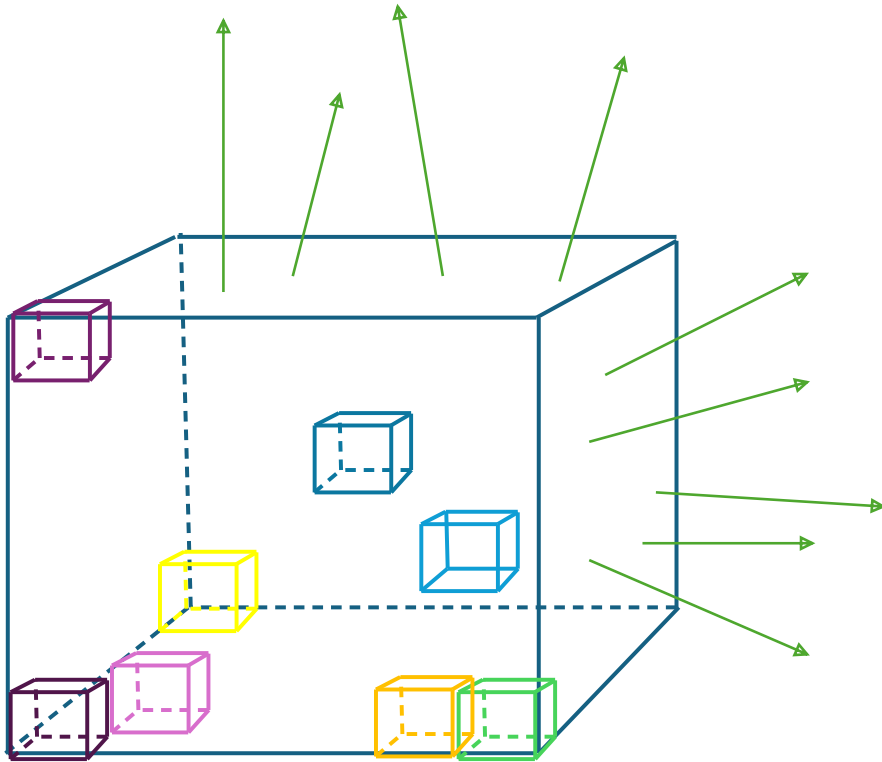


Divergence and Divergence Theorem

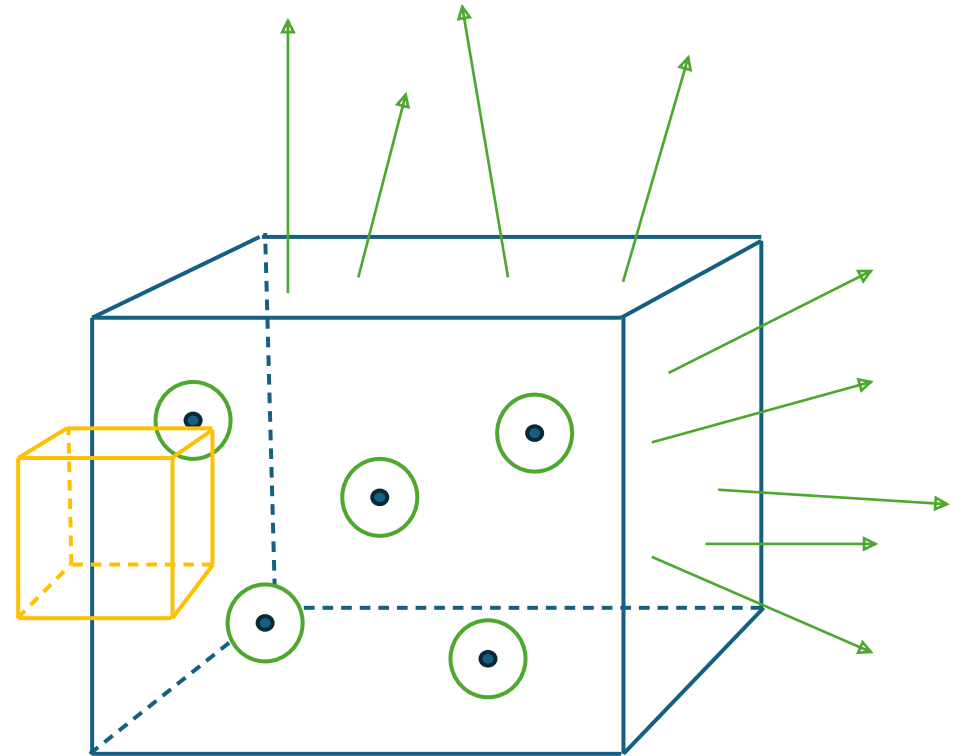


$$\int_V \nabla \cdot \mathbf{A} \, dv$$

Divergence and Divergence Theorem

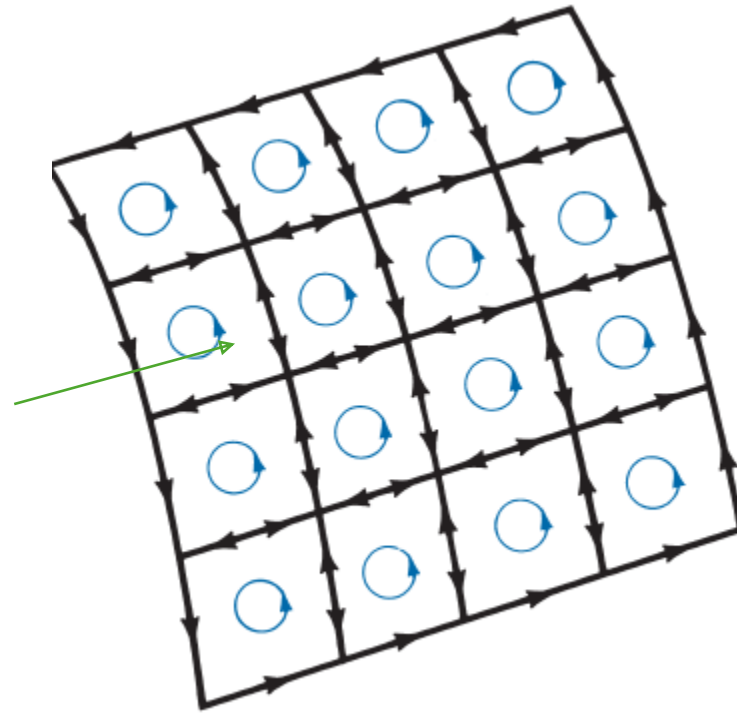


$$\int_V \nabla \cdot \mathbf{A} \, dv$$



$$\oint_S \mathbf{A} \cdot d\mathbf{S}$$

Stokes's Theorem



$$\oint_L \mathbf{A} \cdot d\mathbf{l} = \int_S (\nabla \times \mathbf{A}) \cdot d\mathbf{S}$$

Electrostatics

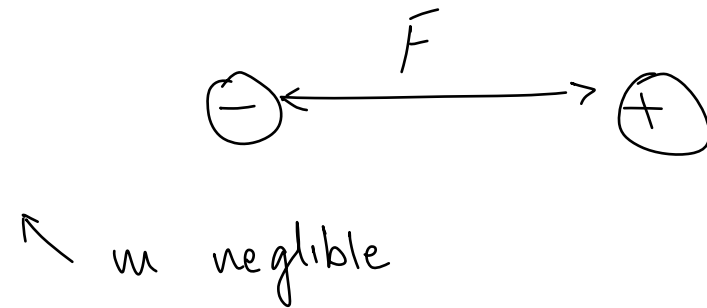
Today's plan

- Coulomb's Law and Field Intensity (4.2 – p. 112)
- Electric Fields Due to Continuous Charge Distribution (4.3 – p. 119)

Coulomb's Law

- There is a force between electrical charges
- This force is called the electric force
- When considering two point charges, Q_1 & Q_2 the force between them, F , has the following properties
 1. The force is along the line joining the charges
 2. The force is proportional to the product of the charges, $Q_1 Q_2$
 3. The force is inversely proportional to the square of the distance between them, R
- The mathematical definition of this force is,

$$F = k \frac{Q_1 Q_2}{R^2}$$



Coulomb's Law

➤ Details about the formula:

The charges can be like or unlike.
The unit of electrical charge is C

$$1\ C = \frac{1.6 \times 10^{+19}}{n_e} e^-$$

$$F = k \frac{Q_1 Q_2}{R^2}$$

The force between the charges, the
unit for force is N .

R is the distance between the
charges, and its unit is m .

$$\frac{1}{k} = \frac{Q_1 Q_2}{F R^2}$$

k is the proportionality constant, and its
unit is $[k] \neq \frac{C^2}{N \cdot m^2}$.

$$k = \frac{N \cdot m^2}{C^2}$$

Coulomb's Law

➤ Details about k :

ϵ_0 - free space permittivity ← lowest permittivity possible
 ϵ_d - any medium permittivity

$$k = \frac{1}{4\pi\epsilon_0}$$

Permittivity of Free Space and is a constant. Its value differs from one medium to another. The unit for this constant is F/m .

The value for k is,
 $9 \times 10^9 m/F$.

The value of the permittivity is,
 $\epsilon_0 = 8.854 \times 10^{-12} \cong \frac{10^{-9}}{36\pi} F/m$

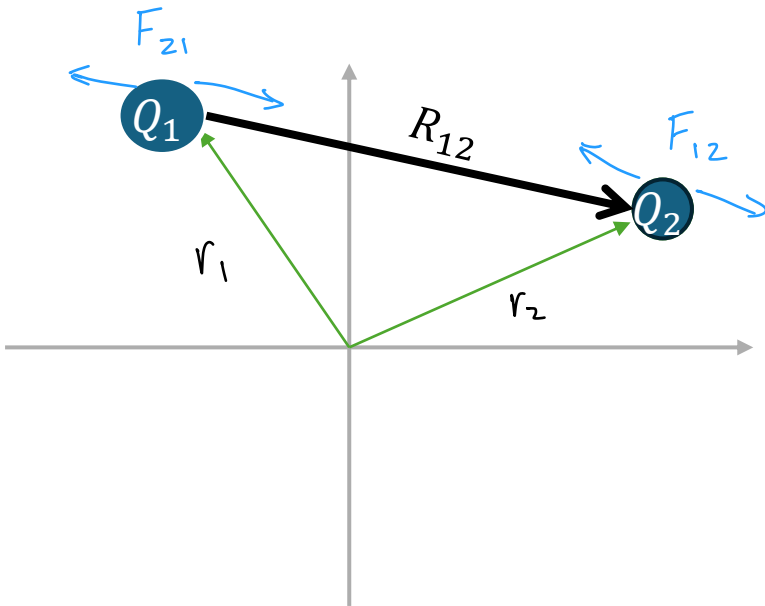
unit = energy stored in
defined space

Farads for capacitance

Coulomb's Law

➤ Electrical Force Equation as a Vector

$$\vec{F}_{21} = -\vec{F}_{12}$$

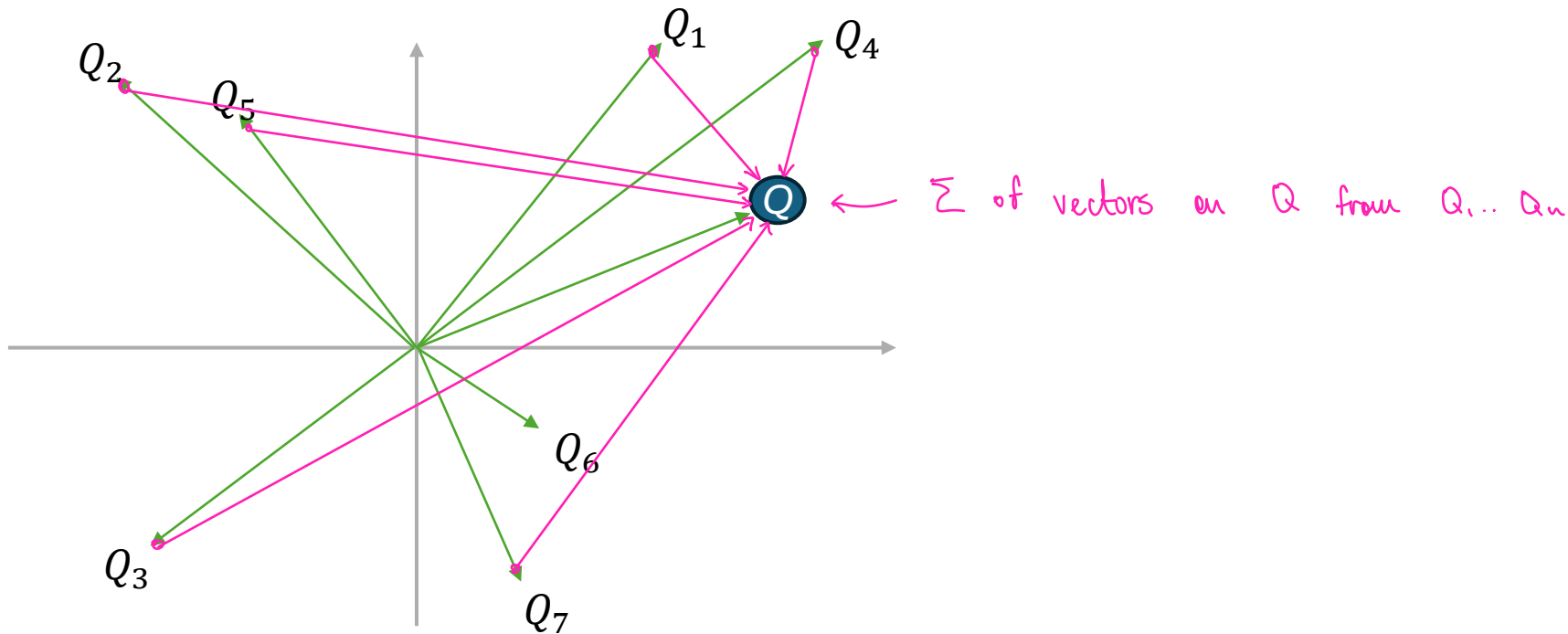


$$\vec{F}_{12} = \frac{Q_1 Q_2}{4\pi\epsilon_0 R^2} \vec{a}_{R_{12}}$$

$$\vec{a}_{R_{12}} = \frac{\vec{R}_{12}}{|\vec{R}_{12}|}$$

Coulomb's Law

- Electrical force due to more than one charge (superposition)



$$\mathbf{F} = \frac{Q}{4\pi\epsilon_0} \sum_{k=1}^N \frac{Q_k(\mathbf{r} - \mathbf{r}_k)}{|\mathbf{r} - \mathbf{r}_k|^3} \rightarrow F = \frac{Q_1 Q_2}{4\pi\epsilon_0 R^2} a_{R12} = \frac{Q_1 Q_2}{4\pi\epsilon_0 R^2} \cdot \frac{\vec{R}_{12}}{R} = \frac{Q_1}{4\pi\epsilon_0} \cdot Q_2$$

Coulomb's Law

➤ Why does the force have this mathematical form?

$$F = \frac{k Q_1 Q_2}{R^2} = \frac{1}{4\pi\epsilon_0} \cdot \frac{Q_1 Q_2}{R^2}$$

$V = \frac{4}{3}\pi R^3$
 $SA = V'$
 $= 4\pi R^2$

The diagram illustrates the derivation of the R^2 term in Coulomb's Law. It shows the formula for the force $F = \frac{k Q_1 Q_2}{R^2}$ and its equivalent form $F = \frac{1}{4\pi\epsilon_0} \cdot \frac{Q_1 Q_2}{R^2}$. Below these, the volume of a sphere is given as $V = \frac{4}{3}\pi R^3$, and its surface area is derived as $SA = V' = 4\pi R^2$. Two blue curved arrows connect the 4π in the denominator of the force equation to the 4π in the surface area equation, indicating that the surface area of a sphere is the geometric basis for the inverse-square relationship.

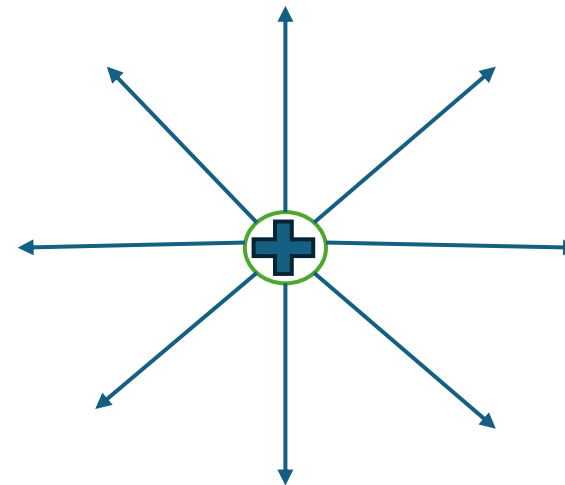
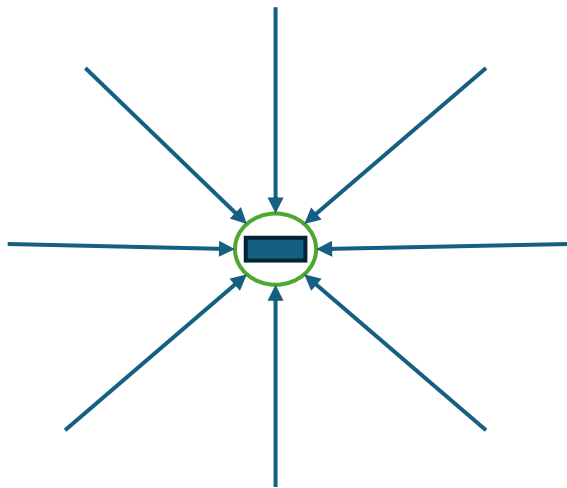
Coulomb's Law

➤ Electric field due to a charge

+ test charge → ideally want $Q = 0$ to find accurate F & Q
→ also don't want test charge to affect surrounding charges
→ should always be +ve

$$E = \lim_{Q \rightarrow 0} \frac{F}{Q}$$

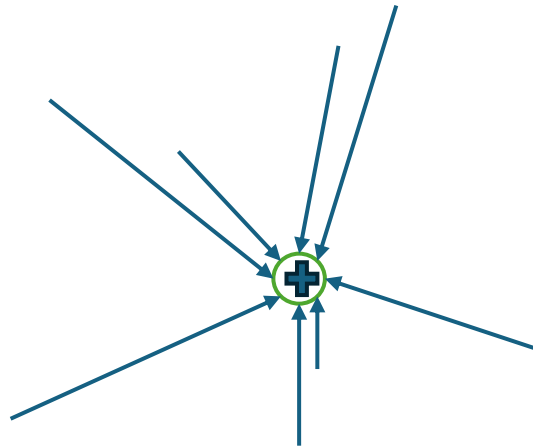
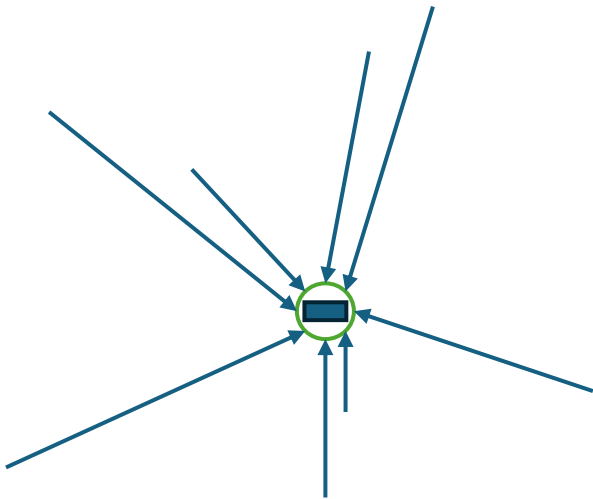
$$E = \frac{F}{Q}$$



Coulomb's Law

- Electric field due to multiple charges (superposition)

$$\mathbf{E} = \frac{1}{4\pi\epsilon_0} \sum_{k=1}^N \frac{Q_k(\mathbf{r} - \mathbf{r}_k)}{|\mathbf{r} - \mathbf{r}_k|^3}$$



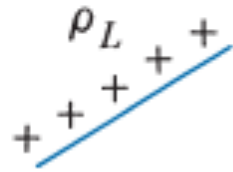
Coulomb's Law

➤ Electrical charge density = ρ ← general density formula



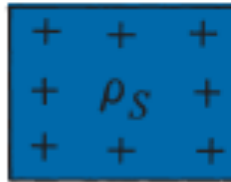
Point
charge

0D



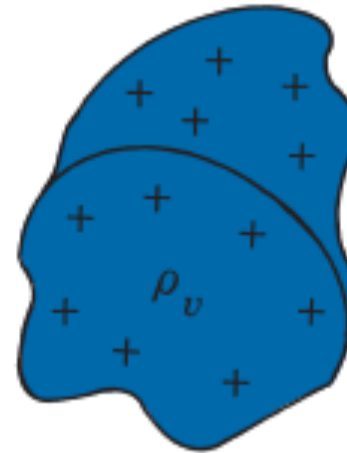
Line
charge

1D



Surface
charge

2D



Volume
charge

3D

Electrostatics

- Electric field of a single point charge and multiple point charges

Q

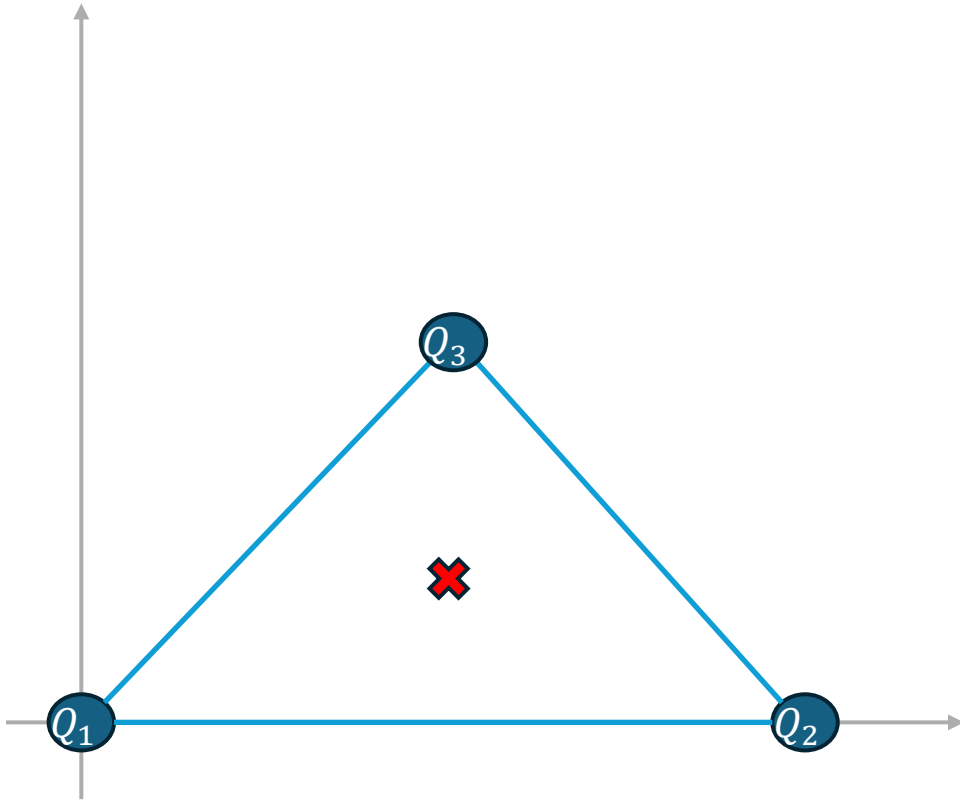
$$\mathbf{E} = \frac{Q}{4\pi\epsilon_0 R^2} \mathbf{a}_R = \frac{Q(\mathbf{r} - \mathbf{r}')}{4\pi\epsilon_0 |\mathbf{r} - \mathbf{r}'|^3} = \frac{Q}{4\pi\epsilon_0 r^2} \mathbf{a}_r$$

$Q_1 \quad Q_2 \quad \dots \quad Q_N$

$$\begin{aligned} \mathbf{E} &= \mathbf{E}_1 + \mathbf{E}_2 + \mathbf{E}_3 + \dots + \mathbf{E}_N \\ &= \frac{Q_1(\mathbf{r} - \mathbf{r}_1)}{4\pi\epsilon_0 |\mathbf{r} - \mathbf{r}_1|^3} + \frac{Q_2(\mathbf{r} - \mathbf{r}_2)}{4\pi\epsilon_0 |\mathbf{r} - \mathbf{r}_2|^3} + \dots + \frac{Q_N(\mathbf{r} - \mathbf{r}_N)}{4\pi\epsilon_0 |\mathbf{r} - \mathbf{r}_N|^3} \\ &= \frac{1}{4\pi\epsilon_0} \sum_{k=1}^N \frac{Q_k(\mathbf{r} - \mathbf{r}_k)}{|\mathbf{r} - \mathbf{r}_k|^3} \end{aligned}$$

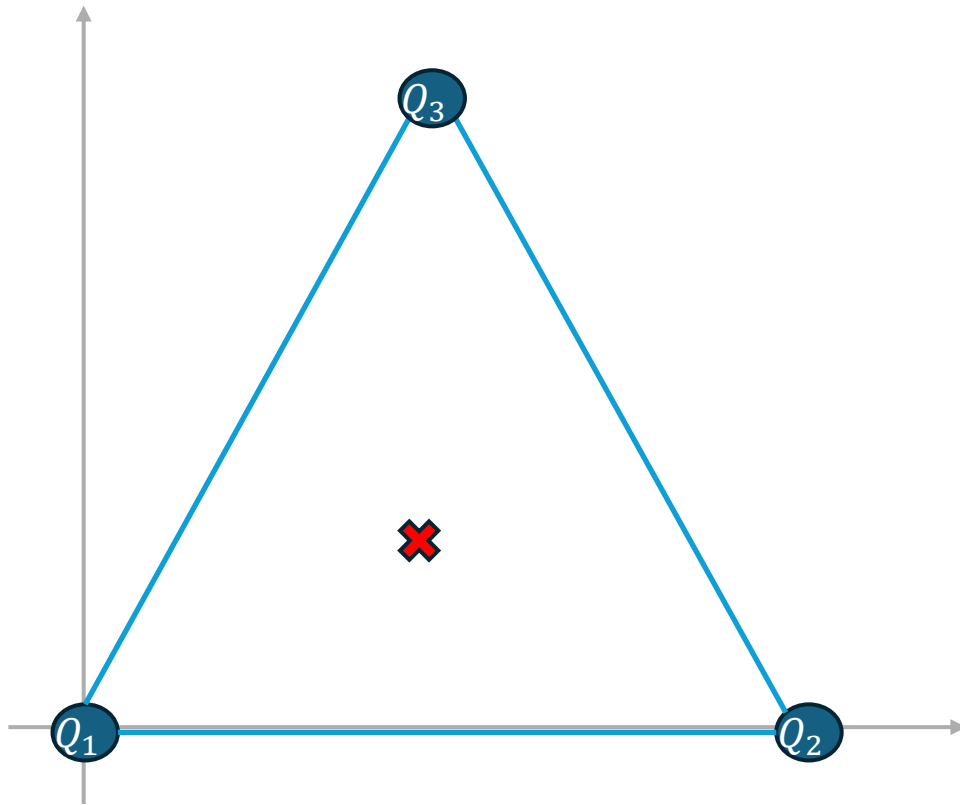
Electrostatics

- Electric field due to a few point charges



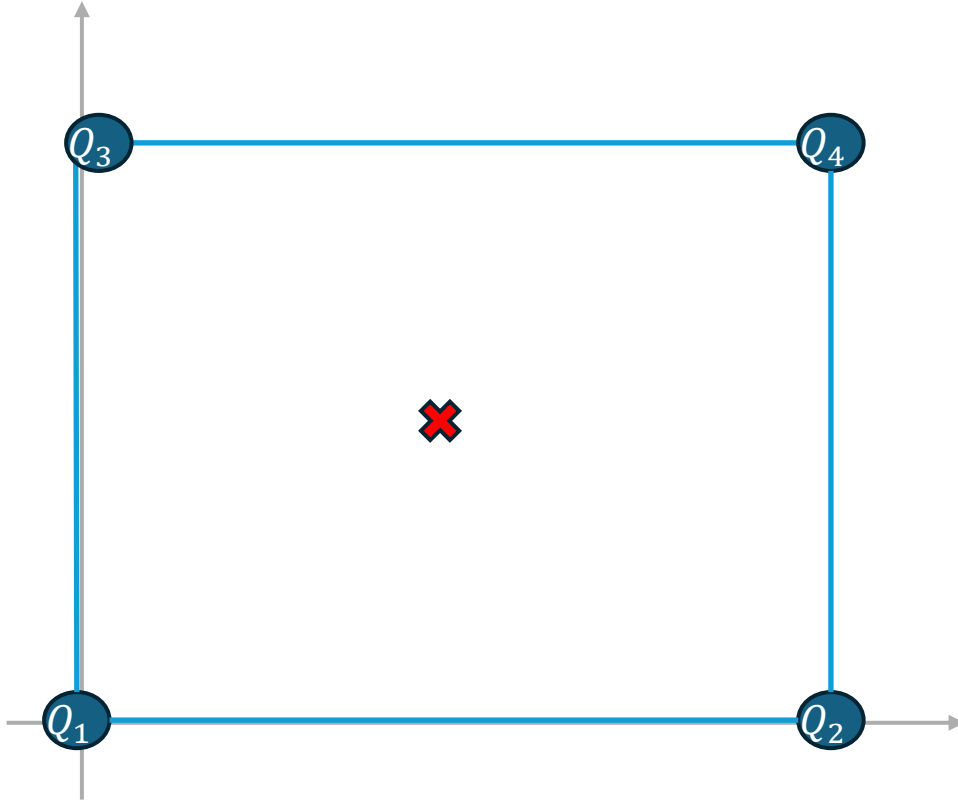
Electrostatics

- Electric field due to a few point charges



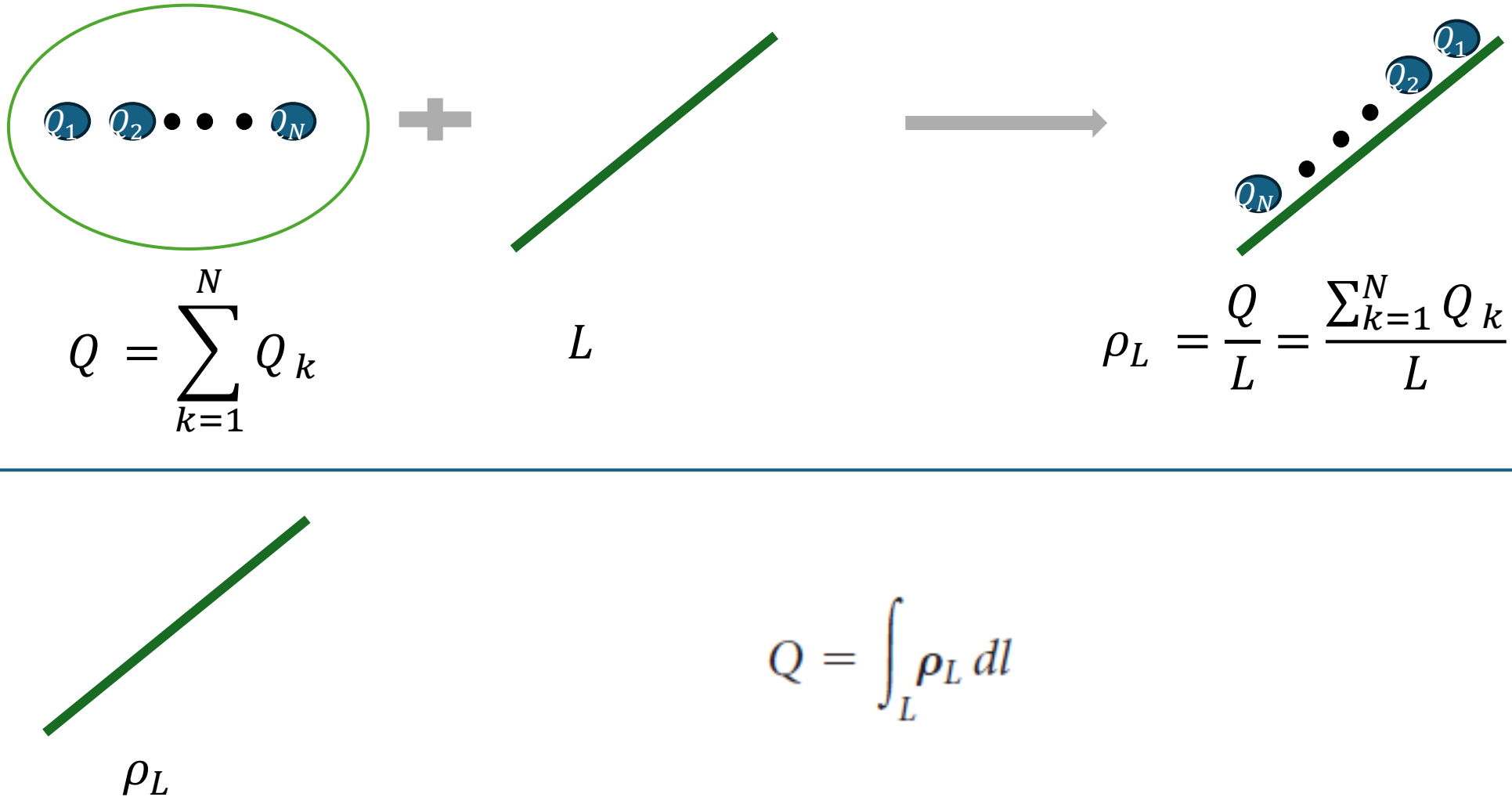
Electrostatics

- Electric field due to a few point charges



Electrostatics

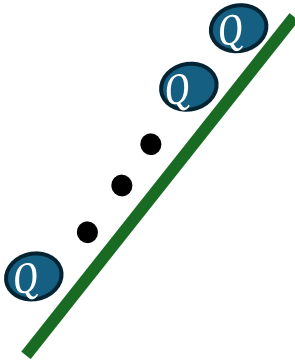
- The electrical field due to a line distribution of charge



Electrostatics

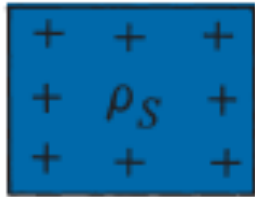
- The electrical field due to a line distribution of charge

$$\mathbf{E} = \frac{Q}{4\pi\epsilon_0 R^2} \mathbf{a}_R \quad \longrightarrow \quad \mathbf{E} = \int_L \frac{\rho_L dl}{4\pi\epsilon_0 R^2} \mathbf{a}_R$$



Electrostatics

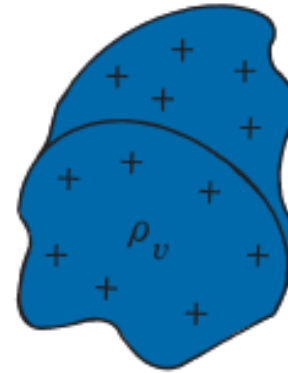
- The electrical field due to a surface and a volume distribution of charge



Surface
charge

$$dQ = \rho_S dS \rightarrow Q = \int_S \rho_S dS$$

$$\mathbf{E} = \int_S \frac{\rho_S dS}{4\pi\epsilon_0 R^2} \mathbf{a}_R$$



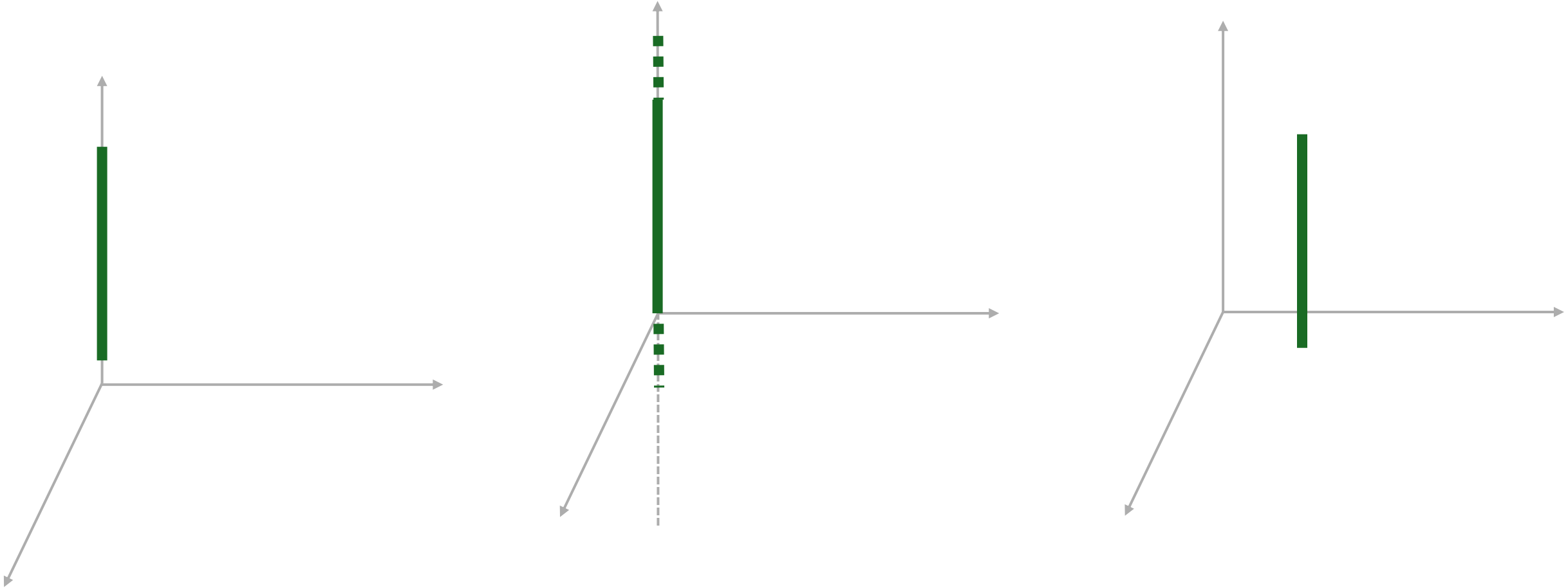
Volume
charge

$$dQ = \rho_v dv \rightarrow Q = \int_v \rho_v dv$$

$$\mathbf{E} = \int_v \frac{\rho_v dv}{4\pi\epsilon_0 R^2} \mathbf{a}_R$$

Electrostatics

➤ A line charge



Electrostatics

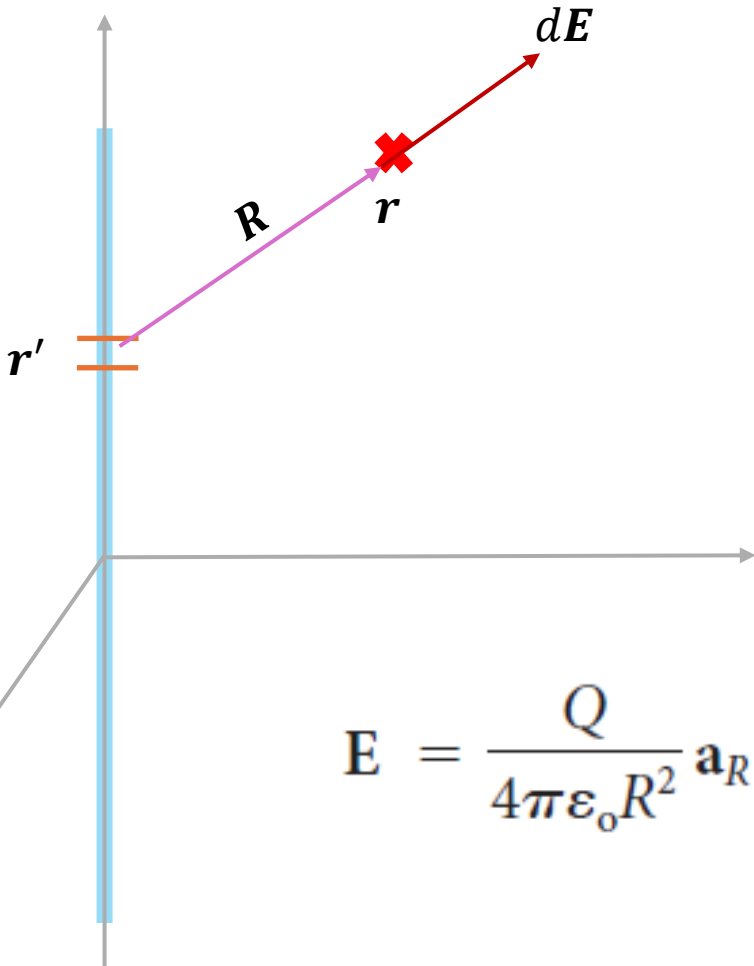
- A line charge, symmetric about xy-plane

$$\mathbf{r} = x\vec{a}_x + y\vec{a}_y + z\vec{a}_z$$

$$\mathbf{r}' = 0\vec{a}_x + 0\vec{a}_y + z'\vec{a}_z$$

$$\mathbf{R} = \mathbf{r} - \mathbf{r}' = x\vec{a}_x + y\vec{a}_y + (z - z')\vec{a}_z$$

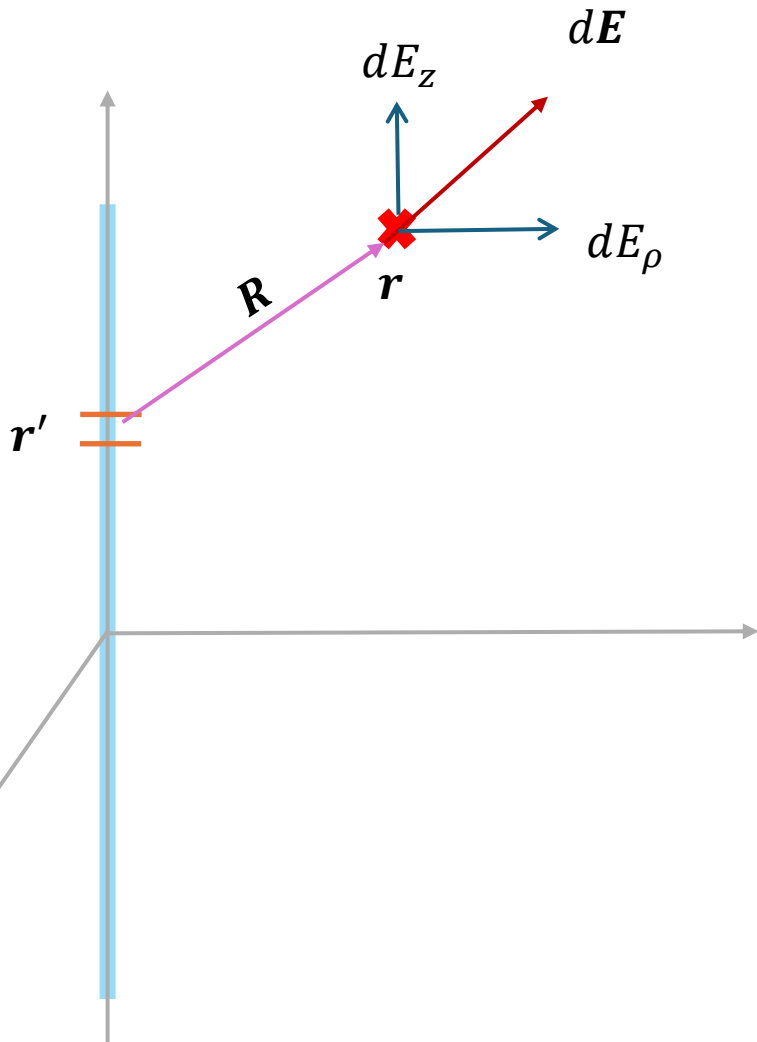
$$R = \sqrt{\rho^2 + z'^2}$$



$$\mathbf{E} = \frac{Q}{4\pi\epsilon_0 R^2} \mathbf{a}_R \xrightarrow{\lambda = \rho_L} d\mathbf{E} = \frac{1}{4\pi\epsilon_0} \frac{\lambda dz'}{(\rho^2 + (z - z')^2)^{3/2}} (\rho\vec{a}_\rho + (z - z')\vec{a}_z)$$

Electrostatics

- A line charge, symmetric about xy-plane

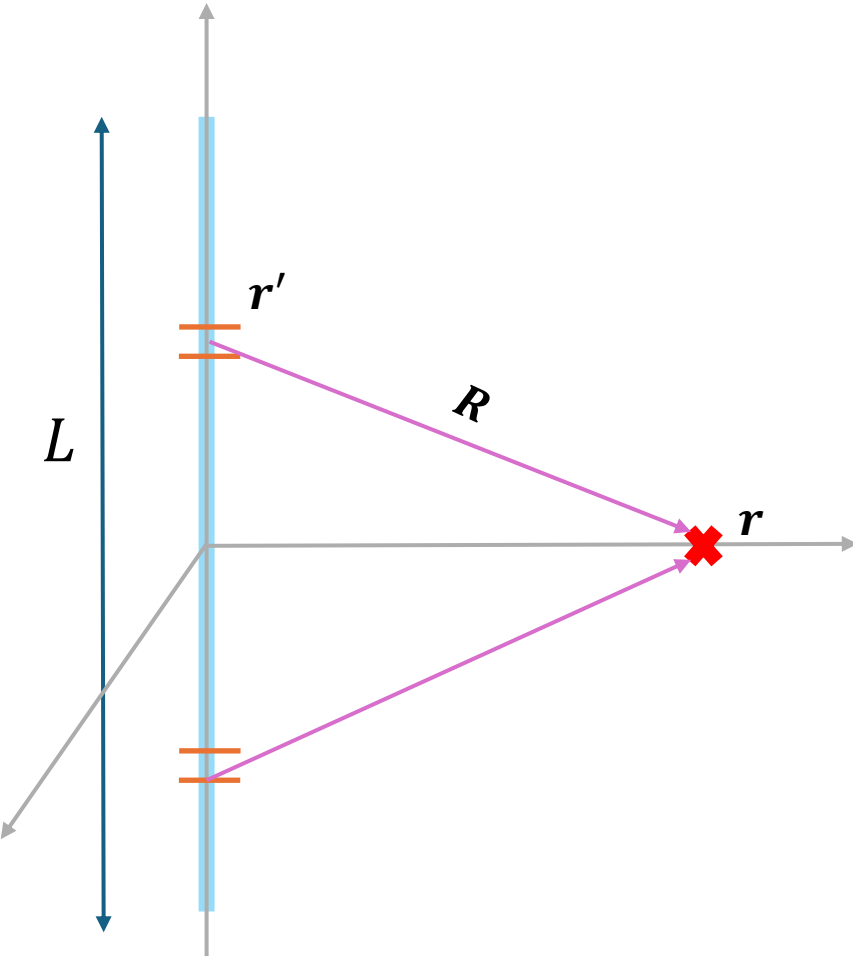


$$d\mathbf{E} = \frac{1}{4\pi\epsilon_0} \frac{\lambda dz'}{(\rho^2 + (z - z')^2)^{3/2}} (\rho \vec{a}_\rho + (z - z') \vec{a}_z)$$

$$\begin{cases} dE_\rho = \frac{\lambda \rho}{4\pi\epsilon_0} \frac{dz'}{(\rho^2 + (z - z')^2)^{3/2}} \\ dE_z = \frac{\lambda}{4\pi\epsilon_0} \frac{(z - z') dz'}{(\rho^2 + (z - z')^2)^{3/2}} \end{cases}$$

Electrostatics

- Observation on xy-plane, symmetric finite line (limited)

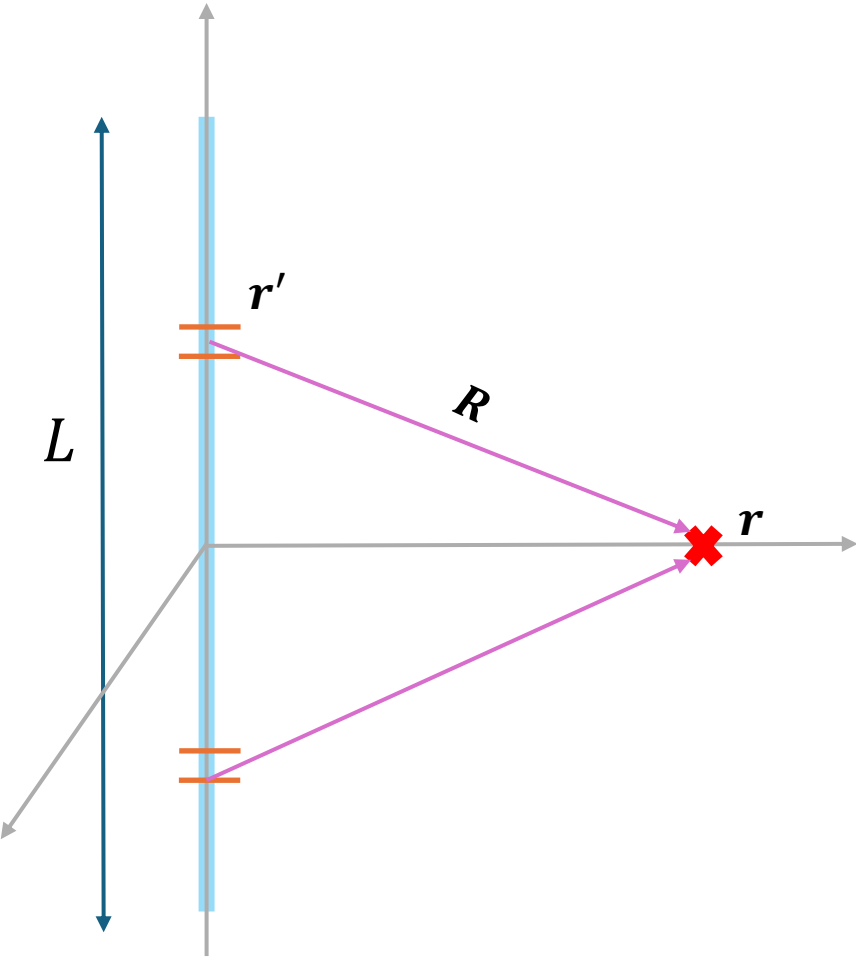


$$\begin{cases} dE_{\rho} = \frac{\lambda \rho}{4\pi\epsilon_0} \frac{dz'}{(\rho^2 + (z - z')^2)^{3/2}} \\ dE_z = \frac{\lambda}{4\pi\epsilon_0} \frac{(z - z')dz'}{(\rho^2 + (z - z')^2)^{3/2}} \end{cases}$$

$$dE_{\rho} = \frac{1}{4\pi\epsilon_0} \frac{\lambda \rho dz'}{(\rho^2 + z'^2)^{\frac{3}{2}}}$$

Electrostatics

- Observation on xy-plane, symmetric finite line (limited)



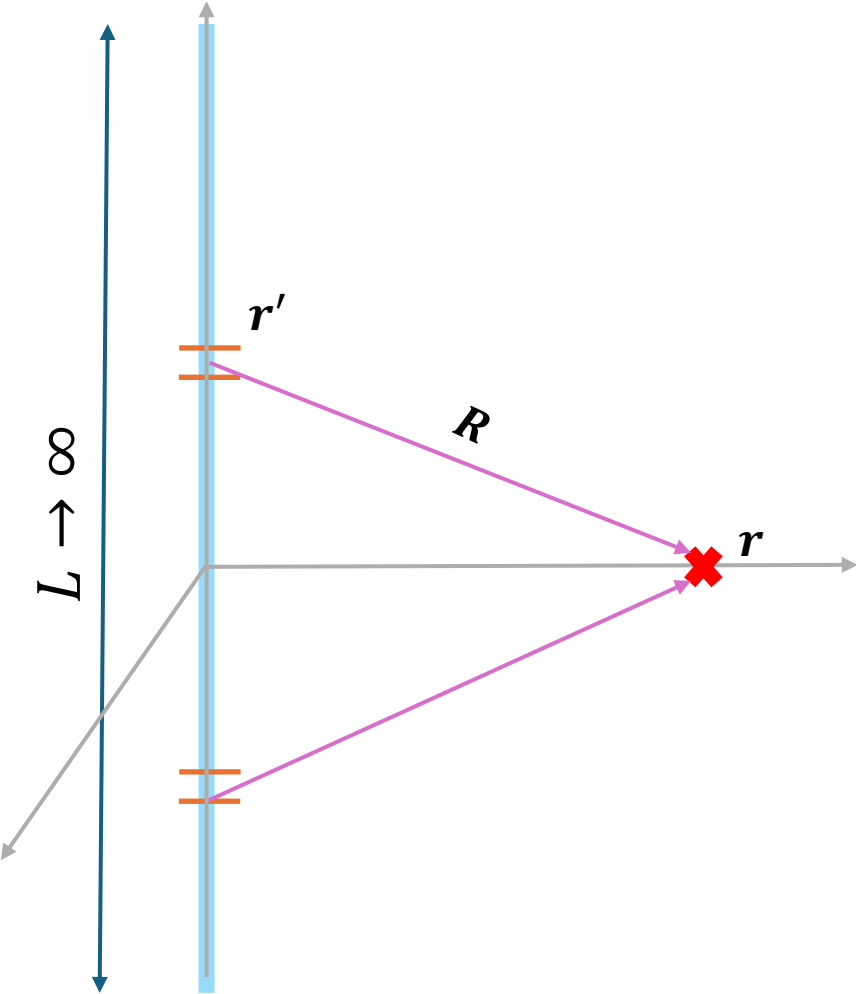
$$dE_{\rho} = \frac{\lambda \rho}{4\pi\epsilon_0} \int_{-L/2}^{L/2} \frac{dz'}{(\rho^2 + z'^2)^{3/2}}$$

$$\int_{-L/2}^{L/2} \frac{dz'}{(\rho^2 + z'^2)^{3/2}} = \frac{z'}{\rho^2 \sqrt{\rho^2 + z'^2}}$$

$$E_{\rho} = \frac{\lambda \rho}{4\pi\epsilon_0} \left[\frac{z'}{\rho^2 \sqrt{\rho^2 + z'^2}} \right]_{-L/2}^{L/2} = \frac{\lambda}{4\pi\epsilon_0 \rho} \frac{L}{\sqrt{L^2 + 4\rho^2}}$$

Electrostatics

- Observation on xy-plane, symmetric infinite line

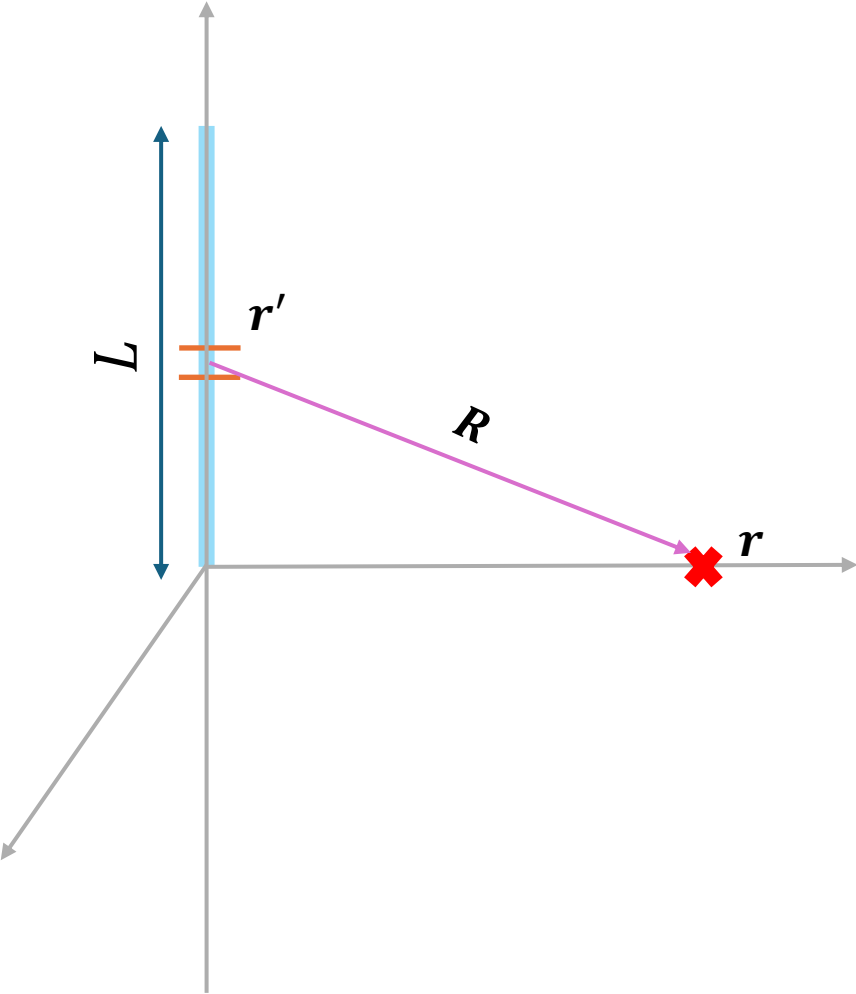


$$E_{\rho} = \frac{\lambda}{4\pi\epsilon_0\rho} \frac{L}{\sqrt{L^2 + 4\rho^2}}$$

$$E_{\rho} = \frac{\lambda}{4\pi\epsilon_0\rho} \lim_{L \rightarrow \infty} \frac{L}{\sqrt{L^2 + 4\rho^2}} = \frac{\lambda}{2\pi\epsilon_0\rho}$$

Electrostatics

- Observation on xy-plane, asymmetric finite line

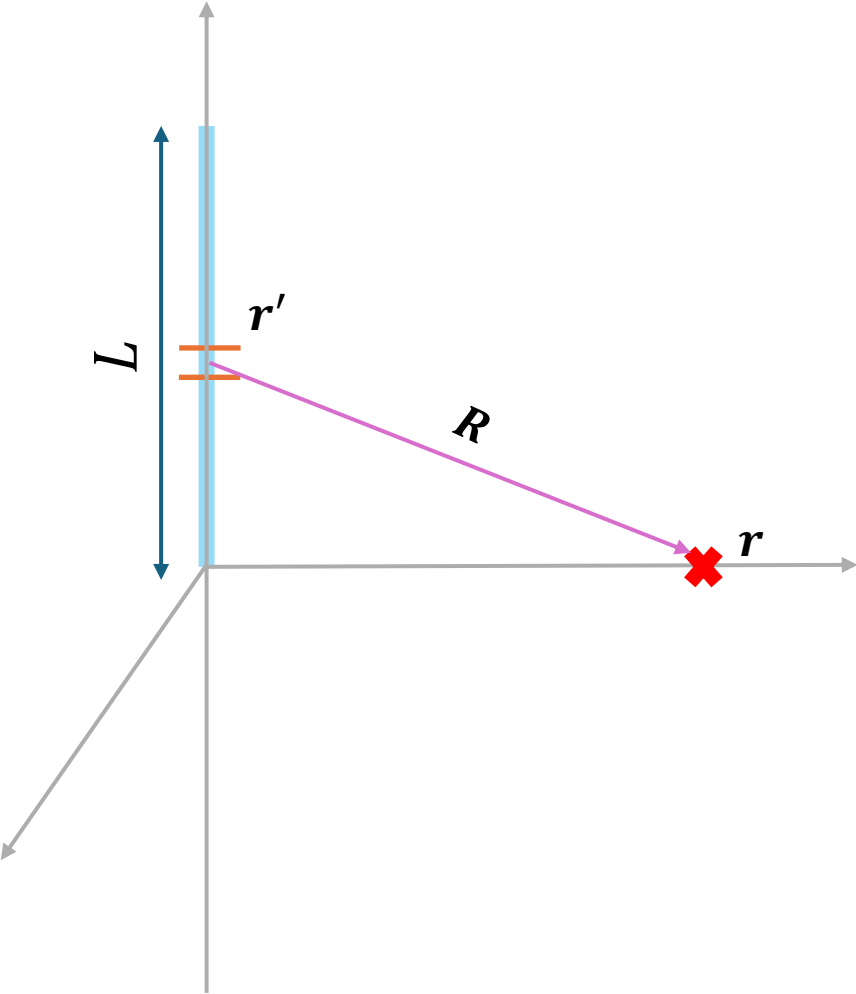


$$dE_{\rho} = \frac{\lambda \rho}{4\pi\epsilon_0} \frac{dz'}{(\rho^2 + (z - z')^2)^{3/2}}$$

$$dE_z = \frac{\lambda}{4\pi\epsilon_0} \frac{-z' dz'}{(\rho^2 + (z - z')^2)^{3/2}}$$

Electrostatics

- Observation on xy-plane, asymmetric finite line

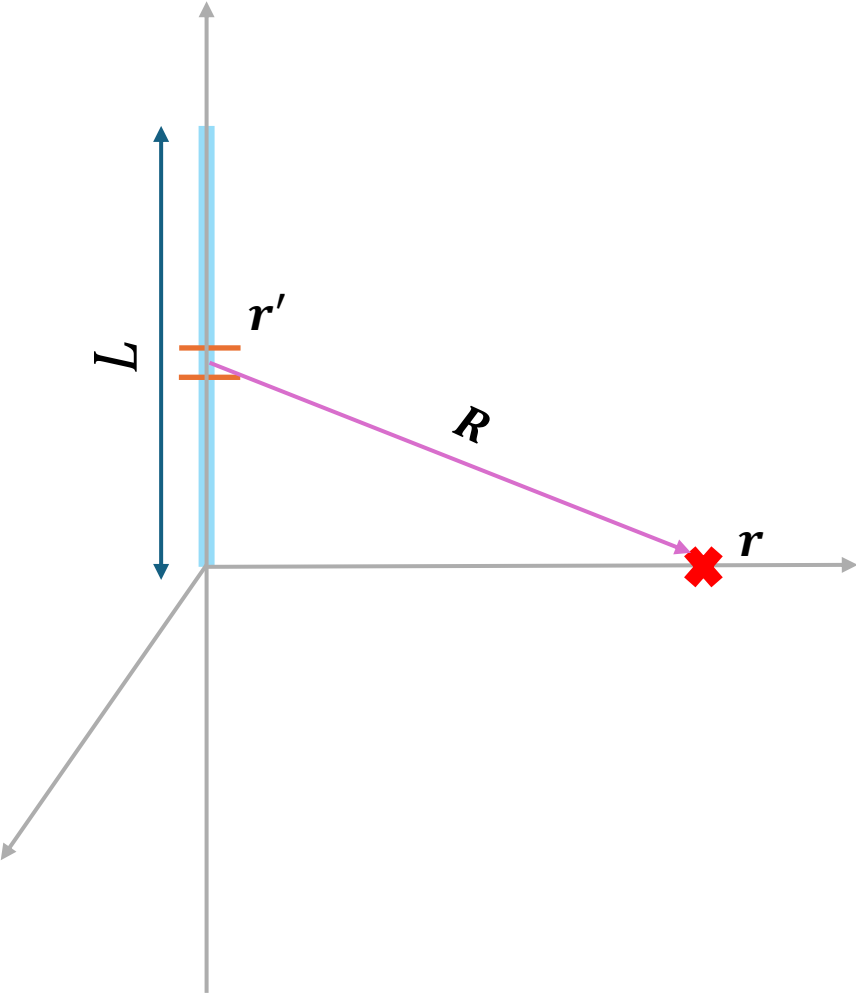


$$E_{\rho} = \frac{\lambda \rho}{4\pi\epsilon_0} \int_0^L \frac{dz'}{(\rho^2 + (z - z')^2)^{3/2}}$$

$$E_z = \frac{\lambda}{4\pi\epsilon_0} \int_0^L \frac{-z' dz'}{(\rho^2 + (z - z')^2)^{3/2}}$$

Electrostatics

- Observation on xy-plane, asymmetric finite line

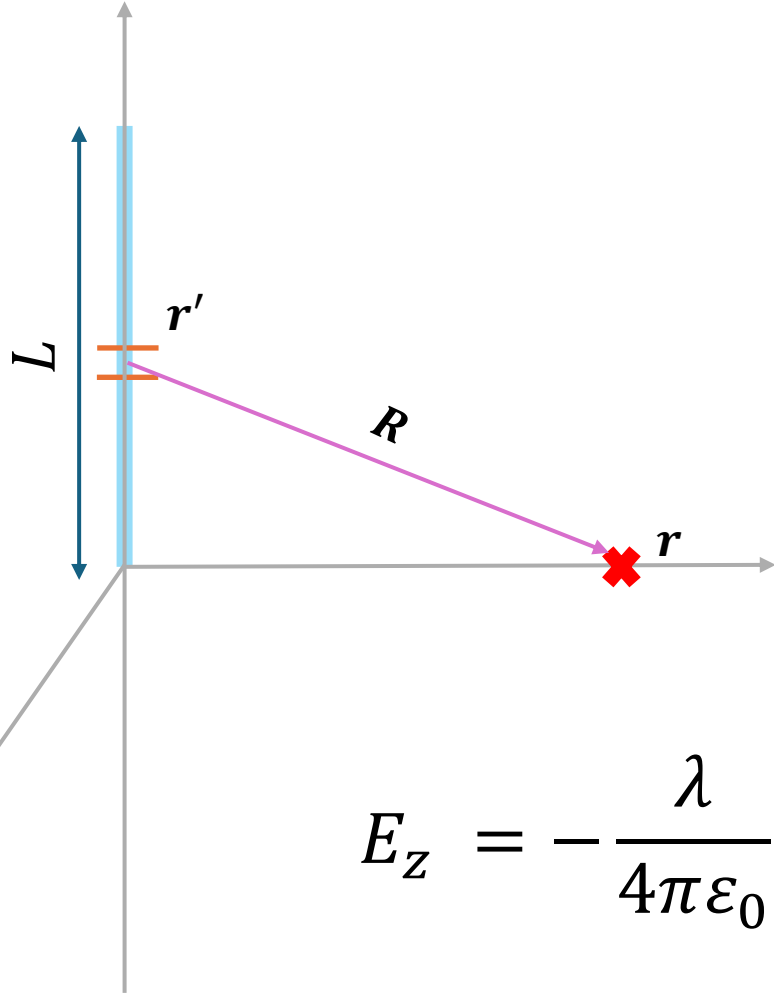


$$E_{\rho} = \frac{\lambda \rho}{4\pi\epsilon_0} \int_0^L \frac{dz'}{(\rho^2 + (z - z')^2)^{3/2}}$$

$$E_{\rho} = \frac{\lambda \rho}{4\pi\epsilon_0} \left[\frac{z'}{\rho^2 \sqrt{\rho^2 + z'^2}} \right]_0^L = \frac{\lambda}{4\pi\epsilon_0 \rho} \frac{L}{\sqrt{\rho^2 + L^2}}$$

Electrostatics

- Observation on xy-plane, asymmetric finite line



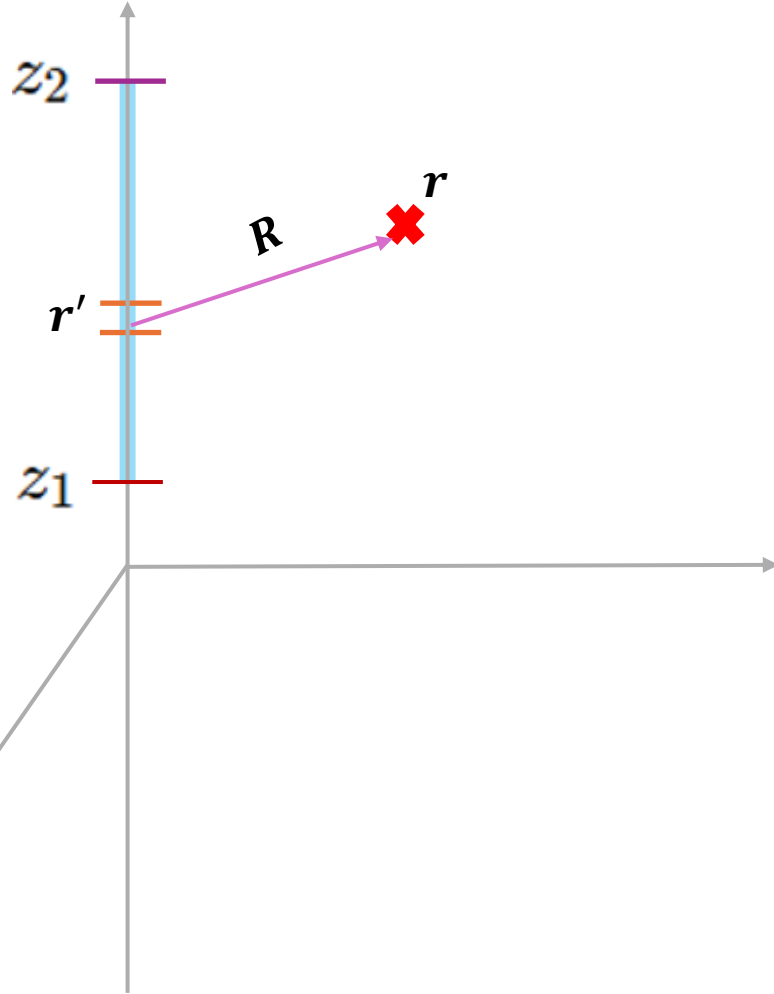
$$E_z = \frac{\lambda}{4\pi\epsilon_0} \int_0^L \frac{-z' dz'}{(\rho^2 + (z - z')^2)^{3/2}}$$

$$\int \frac{z' dz'}{(\rho^2 + (z - z')^2)^{3/2}} = \frac{-1}{\sqrt{\rho^2 + z'^2}}$$

$$E_z = -\frac{\lambda}{4\pi\epsilon_0} \left[\frac{-1}{\sqrt{\rho^2 + z'^2}} \right]_0^L = \frac{\lambda}{4\pi\epsilon_0} \left(\frac{1}{\sqrt{\rho^2 + L^2}} - \frac{1}{\rho} \right)$$

Electrostatics

- Random observation point (general 3D), finite line



$$E_\rho = \frac{\lambda}{4\pi\epsilon_0} \int_{z_1}^{z_2} \frac{\rho}{(\rho^2 + (z - z')^2)^{3/2}} dz'$$

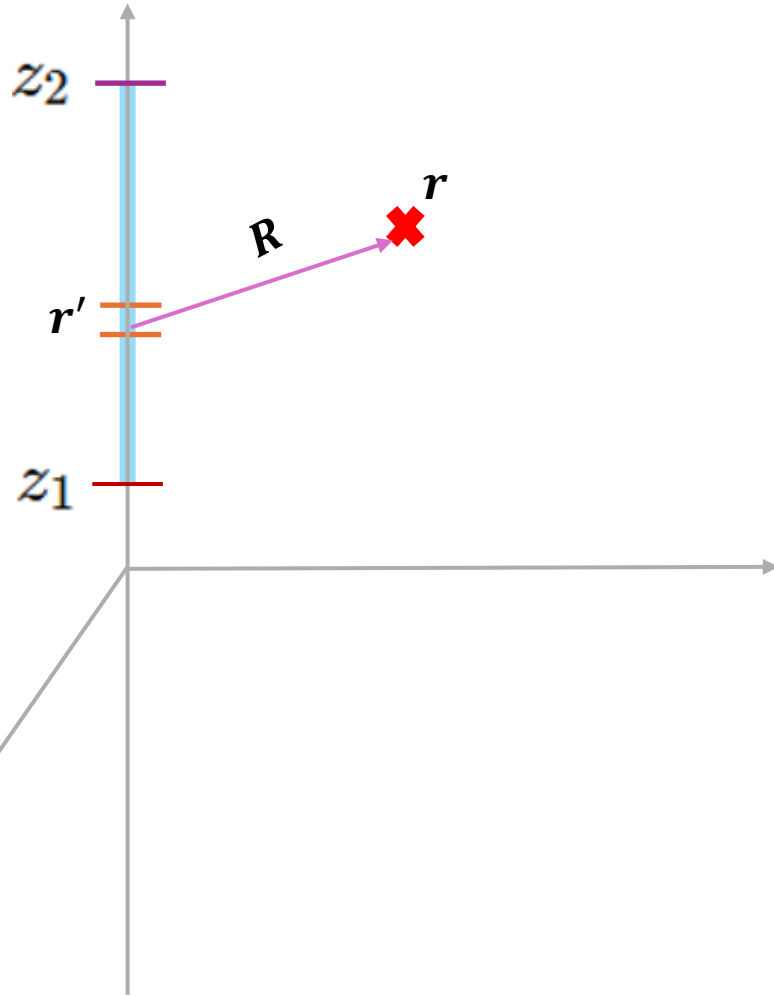
$$E_z = \frac{\lambda}{4\pi\epsilon_0} \int_{z_1}^{z_2} \frac{(z - z')}{(\rho^2 + (z - z')^2)^{3/2}} dz'$$

$$u = z - z' \Rightarrow dz' = -du.$$

$$u_1 = z - z_1, \quad u_2 = z - z_2$$

Electrostatics

- Random observation point (general 3D), finite line



$$E_\rho = \frac{\lambda}{4\pi\epsilon_0} \rho \int_{u_2}^{u_1} \frac{du}{(\rho^2 + u^2)^{3/2}}$$

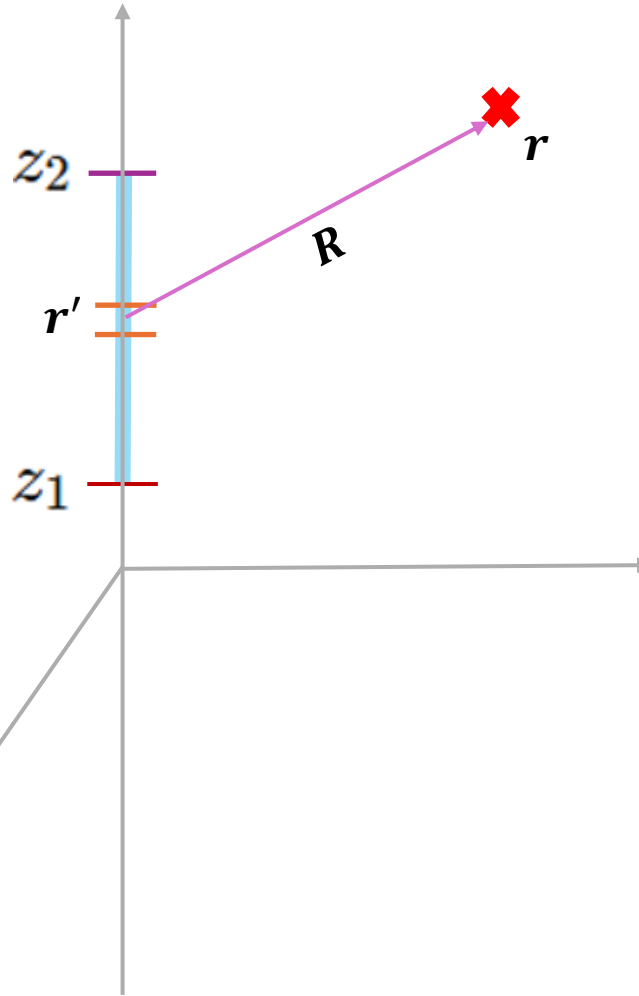
$$\int \frac{du}{(\rho^2 + u^2)^{3/2}} = \frac{u}{\rho^2 \sqrt{\rho^2 + u^2}}$$

$$E_\rho = \frac{\lambda}{4\pi\epsilon_0\rho} \left(\frac{u_1}{\sqrt{\rho^2 + u_1^2}} - \frac{u_2}{\sqrt{\rho^2 + u_2^2}} \right)$$

$$E_\rho = \frac{\lambda}{4\pi\epsilon_0\rho} \left(\frac{z - z_1}{\sqrt{\rho^2 + (z - z_1)^2}} - \frac{z - z_2}{\sqrt{\rho^2 + (z - z_2)^2}} \right)$$

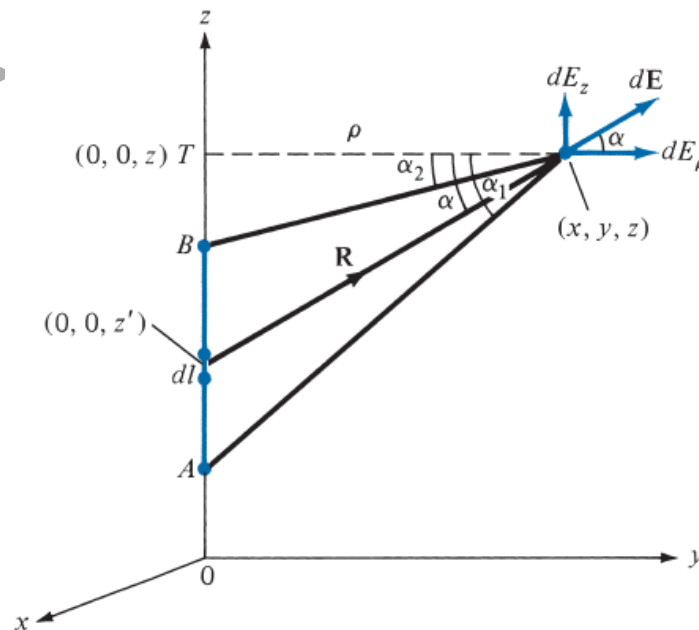
Electrostatics

- Random observation point (general 3D), finite line



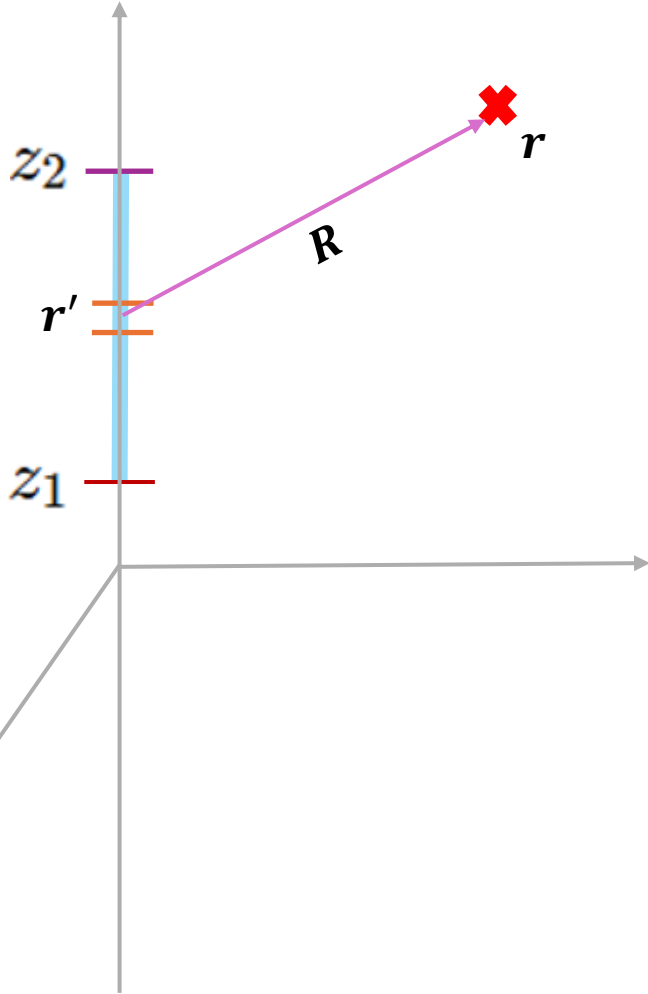
$$E_\rho = \frac{\lambda}{4\pi\epsilon_0\rho} \left(\frac{z - z_1}{\sqrt{\rho^2 + (z - z_1)^2}} - \frac{z - z_2}{\sqrt{\rho^2 + (z - z_2)^2}} \right)$$

$$\mathbf{E} = \frac{\rho L}{4\pi\epsilon_0\rho} [-(\sin \alpha_2 - \sin \alpha_1)\mathbf{a}_\rho + (\cos \alpha_2 - \cos \alpha_1)\mathbf{a}_z]$$



Electrostatics

- Random observation point (general 3D), finite line



$$E_z = \frac{\lambda}{4\pi\epsilon_0} \int_{z_1}^{z_2} \frac{(z - z')}{(\rho^2 + (z - z')^2)^{3/2}} dz'$$

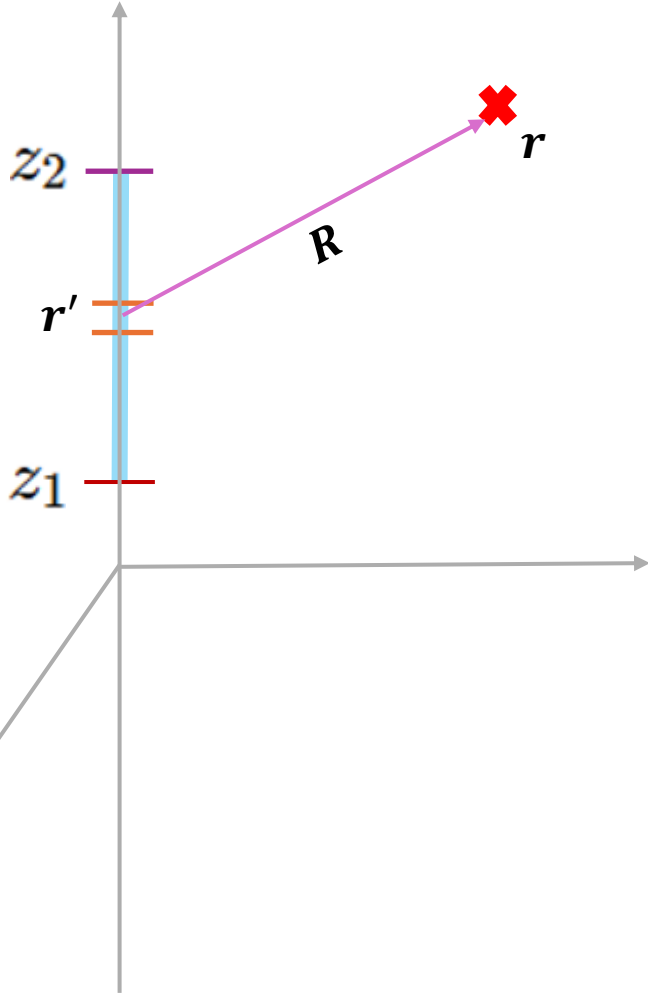
$$E_z = \frac{\lambda}{4\pi\epsilon_0} \int_{u_2}^{u_1} \frac{u}{(\rho^2 + u^2)^{3/2}} du$$

$$\int \frac{u du}{(\rho^2 + u^2)^{3/2}} = -\frac{1}{\sqrt{\rho^2 + u^2}}$$

$$E_z = \frac{\lambda}{4\pi\epsilon_0} \left(\frac{1}{\sqrt{\rho^2 + u_2^2}} - \frac{1}{\sqrt{\rho^2 + u_1^2}} \right)$$

Electrostatics

- Random observation point (general 3D), finite line

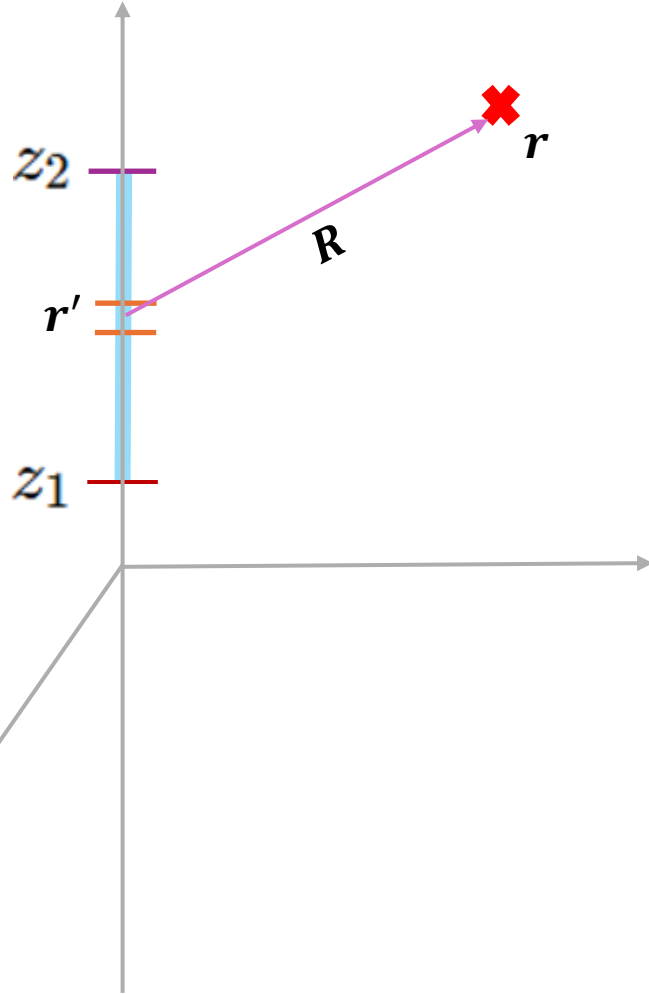


$$E_z = \frac{\lambda}{4\pi\epsilon_0} \left(\frac{1}{\sqrt{\rho^2 + u_2^2}} - \frac{1}{\sqrt{\rho^2 + u_1^2}} \right)$$

$$E_z = \frac{\lambda}{4\pi\epsilon_0} \left(\frac{1}{\sqrt{\rho^2 + (z - z_2)^2}} - \frac{1}{\sqrt{\rho^2 + (z - z_1)^2}} \right)$$

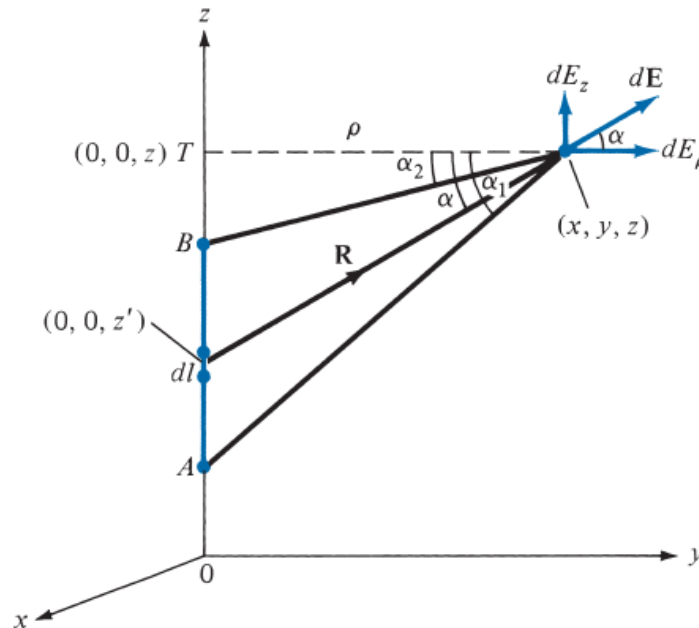
Electrostatics

- Random observation point (general 3D), finite line



$$E_z = \frac{\lambda}{4\pi\epsilon_0} \left(\frac{1}{\sqrt{\rho^2 + (z - z_2)^2}} - \frac{1}{\sqrt{\rho^2 + (z - z_1)^2}} \right)$$

$$\mathbf{E} = \frac{\rho L}{4\pi\epsilon_0\rho} [-(\sin \alpha_2 - \sin \alpha_1)\mathbf{a}_\rho + (\cos \alpha_2 - \cos \alpha_1)\mathbf{a}_z]$$



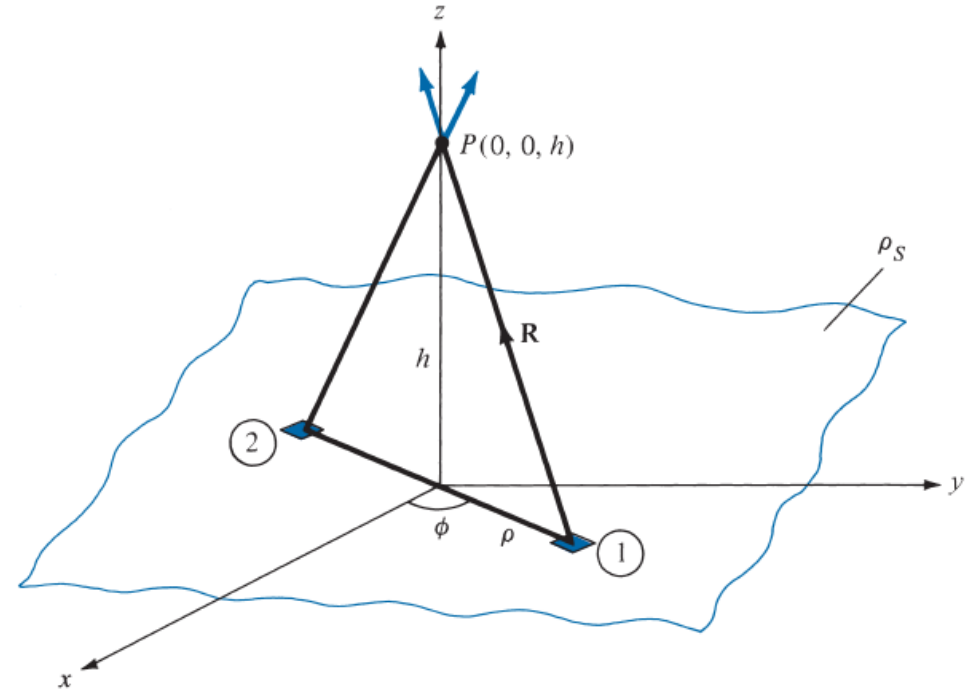
Electrostatics

- Infinite sheet of charge in the xy -plane

$$d\mathbf{E} = \frac{1}{4\pi\epsilon_0} \frac{dQ}{R^2} \mathbf{a}_R$$

$$dQ = \sigma dS$$

$$dS = \rho d\rho d\phi$$



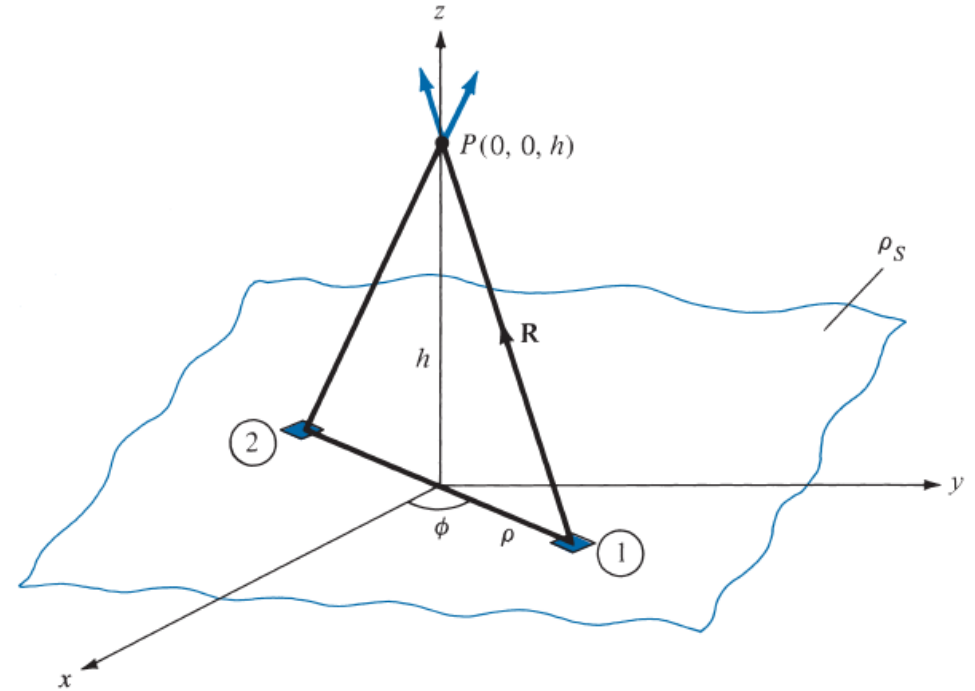
Electrostatics

- Infinite sheet of charge in the xy -plane

$$\mathbf{r}' = \rho \mathbf{a}_\rho$$

$$\mathbf{r} = h \mathbf{a}_z$$

$$\mathbf{R} = \mathbf{r} - \mathbf{r}' = -\rho \mathbf{a}_\rho + h \mathbf{a}_z \longrightarrow R = |\mathbf{R}| = \sqrt{\rho^2 + h^2}$$

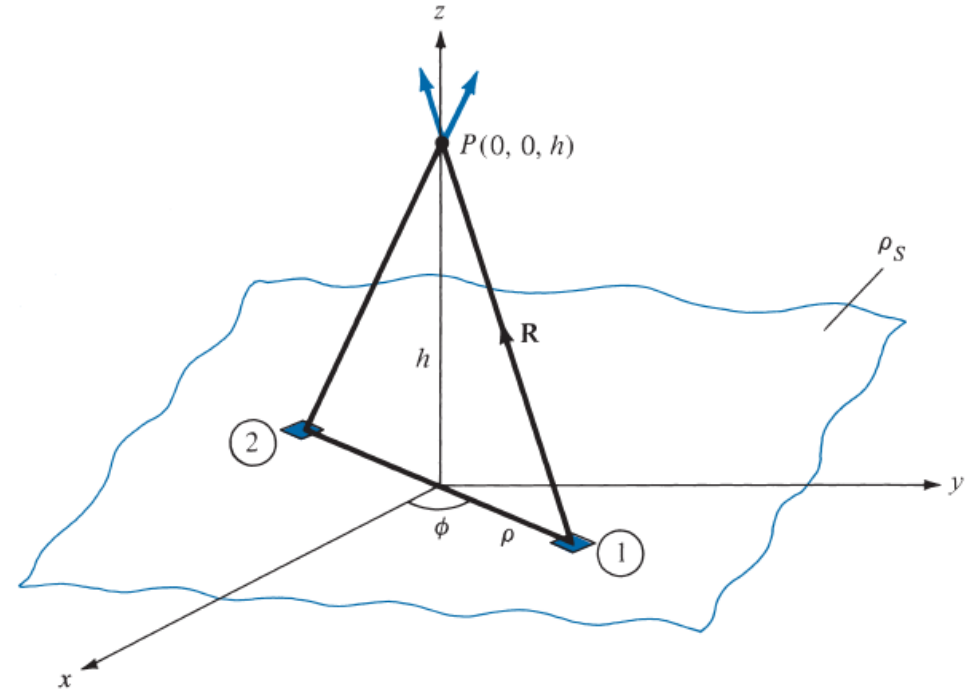


Electrostatics

- Infinite sheet of charge in the xy -plane

$$d\mathbf{E} = \frac{\sigma \rho d\rho d\phi}{4\pi\epsilon_0(\rho^2 + h^2)^{3/2}} (-\rho \mathbf{a}_\rho + h \mathbf{a}_z)$$

$$\mathbf{E} = \mathbf{a}_z \frac{\sigma}{4\pi\epsilon_0} \int_0^{2\pi} \int_0^\infty \frac{h \rho}{(\rho^2 + h^2)^{3/2}} d\rho d\phi$$

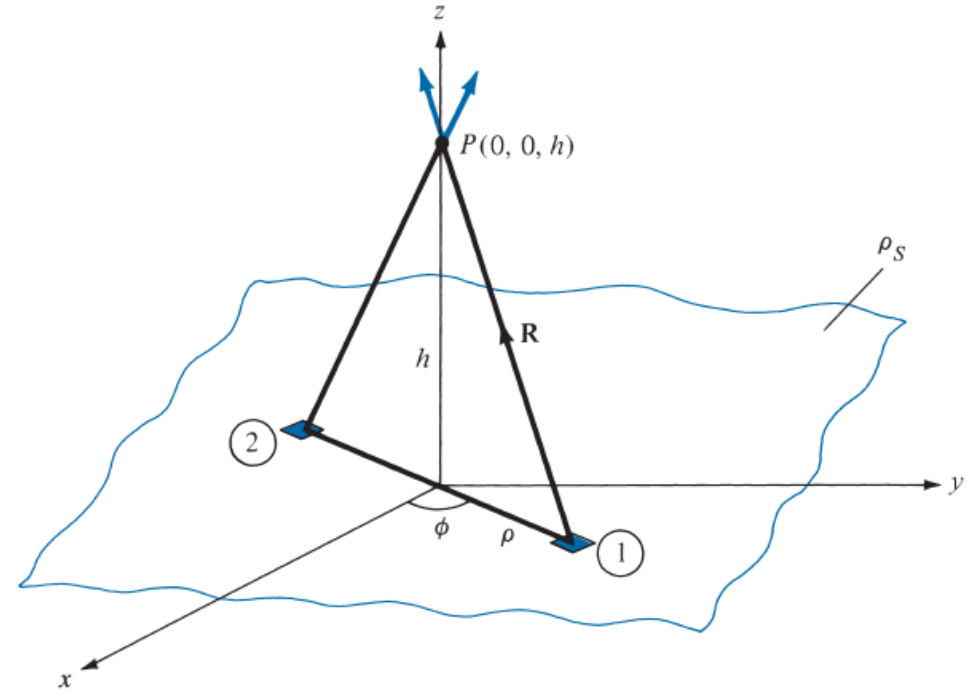


Electrostatics

- Infinite sheet of charge in the xy -plane

$$d\mathbf{E} = \frac{\sigma \rho d\rho d\phi}{4\pi\epsilon_0(\rho^2 + h^2)^{3/2}} (-\rho \mathbf{a}_\rho + h \mathbf{a}_z)$$

$$\mathbf{E} = \mathbf{a}_z \frac{\sigma}{4\pi\epsilon_0} \int_0^{2\pi} \int_0^\infty \frac{h \rho}{(\rho^2 + h^2)^{3/2}} d\rho d\phi$$



$$\int_0^\infty \frac{h \rho}{(\rho^2 + h^2)^{3/2}} d\rho = \frac{h}{2} \int_{h^2}^\infty u^{-3/2} du = \frac{h}{2} \left[-2u^{-1/2} \right]_{h^2}^\infty = 1$$

Electrostatics

- Infinite sheet of charge in the xy -plane

$$h > 0 \longrightarrow \mathbf{E} = \frac{\sigma}{2\epsilon_0} \mathbf{a}_z$$

$$h < 0 \longrightarrow \mathbf{E} = -\frac{\sigma}{2\epsilon_0} \mathbf{a}_z$$

